

17

Multiple Integrals

The concept of the definite integral, which was introduced in Chapter 5, may be extended to functions of several variables. In this chapter we shall define double

and triple integrals and discuss some of their fundamental properties and applications.

17.1 Double Integrals

Most of the integrals considered previously were of the form $\int_a^b f(x) dx$, where f is continuous on the interval $[a, b]$. There are many other types of integrals. Among those we shall study are *double integrals*, *triple integrals*, *surface integrals*, and *line integrals*. Each of these integrals is obtained in similar fashion. The principle difference is the domain of the integrand. Recall that $\int_a^b f(x) dx$ may be defined by applying the following four steps.

- (i) Partition $[a, b]$ by choosing $a = x_0 < x_1 < x_2 < \cdots < x_n = b$.
- (ii) For each i , select any number w_i in the subinterval $[x_{i-1}, x_i]$.
- (iii) Consider the Riemann sum $\sum_i f(w_i) \Delta x_i$, where $\Delta x_i = x_i - x_{i-1}$.
- (iv) If $\lim_{\|\Delta\| \rightarrow 0} \sum_i f(w_i) \Delta x_i$ exists, where $\|\Delta\|$ is the norm of the partition, then this limit equals $\int_a^b f(x) dx$.

Suppose next that f is a function of *two* variables and that $f(x, y)$ exists throughout a region R in the xy -plane. Our objective in this section is to define the *double integral* $\iint_R f(x, y) dA$. As will be seen, the procedure parallels steps (i)–(iv). To define this double integral it is essential to place restrictions on the region R . Although the restrictions may appear strong, most regions encountered in applications are included among those we shall

consider. Of primary importance are regions in the xy -plane of the types illustrated in Figure 17.1, where the functions g_1, g_2 and h_1, h_2 are continuous on the intervals $[a, b]$ and $[c, d]$, respectively, and where $g_1(x) \leq g_2(x)$ for all x in $[a, b]$ and $h_1(y) \leq h_2(y)$ for all y in $[c, d]$. The region illustrated in (i) of Figure 17.1 will be called a **region of Type I**, whereas that illustrated in (ii) of the figure will be called a **region of Type II**.

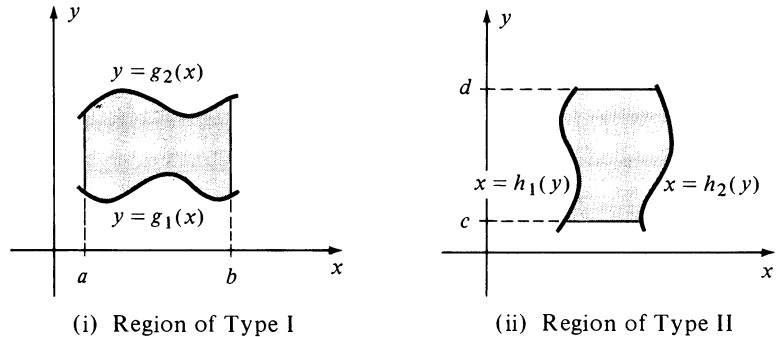


FIGURE 17.1

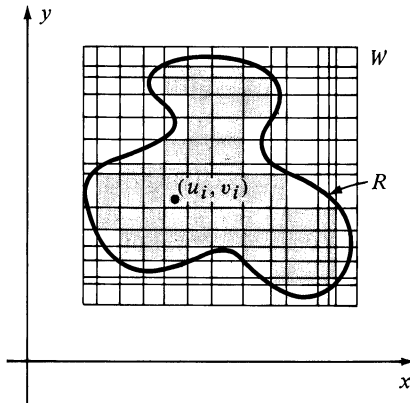


FIGURE 17.2

Throughout the remainder of this chapter, the symbol R will denote a region that can be decomposed into a finite number of subregions of Type I or II. Every such region R is a subset of a closed rectangular region W . If W is divided into smaller rectangles by means of a finite number of horizontal and vertical lines, then the totality of closed rectangular subregions that lie completely *within* R is called an **inner partition** of R . The shaded rectangles in Figure 17.2 illustrate an inner partition. If these shaded rectangular subregions are labeled $R_1, R_2, \dots, R_n$, then the inner partition is $P = \{R_i: i = 1, \dots, n\}$. The length of the longest diagonal of all the R_i will be denoted by $\|P\|$ and called the **norm of the partition** P . The symbol ΔA_i will be used for the area of R_i . Next, for each i , we choose any point (u_i, v_i) in R_i as illustrated in Figure 17.2. This corresponds to step (ii) of the four step process. Riemann sums (cf. step (iii)) are defined as follows.

Definition (17.1)

Let f be a function of two variables that is defined on a region R and let $P = \{R_i: i = 1, \dots, n\}$ be an inner partition of R . A **Riemann sum** of f for P is any sum of the form

$$\sum_{i=1}^n f(u_i, v_i) \Delta A_i$$

where (u_i, v_i) is in R_i and ΔA_i is the area of R_i .

The fourth step is to consider limits of such sums. First, however, let us consider a specific example of Definition (17.1).

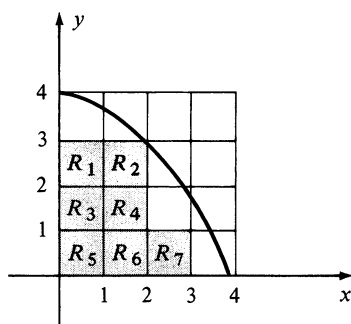


FIGURE 17.3

Example Suppose $f(x, y) = 4x + 2y + 1$ and

$$R = \{(x, y): 0 \leq x \leq 4, 0 \leq y \leq \frac{1}{4}(16 - x^2)\}.$$

Let $P = \{R_i: i = 1, \dots, 7\}$ be the inner partition of R determined by vertical and horizontal lines with integral intercepts, as shown in Figure 17.3. If the point (u_i, v_i) is the centroid of R_i , calculate the Riemann sum (17.1).

Solution Referring to Figure 17.3 we see that the points (u_i, v_i) should be chosen as follows: $(\frac{1}{2}, \frac{5}{2})$ in R_1 ; $(\frac{3}{2}, \frac{5}{2})$ in R_2 ; $(\frac{1}{2}, \frac{3}{2})$ in R_3 ; $(\frac{3}{2}, \frac{3}{2})$ in R_4 ; $(\frac{1}{2}, \frac{1}{2})$ in R_5 ; $(\frac{3}{2}, \frac{1}{2})$ in R_6 ; and $(\frac{5}{2}, \frac{1}{2})$ in R_7 . Since the area ΔA_i of each R_i is 1, the Riemann sum is

$$\begin{aligned} & f\left(\frac{1}{2}, \frac{5}{2}\right) \cdot 1 + f\left(\frac{3}{2}, \frac{5}{2}\right) \cdot 1 + f\left(\frac{1}{2}, \frac{3}{2}\right) \cdot 1 + f\left(\frac{3}{2}, \frac{3}{2}\right) \cdot 1 \\ & + f\left(\frac{1}{2}, \frac{1}{2}\right) \cdot 1 + f\left(\frac{3}{2}, \frac{1}{2}\right) \cdot 1 + f\left(\frac{5}{2}, \frac{1}{2}\right) \cdot 1. \end{aligned}$$

This simplifies to

$$8 + 12 + 6 + 10 + 4 + 8 + 12 = 60. \quad \blacksquare$$

In the preceding example, the point (u_i, v_i) was the centroid of R_i . It is important to remember that when working with *arbitrary* Riemann sums, *any* pair (u_i, v_i) in R_i may be chosen (see Exercise 1). For brevity we shall often denote inner partitions of R by $P = \{R_i\}$ without specifying the domain of the variable i , and the symbol $\sum_i$ will signify that the summation takes place over all rectangles R_i in the partition. The next definition of **limit of a sum** is patterned after Definition (5.7).

Definition (17.2)

Let f be a function of two variables that is defined on a region R and let L be a real number. The statement

$$\lim_{\|P\| \rightarrow 0} \sum_i f(u_i, v_i) \Delta A_i = L$$

means that for every $\varepsilon > 0$ there exists a $\delta > 0$ such that if $P = \{R_i\}$ is an inner partition of R with $\|P\| < \delta$, then

$$\left| \sum_i f(u_i, v_i) \Delta A_i - L \right| < \varepsilon$$

for every choice of (u_i, v_i) in R_i .

Intuitively, Definition (17.2) states that the Riemann sums of f get closer and closer to L as the norm $\|P\|$ approaches zero. If the limit L exists, it is unique. The next definition is analogous to (5.8).

Definition (17.3)

Let f be a function of two variables that is defined on a region R . The **double integral of f over R** , denoted by $\iint_R f(x, y) dA$, is

$$\iint_R f(x, y) dA = \lim_{\|P\| \rightarrow 0} \sum_i f(u_i, v_i) \Delta A_i$$

provided the limit exists.

If the double integral of f over R exists, then f is said to be **integrable over R** . It can be proved that if f is continuous on R , then f is integrable over R .

There is a useful geometric interpretation for double integrals whenever f is continuous and $f(x, y) \geq 0$ throughout R . Let S denote the graph of f and T the solid that lies under S and over R , as illustrated in Figure 17.4.

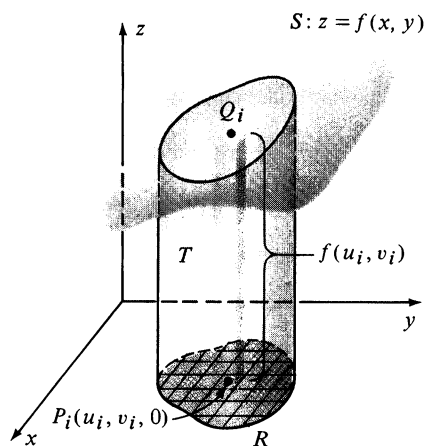


FIGURE 17.4

If $P_i(u_i, v_i, 0)$ is a point in the subregion R_i of an inner partition P of R , then $f(u_i, v_i)$ is the distance from the xy -plane to the point Q_i on S directly above P_i . The number $f(u_i, v_i) \Delta A_i$ is the volume of the prism with the rectangular base illustrated in Figure 17.4. The sum of all such volumes is an approximation to the volume V of T . Since it appears that this approximation improves as $\|P\|$ approaches zero, we define V as the limit of a sum of the numbers $f(u_i, v_i) \Delta A_i$. Applying (17.3) gives us the following.

Definition (17.4)

Let f be a function of two variables such that $f(x, y) \geq 0$ for all (x, y) in a region R . The **volume V** of the solid that lies under the graph of $z = f(x, y)$ and over R is

$$V = \iint_R f(x, y) dA.$$

Evidently, if $f(x, y) \leq 0$ throughout R , then the double integral of f over R is the *negative* of the volume of the solid that lies *over* the graph of f and *under* the region R .

We shall conclude this section by listing, without proof, some properties of double integrals that correspond to those given for definite integrals in Chapter 5. It is assumed that all regions and functions are suitably restricted so that the indicated integrals exist.

$$(17.5) \quad (i) \quad \iint_R cf(x, y) \, dA = c \iint_R f(x, y) \, dA, \quad \text{if } c \text{ is any real number.}$$

$$(ii) \quad \iint_R [f(x, y) + g(x, y)] \, dA = \iint_R f(x, y) \, dA + \iint_R g(x, y) \, dA.$$

(iii) If R is the union of two nonoverlapping regions R_1 and R_2 ,

$$\iint_R f(x, y) \, dA = \iint_{R_1} f(x, y) \, dA + \iint_{R_2} f(x, y) \, dA.$$

(iv) If $f(x, y) \geq 0$ throughout R , then $\iint_R f(x, y) \, dA \geq 0$.

17.1 Exercises

- 1 If f , R , and P are as in the Example of this section, find the Riemann sum of f for P if (u_i, v_i) is
 - (a) the point in the lower left-hand corner of R_i .
 - (b) the point in the upper right-hand corner of R_i .
- 2 Suppose f and R are as in the Example of this section and $P = \{R_i\}$ is the inner partition determined by horizontal and vertical lines having intercepts $0, \frac{1}{2}, \frac{3}{2}, \frac{5}{2}$, and 4. Find the Riemann sum of f for P if (u_i, v_i) is the centroid of R_i .
- 3 Let R be the triangular region with vertices $(0, 0)$, $(6, 0)$, $(6, 12)$ and let P be the inner partition of R determined by vertical lines with x -intercepts $0, 1, 3, 4, 6$ and horizontal lines with y -intercepts $0, 2, 5, 7, 8, 12$. If $f(x, y) = x^2y$, find the Riemann sum of f for P if (u_i, v_i) is the centroid of R_i .
- 4 Let $R = \{(x, y): 0 \leq x \leq 5, 0 \leq y \leq \sqrt{25 - x^2}\}$. Let P be the partition of R determined by vertical lines with x -intercepts $0, 2, 3, 4, 5$ and by horizontal lines with y -intercepts $0, 2, 4, 5$. If $f(x, y) = 4 - xy$, find the Riemann sum of f for P if (u_i, v_i) is the point in the lower right-hand corner of R_i .
- 5 Let R be the region bounded by the trapezoid with vertices $(0, 0)$, $(4, 4)$, $(8, 4)$, and $(12, 0)$. Let P be the partition of R determined by vertical lines with x -intercepts $0, 2, 4, 6, 8, 10, 12$ and by horizontal lines with y -intercepts $0, 2, 4$. If $f(x, y) = xy$, find the Riemann sum of f for P if (u_i, v_i) is the centroid of R_i .
- 6 Let R be the region bounded by the triangle with vertices $(-4, 0)$, $(0, 8)$, and $(4, 0)$. Let P be the partition of R determined by vertical lines with x -intercepts $-4, -2, 0, 1, 3, 4$ and by horizontal lines with y -intercepts $0, 2, 4, 6, 8$. If $f(x, y) = x + y$, find the Riemann sum of f for P if (u_i, v_i) is the upper right-hand corner of R_i .
- 7 If $R = \{(x, y): a \leq x \leq b, c \leq y \leq d\}$ and $f(x, y) = k$ for all (x, y) in R , prove that the double integral of f over R equals $k(b - a)(d - c)$ by using (a) Definition (17.3), and (b) Definition (17.4).
- 8 Suppose that the functions f and g are integrable over R and $f(x, y) \geq g(x, y)$ for all (x, y) in R . Use properties of double integrals to prove that

$$\iint_R f(x, y) \, dA \geq \iint_R g(x, y) \, dA.$$
- 9 Prove (i) of (17.5). (*Hint*: See the proof of (5.14) in Appendix II.)
- 10 Prove (ii) of (17.5). (*Hint*: See the proof of (5.15) in Appendix II.)

17.2 Evaluation of Double Integrals

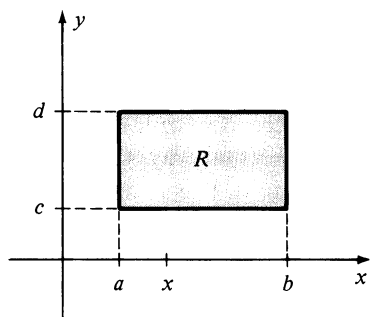


FIGURE 17.5

Except for elementary cases, it is virtually impossible to find the value of a double integral directly from Definition (17.3). This section contains methods of evaluation that make use of the Fundamental Theorem of Calculus.

Suppose f is a function of two variables that is continuous on a closed rectangular region R of the type illustrated in Figure 17.5. We shall use the symbol $\int_c^d f(x, y) dy$ to mean that x is held fixed and the integration is performed with respect to y . This is sometimes called a **partial integration with respect to y** . To each x in the interval $[a, b]$ there corresponds a unique value for this integral and hence a function A is determined, where the value $A(x)$ is given by

$$A(x) = \int_c^d f(x, y) dy.$$

As an illustration, if $f(x, y) = x^3 + 4y$, $c = 1$, and $d = 2$, then

$$\begin{aligned} A(x) &= \int_1^2 (x^3 + 4y) dy = x^3 y + 2y^2 \Big|_1^2 \\ &= (2x^3 + 8) - (x^3 + 2) \\ &= x^3 + 6. \end{aligned}$$

It can be proved that the function A is continuous on $[a, b]$ and hence has a definite integral which may be written

$$\int_a^b A(x) dx = \int_a^b \left[\int_c^d f(x, y) dy \right] dx.$$

The expression on the right side of this equation is called an **iterated (double) integral**. Ordinarily the notation is shortened by omitting the brackets, as in the following definition.

Definition (17.6)

$$\int_a^b \int_c^d f(x, y) dy dx = \int_a^b \left[\int_c^d f(x, y) dy \right] dx.$$

Example 1 Evaluate $\int_1^4 \int_{-1}^2 (2x + 6x^2 y) dy dx$.

Solution As in Definition (17.6), the integral equals

$$\begin{aligned} \int_1^4 \left[\int_{-1}^2 (2x + 6x^2 y) dy \right] dx &= \int_1^4 \left[2xy + 3x^2 y^2 \right]_{-1}^2 dx \\ &= \int_1^4 [(4x + 12x^2) - (-2x + 3x^2)] dx \\ &= \int_1^4 (6x + 9x^2) dx \\ &= 3x^2 + 3x^3 \Big|_1^4 = 234. \end{aligned}$$

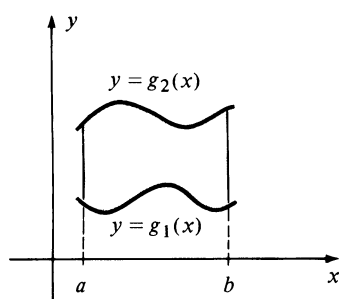
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Similarly, we may consider iterated integrals of the following form.

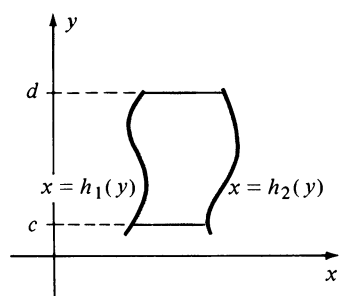
Definition (17.7)

$$\int_c^d \int_a^b f(x, y) dx dy = \int_c^d \left[\int_a^b f(x, y) dx \right] dy.$$

In Definition (17.7) we integrate first with respect to x (with y held fixed). After substituting the limits b and a for x in the usual way, the resulting expression is integrated with respect to y from c to d . It is proved in more advanced texts that the integrals in Definitions (17.6) and (17.7) are equal. This is illustrated in the next example where we reevaluate the integral in Example 1.



(i) Region of Type I



(ii) Region of Type II

FIGURE 17.6

Example 2 Evaluate $\int_{-1}^2 \int_1^4 (2x + 6x^2 y) dx dy$.

Solution Applying Definition (17.7),

$$\begin{aligned} \int_{-1}^2 \left[\int_1^4 (2x + 6x^2 y) dx \right] dy &= \int_{-1}^2 \left[x^2 + 2x^3 y \right]_1^4 dy \\ &= \int_{-1}^2 [(16 + 128y) - (1 + 2y)] dy \\ &= \int_{-1}^2 (126y + 15) dy \\ &= \left[63y^2 + 15y \right]_{-1}^2 = 234. \end{aligned}$$

Iterated integrals may be defined over nonrectangular regions. In particular, suppose f is continuous on a region of Type I or II, as shown in Figure 17.6. We define iterated integrals of f over the regions illustrated in (i) and (ii) of the figure as follows.

Definition (17.8)

$$\begin{aligned} \text{(i)} \quad & \int_a^b \int_{g_1(x)}^{g_2(x)} f(x, y) dy dx = \int_a^b \left[\int_{g_1(x)}^{g_2(x)} f(x, y) dy \right] dx \\ \text{(ii)} \quad & \int_c^d \int_{h_1(y)}^{h_2(y)} f(x, y) dx dy = \int_c^d \left[\int_{h_1(y)}^{h_2(y)} f(x, y) dx \right] dy \end{aligned}$$

In (i) of Definition (17.8) we perform a partial integration with respect to y and then substitute $g_2(x)$ and $g_1(x)$ for y in the usual way. The resulting expression in x is then integrated from a to b . Similar remarks hold for (ii).

Example 3 Evaluate $\int_0^2 \int_{x^2}^{2x} (x^3 + 4y) dy dx$.

Solution By (i) of Definition (17.8) the integral equals

$$\begin{aligned} \int_0^2 \left[\int_{x^2}^{2x} (x^3 + 4y) dy \right] dx &= \int_0^2 \left[x^3 y + 2y^2 \right]_{x^2}^{2x} dx \\ &= \int_0^2 [(2x^4 + 8x^2) - (x^5 + 2x^4)] dx \\ &= \left[\frac{8}{3}x^3 - \frac{1}{6}x^6 \right]_0^2 = \frac{32}{3}. \end{aligned}$$

Example 4 Evaluate $\int_1^3 \int_{\pi/6}^{y^2} 2y \cos x dx dy$.

Solution By (ii) of Definition (17.8) the integral equals

$$\begin{aligned} \int_1^3 \left[\int_{\pi/6}^{y^2} 2y \cos x dx \right] dy &= \int_1^3 \left[2y \sin x \right]_{\pi/6}^{y^2} dy \\ &= \int_1^3 (2y \sin y^2 - y) dy \\ &= \left[-\cos y^2 - \frac{1}{2}y^2 \right]_1^3 \\ &= (-\cos 9 - \frac{9}{2}) - (-\cos 1 - \frac{1}{2}) \\ &= \cos 1 - \cos 9 - 4 \approx -2.55. \end{aligned}$$

The next theorem indicates that if the region R is of Type I or II, then the double integral defined in Section 17.1 may be evaluated by means of an iterated integral.

Theorem (17.9)

- (i) Let R be a region of Type I that lies between the graphs of $y = g_1(x)$ and $y = g_2(x)$, where g_1 and g_2 are continuous on $[a, b]$. If f is continuous on R , then

$$\iint_R f(x, y) dA = \int_a^b \int_{g_1(x)}^{g_2(x)} f(x, y) dy dx.$$

- (ii) Let R be a region of Type II that lies between the graphs of $x = h_1(y)$ and $x = h_2(y)$, where h_1 and h_2 are continuous on $[c, d]$. If f is continuous on R then

$$\iint_R f(x, y) dA = \int_c^d \int_{h_1(y)}^{h_2(y)} f(x, y) dx dy.$$

It is important to note that in order to use (i) of Theorem 17.9, the region R must have a **lower boundary**, consisting of the graph of an equation $y = g_1(x)$ and an **upper boundary** consisting of the graph of an equation $y = g_2(x)$ (see (i) of Fig. 17.6). To use (ii), R must have a **left-hand boundary** consisting of the graph of an equation $x = h_1(y)$ and a **right-hand boundary** consisting of the graph of an equation $x = h_2(y)$ (see (ii) of Fig. 17.6). If the region is more complicated, then it is often possible to divide R into subregions of the required type and then apply (i) or (ii) to each subregion.

Instead of giving a rigorous proof of Theorem (17.9), we shall present an intuitive discussion that makes the result plausible, at least for nonnegative-valued functions. Suppose $f(x, y) \geq 0$ throughout the region R of Type I illustrated in (i) of Figure 17.6. Let S denote the graph of f , T the solid that lies under S and over R , and V the volume of T . We shall begin by considering the plane Γ that is parallel to the yz -plane and intersects the x -axis at the point $(x, 0, 0)$, where $a \leq x \leq b$. As illustrated in Figure 17.7, C will denote the trace of S on Γ . The points $P(x, g_1(x), 0)$ and $Q(x, g_2(x), 0)$ indicate where Γ intersects the boundaries $y = g_1(x)$ and $y = g_2(x)$ of R . In the figure we have shown the part of Γ that lies above the segment PQ and under the surface S .

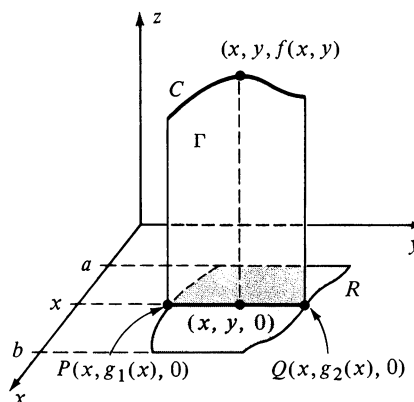


FIGURE 17.7

From our work in Chapter 6, the area $A(x)$ of this part of Γ is

$$A(x) = \int_{g_1(x)}^{g_2(x)} f(x, y) dy.$$

Since $A(x)$ is the area of a typical cross section of T , it follows from (6.8) that

$$V = \int_a^b A(x) dx = \int_a^b \int_{g_1(x)}^{g_2(x)} f(x, y) dy dx.$$

Using the fact that V is also given by Definition (17.4) we obtain the evaluation formula in part (i) of Theorem (17.9). A similar discussion can be given for part (ii).

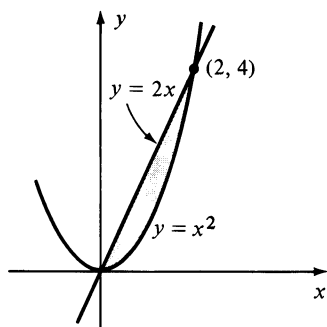


FIGURE 17.8

Example 5 Evaluate $\iint_R (x^3 + 4y) dA$, where R is the region in the xy -plane bounded by the graphs of the equations $y = x^2$ and $y = 2x$.

Solution The graph of R is sketched in Figure 17.8. Note that R is a region both of Type I and Type II. If we regard R as a region of Type I with lower boundary $y = x^2$ and upper boundary $y = 2x$, where $0 \leq x \leq 2$, then by (i) of Theorem 17.9,

$$\iint_R (x^3 + 4y) dA = \int_0^2 \int_{x^2}^{2x} (x^3 + 4y) dy dx.$$

From Example 3 we know that the last integral equals $\frac{32}{3}$.

If we regard R as a region of Type II, then the left-hand boundary is the graph of $x = \frac{1}{2}y$ and the right-hand boundary is the graph of $x = \sqrt{y}$, where $0 \leq y \leq 4$. Hence by (ii) of Theorem 17.9,

$$\begin{aligned} \iint_R f(x, y) dA &= \int_0^4 \int_{(1/2)y}^{\sqrt{y}} (x^3 + 4y) dx dy \\ &= \int_0^4 \left[\frac{1}{4}x^4 + 4yx \right]_{(1/2)y}^{\sqrt{y}} dy \\ &= \int_0^4 \left[\left(\frac{1}{4}y^2 + 4y^{3/2} \right) - \left(\frac{1}{64}y^4 + 2y^2 \right) \right] dy = \frac{32}{3}. \quad \blacksquare \end{aligned}$$

Example 6 Let R be the region bounded by the graphs of the equations $y = \sqrt{x}$, $y = \sqrt{3x - 18}$ and $y = 0$. If f is continuous on R , express the double integral $\iint_R f(x, y) dA$ in terms of iterated integrals by

- using only part (i) of Theorem 17.9.
- using only part (ii) of Theorem 17.9.

Solution The graphs of $y = \sqrt{x}$ and $y = \sqrt{3x - 18}$ are the top halves of the parabolas $y^2 = x$ and $y^2 = 3x - 18$. The region R is sketched in Figure 17.9.

(a) If we wish to use only part (i) of Theorem 17.9, then it is necessary to employ two iterated integrals, because if $0 \leq x \leq 6$, then the lower boundary of the region is the graph of $y = 0$, whereas if $6 \leq x \leq 9$, the lower boundary is the graph of $y = \sqrt{3x - 18}$. If R_1 denotes the part of the region R that lies between $x = 0$ and $x = 6$, and if R_2 denotes the part between $x = 6$ and $x = 9$, then both R_1 and R_2 are regions of Type I. Hence

$$\begin{aligned} \iint_R f(x, y) dA &= \iint_{R_1} f(x, y) dA + \iint_{R_2} f(x, y) dA \\ &= \int_0^6 \int_0^{\sqrt{x}} f(x, y) dy dx + \int_6^9 \int_{\sqrt{3x-18}}^{\sqrt{x}} f(x, y) dy dx. \end{aligned}$$

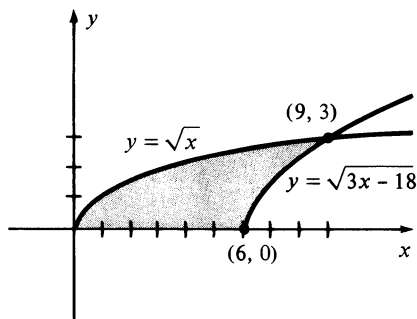


FIGURE 17.9

(b) To use (ii) of Theorem 17.9 we must solve each of the given equations for x in terms of y , obtaining

$$x = y^2 \quad \text{and} \quad x = \frac{y^2 + 18}{3} = \frac{1}{3}y^2 + 6.$$

Only one iterated integral is required in this case since R is a region of Type II. Thus

$$\iint_R f(x, y) dA = \int_0^3 \int_{y^2}^{(1/3)y^2 + 6} f(x, y) dx dy. \quad \blacksquare$$

Examples 5 and 6 indicate how certain double integrals may be evaluated by using either (i) or (ii) of Theorem 17.9. Generally, the choice of the *order of integration* $dy dx$ or $dx dy$ often depends on the form of $f(x, y)$ and the region R . Sometimes it is extremely difficult, or even impossible, to evaluate a given iterated double integral. However, by **reversing the order of integration** from $dy dx$ to $dx dy$, or vice versa, it may be possible to find an equivalent iterated double integral that is easy to evaluate. This technique is illustrated in the next example.

Example 7 Given $\int_0^4 \int_{\sqrt{y}}^2 y \cos x^5 dx dy$, reverse the order of integration and evaluate the resulting integral.

Solution Since the given order of integration is $dx dy$, the region R is of Type II. As illustrated in Figure 17.10, the left-hand and right-hand boundaries are the graphs of $x = \sqrt{y}$ and $x = 2$, respectively, and $0 \leq y \leq 4$.

Note that R is also a region of Type I whose lower and upper boundaries are given by $y = 0$ and $y = x^2$, respectively, and where $0 \leq x \leq 2$. Hence by Theorem 17.9,

$$\begin{aligned} \int_0^4 \int_{\sqrt{y}}^2 y \cos x^5 dx dy &= \iint_R y \cos x^5 dA = \int_0^2 \int_0^{x^2} y \cos x^5 dy dx \\ &= \int_0^2 \left[\frac{y^2}{2} \cos x^5 \right]_0^{x^2} dx = \int_0^2 \frac{x^4}{2} \cos x^5 dx \\ &= \frac{1}{10} \int_0^2 \cos x^5 (5x^4) dx \\ &= \frac{1}{10} \sin x^5 \Big|_0^2 = \frac{1}{10} \sin 32 \approx 0.055. \quad \blacksquare \end{aligned}$$

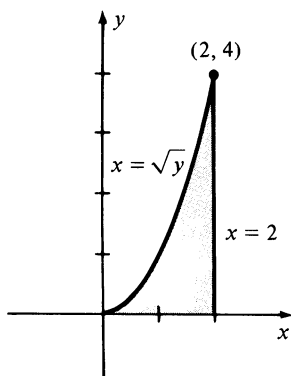


FIGURE 17.10

On page 788 we saw that the volume V of a solid that lies under the graph of $z = f(x, y)$ and over a region of Type I in the xy -plane can be expressed as

$$V = \int_a^b A(x) dx = \int_a^b \int_{g_1(x)}^{g_2(x)} f(x, y) dy dx$$

where $A(x)$ is the area of the typical cross section of the solid shown in Figure 17.7. It is often useful to regard these two integrals as limits of sums. Thus, by (6.8), we may write

$$V = \int_a^b A(x) dx = \lim_{\|P'\| \rightarrow 0} \sum_i A(u_i) \Delta x_i$$

where P' is a partition of the interval $[a, b]$, u_i is any number in the i th subinterval $[x_{i-1}, x_i]$ of P' , and $\Delta x_i = x_i - x_{i-1}$. As illustrated in Figure 17.11, $A(u_i) \Delta x_i$ is the volume of a lamina L_i whose face is parallel to the yz -plane and whose base is a rectangle of width Δx_i in the xy -plane. Consequently, the volume V may be regarded as a limit of a sum of volumes of such laminas.

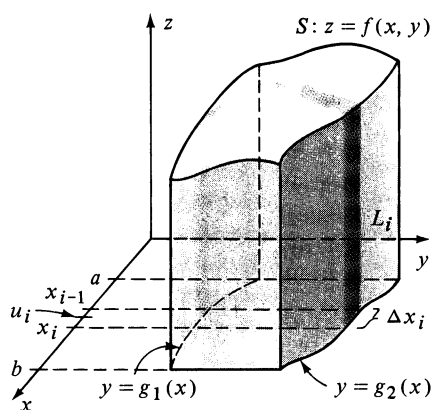


FIGURE 17.11

The *iterated* integral $V = \int_a^b \int_{g_1(x)}^{g_2(x)} f(x, y) dy dx$ may also be expressed in terms of limits of sums. First, applying the definition of the definite integral,

$$A(x) = \int_{g_1(x)}^{g_2(x)} f(x, y) dy = \lim_{\|P''\| \rightarrow 0} \sum_j f(x, v_j) \Delta y_j$$

where, for each x , P'' is a partition of the y -interval $[g_1(x), g_2(x)]$, v_j is a number in the j th subinterval $[y_{j-1}, y_j]$ of P'' , and $\Delta y_j = y_j - y_{j-1}$. Hence, for each u_i in $[a, b]$,

$$A(u_i) = \lim_{\|P''\| \rightarrow 0} \sum_j f(u_i, v_j) \Delta y_j$$

and, therefore,

$$\begin{aligned} V &= \int_a^b A(x) dx = \lim_{\|P'\| \rightarrow 0} \sum_i A(u_i) \Delta x_i \\ &= \lim_{\|P'\| \rightarrow 0} \sum_i \left[\lim_{\|P''\| \rightarrow 0} \sum_j f(u_i, v_j) \Delta y_j \right] \Delta x_i. \end{aligned}$$

If we ignore the limits in this formula, there results what is called a **double sum**, written

$$\sum_i \sum_j f(u_i, v_j) \Delta y_j \Delta x_i.$$

Referring to Figure 17.12, we see that the expression $f(u_i, v_j) \Delta y_j \Delta x_i$ may be regarded as the volume of a prism having base area $\Delta y_j \Delta x_i$ and altitude $f(u_i, v_j)$.

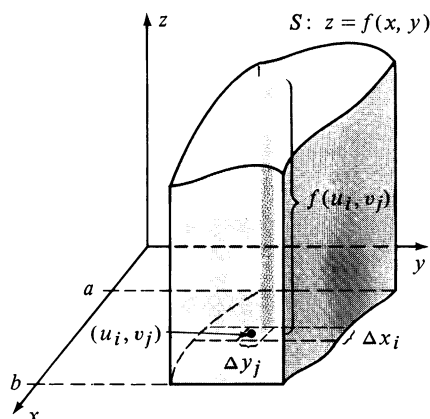


FIGURE 17.12

The double sum may then be interpreted as follows. *Holding x fixed* we sum the volumes of the prisms in the direction of the y -axis, thereby obtaining the volume of the lamina illustrated in Figure 17.11. The volumes of all such laminae are then added by summing in the direction of the x -axis. In a certain sense, the iterated integral does this and at the same time takes limits as the bases of the prisms shrink to points. This interpretation is often useful in visualizing applications of the double integral. To emphasize this point of view we sometimes write

$$(17.10) \quad V = \lim_{\|\Delta\| \rightarrow 0} \sum_i \sum_j f(u_i, v_j) \Delta y_j \Delta x_i = \int_a^b \int_{g_1(x)}^{g_2(x)} f(x, y) dy dx$$

where the symbol $\|\Delta\| \rightarrow 0$ is used to signify that all the Δx_i and Δy_j approach 0.

17.2 Exercises

Using the values given in Exercises 1 and 2, verify that

$$\int_a^b \int_c^d f(x, y) dy dx = \int_c^d \int_a^b f(x, y) dx dy.$$

1 $a = 1, b = 2, c = -1, d = 2; f(x, y) = 12xy^2 - 8x^3$

2 $a = -2, b = -1, c = 0, d = 3; f(x, y) = 4xy^3 + y$

Evaluate the iterated integrals in Exercises 3–12.

3 $\int_1^2 \int_{1-x}^{\sqrt{x}} x^2 y dy dx$

4 $\int_{-1}^1 \int_{x^3}^{x+1} (3x + 2y) dy dx$

5 $\int_0^2 \int_{y^2}^{2y} (4x - y) dx dy$

$$6 \int_0^1 \int_{-y-1}^{y-1} (x^2 + y^2) dx dy$$

$$7 \int_1^2 \int_{x^3}^x e^{y/x} dy dx$$

$$8 \int_0^{\pi/6} \int_0^{\pi/2} (x \cos y - y \cos x) dy dx$$

$$9 \int_1^e \int_0^x \ln x dy dx$$

$$10 \int_0^1 \int_y^1 \frac{1}{1+y^2} dx dy$$

$$11 \int_{\pi/6}^{\pi/4} \int_{\tan x}^{\sec x} (y + \sin x) dy dx$$

$$12 \int_{\pi/6}^{\pi/4} \int_0^{\sin x} e^y \cos x dy dx$$

In Exercises 13–18 sketch the region R bounded by the graphs of the given equations and express $\iint_R f(x, y) dA$ as an iterated integral by using (a) part (i) of Theorem (17.9) and (b) part (ii) of Theorem (17.9).

$$13 \quad y = \sqrt{x}, x = 4, y = 0 \qquad 14 \quad y = \sqrt{x}, x = 0, y = 2$$

$$15 \quad y = x^3, x = 0, y = 8 \qquad 16 \quad y = x^3, x = 2, y = 0$$

$$17 \quad y = \sqrt{x}, y = x^3 \qquad 18 \quad y = \sqrt{1-x^2}, y = 0$$

In Exercises 19–24 express the double integral over R as an iterated integral and find its value.

$$19 \quad \iint_R (y + 2x) dA, \text{ where } R \text{ is the rectangular region with vertices } (-1, -1), (2, -1), (2, 4), \text{ and } (-1, 4).$$

$$20 \quad \iint_R (x - y) dA, \text{ where } R \text{ is the triangular region with vertices } (2, 9), (2, 1), \text{ and } (-2, 1).$$

$$21 \quad \iint_R xy^2 dA, \text{ where } R \text{ is the triangular region with vertices } (0, 0), (3, 1), \text{ and } (-2, 1).$$

$$22 \quad \iint_R (y + 1) dA, \text{ where } R \text{ is the region between the graphs of } y = \sin x \text{ and } y = \cos x \text{ from } x = 0 \text{ to } x = \pi/4.$$

$$23 \quad \iint_R x^3 \cos xy dA, \text{ where } R \text{ is the region bounded by the graphs of } y = x^2, y = 0, \text{ and } x = 2.$$

$$24 \quad \iint_R e^{x/y} dA, \text{ where } R \text{ is the region bounded by the graphs of } y = 2x, y = -x, \text{ and } y = 4.$$

In Exercises 25–30 sketch the region R bounded by the graphs of the given equations. If f is continuous on R , use Definition (17.4) to express $\iint_R f(x, y) dA$ as a sum of two iterated integrals of the type used in (a) part (i) of Theorem (17.9); (b) part (ii) of Theorem (17.9).

$$25 \quad 8y = x^3, y - x = 4, 4x + y = 9$$

$$26 \quad x = 2\sqrt{y}, \sqrt{3}x = \sqrt{y}, y = 2x + 5$$

$$27 \quad x = \sqrt{3-y}, y = 2x, x + y + 3 = 0$$

$$28 \quad x + 2y = 5, x - y = 2, 2x + y = -2$$

$$29 \quad y = e^x, y = \ln x, x + y = 1, x + y = 1 + e$$

$$30 \quad y = \sin x, \pi y = 2x$$

In Exercises 31–38 sketch the region of integration for the iterated integral.

$$31 \quad \int_{-1}^2 \int_{-\sqrt{4-x^2}}^{4-x^2} f(x, y) dy dx$$

$$32 \quad \int_{-1}^2 \int_{x^2-4}^{x-2} f(x, y) dy dx$$

$$33 \quad \int_0^1 \int_{\sqrt{y}}^{3\sqrt{y}} f(x, y) dx dy$$

$$34 \quad \int_{-2}^{-1} \int_{3y}^{2y} f(x, y) dx dy$$

$$35 \quad \int_{-3}^1 \int_{\tan^{-1}x}^{e^x} f(x, y) dy dx$$

$$36 \quad \int_{-1}^2 \int_{\sinh x}^{\cosh x} f(x, y) dy dx$$

$$37 \quad \int_{-1}^1 \int_{1-y^4}^{\cosh y} f(x, y) dx dy$$

$$38 \quad \int_{\pi}^{2\pi} \int_{\sin y}^{\ln y} f(x, y) dx dy$$

In Exercises 39–44 reverse the order of integration and evaluate the resulting integral.

$$39 \quad \int_0^1 \int_{2x}^2 e^{(y^2)} dy dx$$

$$40 \quad \int_0^9 \int_{\sqrt{y}}^3 \sin x^3 dx dy$$

$$41 \quad \int_0^2 \int_{y^2}^4 y \cos x^2 dx dy$$

$$42 \quad \int_1^e \int_0^{\ln x} y dy dx$$

$$43 \quad \int_0^8 \int_{3\sqrt{y}}^2 \frac{y}{\sqrt{16+x^7}} dx dy$$

$$44 \quad \int_0^1 \int_x^1 \frac{1}{y} \sin y \cos \frac{x}{y} dy dx$$

$$45 \quad \text{Interpret the iterated integral in part (ii) of Theorem (17.9) as a limit of a double sum in a manner analogous to (17.10).}$$

17.3 Areas and Volumes

It is customary to denote the double integral $\iint_R 1 \, dA$ by $\iint_R dA$. If R is a region of Type I, as illustrated in (i) of Figure 17.6, then by Theorem (17.9),

$$\begin{aligned}\iint_R dA &= \int_a^b \int_{g_1(x)}^{g_2(x)} 1 \, dy \, dx = \int_a^b \left[y \right]_{g_1(x)}^{g_2(x)} dx \\ &= \int_a^b [g_2(x) - g_1(x)] \, dx\end{aligned}$$

which, according to Theorem (6.1), equals the area A of R . The same is true if R is a region of Type II, as illustrated in (ii) of Figure 17.6. These facts are also evident from the definition of the double integral, for if $f(x, y) = 1$ throughout R , then the Riemann sum in Definition (17.3) is a sum of areas of rectangles in an inner partition P of R . As the norm $\|P\|$ approaches zero, these rectangles cover more of R and it is apparent that the limit equals the area of R .

If an iterated integral is used to find an area it may be regarded as a limit of a double sum in a manner similar to that used for volumes at the end of the previous section. Specifically, as in (17.10), with $f(x, y) = 1$, we write

$$A = \iint_R dA = \int_a^b \int_{g_1(x)}^{g_2(x)} dy \, dx = \lim_{\|\Delta\| \rightarrow 0} \sum_i \sum_j \Delta y_j \Delta x_i.$$

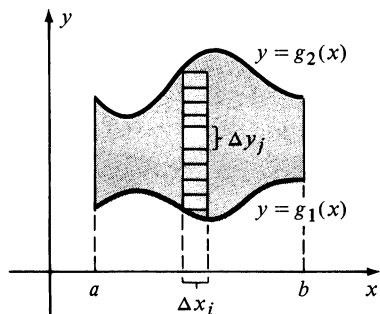


FIGURE 17.13

The double sum may be interpreted as follows. First, *holding x fixed* we sum over rectangles of areas $\Delta y_j \Delta x_i$ in the direction of the y -axis, from the graph of $y = g_1(x)$ to the graph of $y = g_2(x)$. This gives us the area of a rectangular strip that is parallel to the y -axis, as illustrated in Figure 17.13. Second, we sweep out the region R by summing over all such strips from $x = a$ to $x = b$. The limit of the double sum (that is, the iterated integral) does this and, at the same time, forces all the Δx_i and Δy_j to approach 0.

Example 1 Find the area A of the region bounded by the graphs of $2y = 16 - x^2$ and $x + 2y - 4 = 0$.

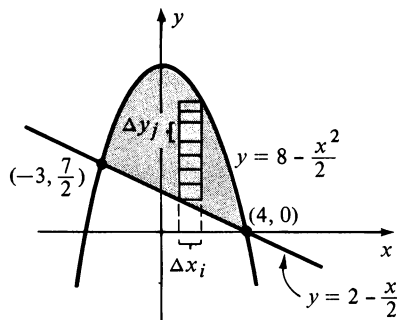


FIGURE 17.14

Solution The region lies under the parabola $y = 8 - (x^2)/2$ and over the line $y = 2 - (x/2)$, as illustrated in Figure 17.14. Also shown in the figure is a typical rectangle of area $\Delta y_j \Delta x_i$ and the strip obtained by summing in the direction of the y -axis. Using an iterated integral we have

$$\begin{aligned}A &= \int_{-3}^4 \int_{2-(x/2)}^{8-(x^2/2)} dy \, dx = \int_{-3}^4 \left[\left(8 - \frac{x^2}{2} \right) - \left(2 - \frac{x}{2} \right) \right] dx \\ &= 6x - \frac{x^3}{6} + \frac{x^2}{4} \Big|_{-3}^4 = \frac{343}{12}.\end{aligned}$$

■

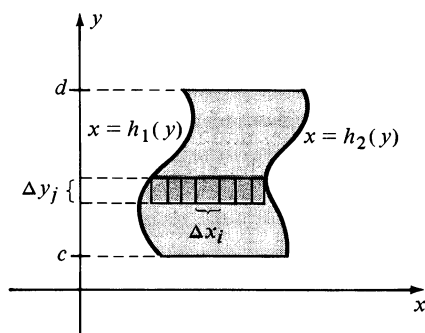


FIGURE 17.15

If part (ii) of Theorem (17.9) is used to find areas, then

$$A = \iint_R dA = \int_c^d \int_{h_1(y)}^{h_2(y)} dx dy = \lim_{\|\Delta\| \rightarrow 0} \sum_j \sum_i \Delta x_i \Delta y_j.$$

The preceding double sum may be interpreted as follows. First, *holding y fixed* we sum over rectangles of areas $\Delta x_i \Delta y_j$ in the direction of the x -axis, from the graph of $x = h_1(y)$ to the graph of $x = h_2(y)$. This gives us the area of a rectangular strip that is parallel to the x -axis, as illustrated in Figure 17.15. Second, we sweep out the region R by summing over all such strips from $y = c$ to $y = d$. As before, the iterated integral accomplishes this and simultaneously forces all the Δx_i and Δy_j to approach 0.

Example 2 Find the area A of the region in the xy -plane bounded by the graphs of $x = y^3$, $x + y = 2$, and $y = 0$.

Solution The region is sketched in Figure 17.16 together with a rectangle of area $\Delta x_i \Delta y_j$ and a strip obtained by summing in the direction of the x -axis.

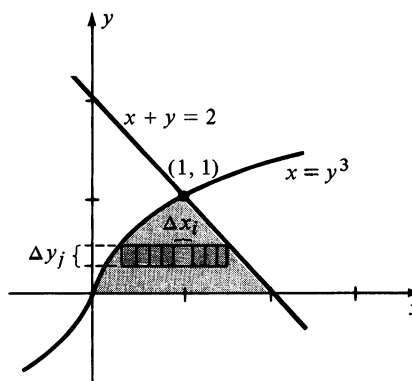


FIGURE 17.16

Using an iterated integral gives us

$$\begin{aligned} A &= \iint_R dA = \int_0^1 \int_{y^3}^{2-y} dx dy = \int_0^1 x \Big|_{y^3}^{2-y} dy \\ &= \int_0^1 (2 - y - y^3) dy = 2y - \frac{y^2}{2} - \frac{y^4}{4} \Big|_0^1 = \frac{5}{4}. \end{aligned}$$

The area can also be found by employing part (i) of Theorem (17.9); however, in this case it is necessary to divide R into two parts by means of the vertical line through $(1, 1)$. We then have

$$A = \int_0^1 \int_0^{\sqrt[3]{x}} dy dx + \int_1^2 \int_0^{2-x} dy dx. \quad \blacksquare$$

We can use Definition (17.4) to express volumes of certain solids in terms of double integrals that may then be evaluated by means of iterated integrals. This is illustrated in the next example.

Example 3 Find the volume V of the solid in the first octant bounded by the coordinate planes and the graphs of the equations $z = x^2 + y^2 + 1$ and $2x + y = 2$.

Solution As illustrated in Figure 17.17, the solid lies under the paraboloid $z = x^2 + y^2 + 1$ and over the triangular region R in the xy -plane bounded by the coordinate axes and the line $y = 2 - 2x$.

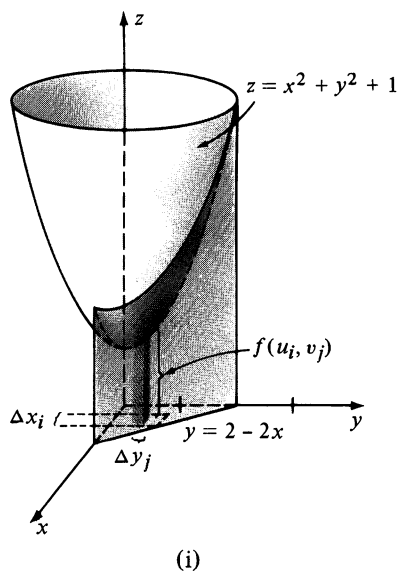
By Definition (17.4), with $f(x, y) = x^2 + y^2 + 1$,

$$V = \iint_R (x^2 + y^2 + 1) dA.$$

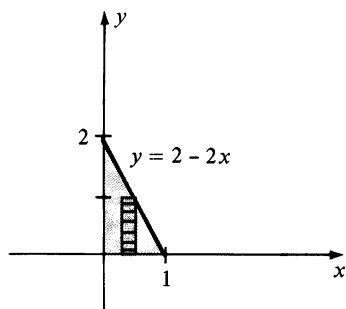
As in (17.10), this integral may be interpreted as the limit of a double sum

$$\sum_i \sum_j f(u_i, v_j) \Delta y_j \Delta x_i = \sum_i \sum_j (u_i^2 + v_j^2 + 1) \Delta y_j \Delta x_i$$

where $(u_i^2 + v_j^2 + 1) \Delta y_j \Delta x_i$ represents the volume of a prism of the type shown in (i) of Figure 17.17, and where we first sum in the direction of the y -axis from $y = 0$ to $y = 2 - 2x$. The sketch in (ii) of the figure shows the region R in the xy -plane and a strip corresponding to the first summation. This first summation gives us the volume of a lamina whose face is parallel to the yz -plane. We then sum these laminar volumes in the direction of the x -axis from $x = 0$ to $x = 1$. Using an iterated integral,



(i)



(ii)

FIGURE 17.17

$$\begin{aligned} V &= \int_0^1 \int_0^{2-2x} (x^2 + y^2 + 1) dy dx = \int_0^1 \left[x^2 y + \frac{1}{3} y^3 + y \right]_0^{2-2x} dx \\ &= \int_0^1 \left(-\frac{14}{3} x^3 + 10x^2 - 10x + \frac{14}{3} \right) dx \\ &= -\frac{7}{6} x^4 + \frac{10}{3} x^3 - 5x^2 + \frac{14}{3} x \Big|_0^1 = \frac{11}{6}. \end{aligned}$$

We may also find V by integrating first with respect to x . The reader should verify that in this case the iterated integral has the form

$$V = \int_0^2 \int_0^{(2-y)/2} (x^2 + y^2 + 1) dx dy. \quad \blacksquare$$

Example 4 Find the volume V of the solid bounded by the graphs of $x^2 + y^2 = 9$ and $y^2 + z^2 = 9$.

Solution The graphs are cylinders of radius 3 whose axes intersect orthogonally. That portion of the solid that lies in the first octant is illustrated in (i) of Figure 17.18. By symmetry it is sufficient to find the volume of this portion and then multiply the result by 8. Referring to (i) of the figure we see that the solid lies under the graph of $z = \sqrt{9 - y^2}$ and over the region R bounded by the x - and y -axes and the graph of $x^2 + y^2 = 9$. Thus, by Definition (17.4),

$$V = 8 \iint_R (9 - y^2)^{1/2} dA.$$

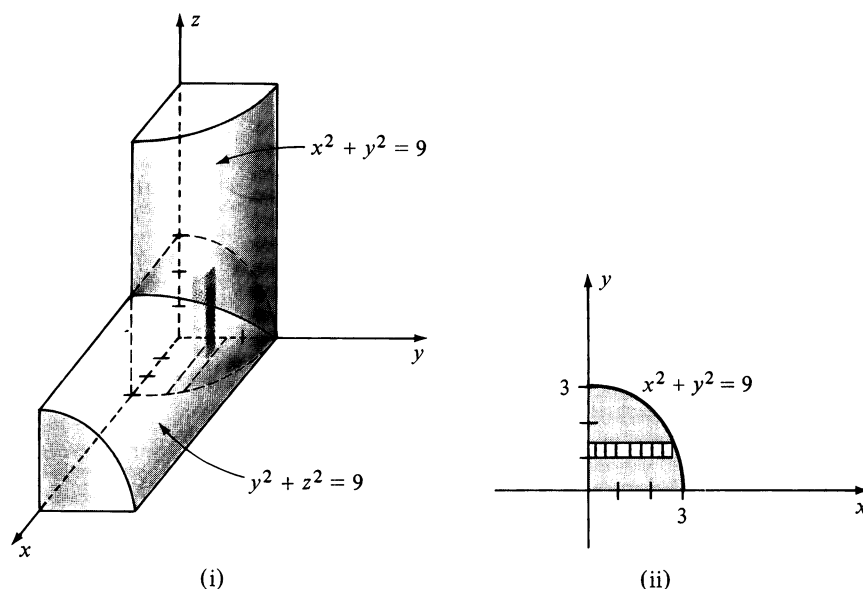


FIGURE 17.18

We shall evaluate the double integral by using (ii) of Theorem (17.9). In this case we regard the integral as a limit of a double sum of volumes of prisms of the type shown in (i) of Figure 17.18, where we first sum in the direction of the x -axis from $x = 0$ to $x = (9 - y^2)^{1/2}$. The sketch in (ii) of Figure 17.18 indicates the region R in the xy -plane and a strip corresponding to this first summation. This gives us the volume of a lamina whose face is parallel to the xz -plane. We then sum these laminar volumes in the direction of the y -axis from $y = 0$ to $y = 3$. Using an iterated integral, we have

$$\begin{aligned} V &= 8 \int_0^3 \int_0^{(9-y^2)^{1/2}} (9-y^2)^{1/2} dx dy = 8 \int_0^3 \left[(9-y^2)^{1/2} x \right]_0^{(9-y^2)^{1/2}} dy \\ &= 8 \int_0^3 (9-y^2) dy = 8 \left[9y - \frac{1}{3}y^3 \right]_0^3 = 144. \end{aligned}$$

The double integral could also be evaluated by integrating first with respect to y ; however, this order of integration is much more complicated. ■

17.3 Exercises

In Exercises 1–10 sketch the region bounded by the graphs of the given equations and find its area by means of double integrals. Interpret each iterated double integral as a limit of a double sum by sketching a typical rectangle of area $\Delta y_j \Delta x_i$ and the rectangular strip that corresponds to the first summation of the double sum.

1 $y = 1/x^2$, $y = -x^2$, $x = 1$, $x = 2$

2 $y = \sqrt{x}$, $y = -x$, $x = 1$, $x = 4$

3 $y^2 = -x$, $x - y = 4$, $y = -1$, $y = 2$

4 $x = y^2$, $y - x = 2$, $y = -2$, $y = 3$

5 $y = x$, $y = 3x$, $x + y = 4$

6 $x - y + 1 = 0$, $7x - y - 17 = 0$, $2x + y + 2 = 0$

7 $y = e^x$, $y = \sin x$, $x = -\pi$, $x = \pi$

8 $y = \ln |x|$, $y = 0$, $y = 1$

- 9 $y = x^2, y' = 1/(1 + x^2)$
- 10 $y = \cosh x, y = \sinh x, x = -1, x = 1$
- 11 Find the volume of the solid that lies under the graph of $z = 4x^2 + y^2$ and over the rectangular region R in the xy -plane having vertices $(0, 0, 0)$, $(0, 1, 0)$, $(2, 0, 0)$, and $(2, 1, 0)$.
- 12 Find the volume of the solid that lies under the graph of $z = x^2 + 4y^2$ and over the triangular region R in the xy -plane having vertices $(0, 0, 0)$, $(1, 0, 0)$, and $(1, 2, 0)$.

In Exercises 13–20 sketch the solid in the first octant that is bounded by the graphs of the given equations and find its volume.

- 13 $x^2 + z^2 = 9, y = 2x, y = 0, z = 0$
- 14 $z = 4 - x^2, x + y = 2, x = 0, y = 0, z = 0$
- 15 $2x + y + z = 4, x = 0, y = 0, z = 0$
- 16 $y^2 = z, y = x, x = 4, z = 0$
- 17 $z = x^2 + y^2, y = 4 - x^2, x = 0, y = 0, z = 0$
- 18 $z = y^3, y = x^3, x = 0, z = 0, y = 1$
- 19 $z = x^3, x = 4y^2, 16y = x^2, z = 0$
- 20 $x^2 + y^2 = 16, x = z, y = 0, z = 0$

In Exercises 21 and 22 find the volume of the solid bounded by the graphs of the given equations.

- 21 $z = x^2 + 4, y = 4 - x^2, x + y = 2, z = 0$
- 22 $y = x^3, y = x^4, z - x - y = 4, z = 0$
- 23 Find the volume of the solid bounded by the graphs of $x^2 + z^2 = a^2$ and $x^2 + y^2 = a^2$, where $a > 0$.
- 24 Find the volume of the solid bounded by the graph of $x^{2/3} + y^{2/3} + z^{2/3} = a^{2/3}$, where $a > 0$.

The iterated integrals in Exercises 25–30 represent volumes of solids. Describe each solid.

- 25 $\int_{-2}^1 \int_{x-1}^{1-x^2} (x^2 + y^2) dy dx$
- 26 $\int_0^1 \int_{3-x}^{3-x^2} \sqrt{25 - x^2 - y^2} dy dx$
- 27 $\int_0^4 \int_{y/4}^{\sqrt{y}} (x + y) dx dy$
- 28 $\int_0^1 \int_{y^{1/2}}^{y^{1/3}} \sqrt{x^2 + y^2} dx dy$
- 29 $\int_0^4 \int_{-1}^2 3 dy dx$
- 30 $\int_0^\pi \int_0^{\sin x} dy dx$

17.4 Moments and Center of Mass

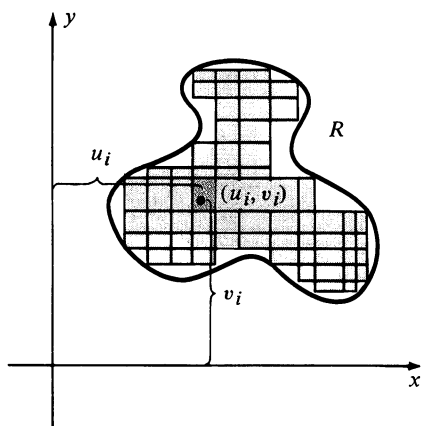


FIGURE 17.19

The first moment and center of mass of a homogeneous lamina were discussed in Chapter 9. We shall employ double integrals to extend these concepts to laminas in which the density is not constant.

Let T denote a lamina that has the shape of a region R in the xy -plane and suppose that the area density, that is, the mass per unit area, at the point (x, y) is given by $\rho(x, y)$, where ρ is a continuous function on R . If $P = \{R_i\}$ is an inner partition of R , then (u_i, v_i) will denote a point in R_i (see Figure 17.19). The symbol T_i will represent the part of the lamina T that corresponds to R_i . Since ρ is continuous, a small change in (x, y) produces a small change in the density $\rho(x, y)$, that is, ρ is almost constant on R_i . Hence, if $\|P\| \approx 0$, then the mass T_i may be approximated by $\rho(u_i, v_i) \Delta A_i$, where ΔA_i is the area of R_i . The sum $\sum_i \rho(u_i, v_i) \Delta A_i$ is an approximation to the mass of the lamina T . The mass M of T is defined as the limit of such sums as $\|P\| \rightarrow 0$. This gives us

$$M = \lim_{\|P\| \rightarrow 0} \sum_i \rho(u_i, v_i) \Delta A_i = \iint_R \rho(x, y) dA.$$

If the mass of T_i is assumed to be concentrated at (u_i, v_i) , then as in Chapter 9, the moment of T_i with respect to the x -axis is the product $v_i \rho(u_i, v_i) \Delta A_i$. The moment M_x of T with respect to the x -axis is defined as the limit of such sums, that is,

$$M_x = \lim_{\|P\| \rightarrow 0} \sum_i v_i \rho(u_i, v_i) \Delta A_i = \iint_R y \rho(x, y) dA.$$

Similarly, the moment M_y of T with respect to the y -axis is

$$M_y = \lim_{\|P\| \rightarrow 0} \sum_i u_i \rho(u_i, v_i) \Delta A_i = \iint_R x \rho(x, y) dA.$$

As usual, the center of mass of the lamina is the point $(\bar{x}, \bar{y})$ such that $\bar{x} = M_y/M$, $\bar{y} = M_x/M$.

We shall list these facts for reference as follows.

Definition (17.11)

Let T be a lamina that has the shape of a region R in the xy -plane. If the area density at (x, y) is $\rho(x, y)$, where ρ is continuous on R , then

- (i) the **mass** of T is $M = \iint_R \rho(x, y) dA$.
- (ii) the **moments** of T with respect to the x - and y -axes are

$$M_x = \iint_R y \rho(x, y) dA; \quad M_y = \iint_R x \rho(x, y) dA.$$

- (iii) the **center of mass** $(\bar{x}, \bar{y})$ of T is given by

$$\bar{x} = \frac{M_y}{M}, \quad \bar{y} = \frac{M_x}{M}.$$

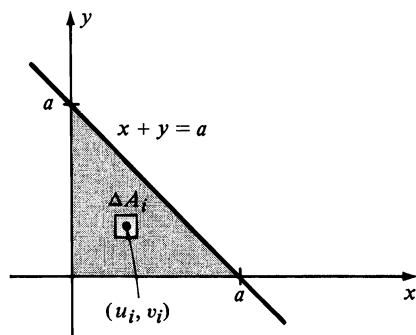


FIGURE 17.20

Example 1 A lamina T has the shape of an isosceles right triangle with equal sides of length a . Find the center of mass if the density at a point Q is directly proportional to the square of the distance from Q to the vertex V that is opposite the hypotenuse.

Solution It is convenient to introduce a coordinate system as illustrated in Figure 17.20, with V at the origin and the hypotenuse of the triangle along the line $x + y = a$. Shown in the figure is a typical rectangle of an inner partition P , where (u_i, v_i) is a point in the rectangle.

By hypothesis, the density at $Q(x, y)$ is given by $\rho(x, y) = k(x^2 + y^2)$ for some constant k . Applying (i) of Definition (17.11) and Theorem (17.9), the mass of T is

$$\begin{aligned} M &= \iint_R k(x^2 + y^2) dA = \int_0^a \int_0^{a-x} k(x^2 + y^2) dy dx \\ &= k \int_0^a \left[x^2 y + \frac{1}{3} y^3 \right]_0^{a-x} dx = k \int_0^a \left[x^2(a-x) + \frac{1}{3}(a-x)^3 \right] dx. \end{aligned}$$

It may be verified that the last integral equals $ka^4/6$. Similarly, by (ii) of Definition (17.11),

$$M_y = \iint_R xk(x^2 + y^2) dA = \int_0^a \int_0^{a-x} xk(x^2 + y^2) dy dx$$

which may be shown to equal $ka^5/15$. From (iii) of (17.11),

$$\bar{x} = \frac{ka^5/15}{ka^4/6} = \frac{2a}{5}.$$

By symmetry, $\bar{y} = 2a/5$. ■

Example 2 A lamina has the shape of the region R in the xy -plane bounded by the graphs of $x = y^2$ and $x = 4$. Find the center of mass if the density at the point $P(x, y)$ is directly proportional to the distance from the y -axis to P .

Solution The region is sketched in Figure 17.21 where, by hypothesis, $\rho(x, y) = kx$ for some constant k . It follows from the form of ρ and the symmetry of the region that the center of mass is on the x -axis, that is, $\bar{y} = 0$.

Using (i) of Definition (17.11) and then integrating first with respect to x as indicated by the strip in Figure 17.21,

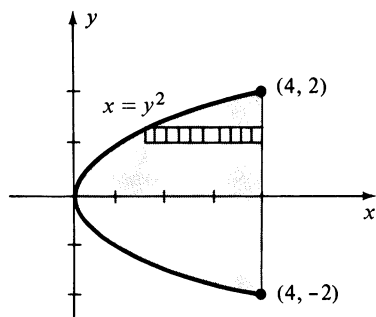


FIGURE 17.21

$$\begin{aligned} M &= \iint_R kx dA = k \int_{-2}^2 \int_{y^2}^4 x dx dy \\ &= k \int_{-2}^2 \left[\frac{x^2}{2} \right]_{y^2}^4 dy = \frac{k}{2} \int_{-2}^2 (16 - y^4) dy = \frac{128}{5} k. \end{aligned}$$

By (ii) of Definition (17.11),

$$\begin{aligned} M_y &= \iint_R x(kx) dA = k \int_{-2}^2 \int_{y^2}^4 x^2 dx dy \\ &= k \int_{-2}^2 \left[\frac{x^3}{3} \right]_{y^2}^4 dy = \frac{k}{3} \int_{-2}^2 (64 - y^6) dy = \frac{512}{7} k. \end{aligned}$$

Consequently,

$$\bar{x} = \frac{M_y}{M} = \frac{512k}{7} \cdot \frac{5}{128k} = \frac{20}{7}$$

and hence the centroid is $(\frac{20}{7}, 0)$. ■

If n particles of masses $m_1, m_2, \dots, m_n$ are located at points $(x_1, y_1), (x_2, y_2), \dots, (x_n, y_n)$, respectively, then the moments of the system with respect to the x - and y -axes were defined in (9.7) by

$$M_x = \sum_{i=1}^n y_i m_i \quad \text{and} \quad M_y = \sum_{i=1}^n x_i m_i.$$

These are also called the *first moments* of the system with respect to the coordinate axes. If we use the *squares* of the distances from the coordinate axes we obtain the *second moments*, or **moments of inertia** I_x and I_y of the system with respect to the x -axis and y -axis, respectively. Thus, by definition,

$$I_x = \sum_{i=1}^n y_i^2 m_i \quad \text{and} \quad I_y = \sum_{i=1}^n x_i^2 m_i.$$

This concept may be extended to laminas by employing the limiting processes used for double integrals. In particular, let us consider a lamina T of the type considered at the beginning of this section. If the area density at (x, y) is $\rho(x, y)$ where ρ is continuous, then it is natural to define the **moment of inertia I_x of T with respect to the x -axis** as follows:

$$(17.12) \quad I_x = \lim_{\|P\| \rightarrow 0} \sum_i v_i^2 \rho(u_i, v_i) \Delta A_i = \iint_R y^2 \rho(x, y) dA.$$

In like manner, the **moment of inertia I_y of T with respect to the y -axis** is given by

$$(17.13) \quad I_y = \lim_{\|P\| \rightarrow 0} \sum_i u_i^2 \rho(u_i, v_i) \Delta A_i = \iint_R x^2 \rho(x, y) dA.$$

If we multiply $\rho(u_i, v_i) \Delta A_i$ by the square $u_i^2 + v_i^2$ of the distance from the origin to (u_i, v_i) and take the limit of a sum of such terms, we obtain the **moment of inertia I_O of T with respect to the origin**. This leads to

$$(17.14) \quad I_O = \iint_R (x^2 + y^2) \rho(x, y) dA.$$

The number I_O is also called the **polar moment of inertia** of T . Observe that $I_O = I_x + I_y$.

Example 3 A lamina T has the semicircular shape illustrated in (i) of Figure 17.22. If the density at a point P is directly proportional to the distance from the diameter AB to P , find the moment of inertia of T with respect to the line through A and B .

Solution Let us introduce a coordinate system as in (ii) of Figure 17.22. Since the density at (x, y) is $\rho(x, y) = ky$ it follows from (17.12) that the desired moment of inertia is

$$\begin{aligned} I_x &= \int_{-a}^a \int_0^{\sqrt{a^2-x^2}} y^2 (ky) dy dx = k \int_{-a}^a \left[\frac{1}{4} y^4 \right]_0^{\sqrt{a^2-x^2}} dx \\ &= \frac{k}{4} \int_{-a}^a (a^4 - 2a^2 x^2 + x^4) dx. \end{aligned}$$

Carrying out the integration we obtain $I_x = 4ka^5/15$. ■

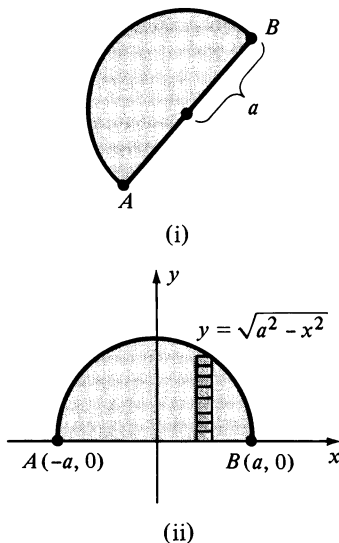


FIGURE 17.22

Moments of inertia are useful in problems that involve the rotation of an object about a fixed axis. For example, suppose a wheel is rotating about an axle. If a particle P of mass m on the wheel is a distance a from the axis of rotation (see Figure 15.12), then the moment of inertia I of P with respect to the axis is ma^2 . If the angular speed is ω , then the speed v of the particle, that is, the distance traveled per unit time, is $a\omega$. By definition, the **kinetic energy** K.E. of P is given by

$$\text{K.E.} = \frac{1}{2}mv^2.$$

Consequently,

$$\text{K.E.} = \frac{1}{2}ma^2\omega^2 = \frac{1}{2}I\omega^2.$$

If we represent the wheel by a flat disc and introduce a limit of a sum, then the same formula may be derived for the kinetic energy of the wheel. The formula can also be extended to laminas having noncircular shapes. The kinetic energy of a rotating object tells the engineer or physicist the amount of work necessary to bring the object to rest. Since $\text{K.E.} = \frac{1}{2}I\omega^2$, the kinetic energy of a rotating lamina is directly proportional to the moment of inertia, and hence for a fixed ω , the larger the moment of inertia, the larger the amount of work required to stop the rotation.

If a lamina T of mass M has moments of inertia I_x and I_y , then the **radius of gyration** $\bar{y}$ of T with respect to the x -axis, and the **radius of gyration** $\bar{x}$ of T with respect to the y -axis are defined by the equations

$$(17.15) \quad \bar{y}^2 M = I_x \quad \text{and} \quad \bar{x}^2 M = I_y.$$

These formulas imply that $(\bar{x}, \bar{y})$ is the point at which all the mass can be concentrated without changing the moments of inertia of T with respect to the coordinate axes.

Example 4 Find the radius of gyration with respect to the x -axis of the lamina described in Example 3.

Solution In Example 3 we found that $I_x = 4ka^5/15$. The mass of T is

$$M = \int_{-a}^a \int_0^{\sqrt{a^2-x^2}} ky \, dy \, dx.$$

Evaluating this integral gives us $M = 2ka^3/3$. Applying (17.15),

$$\bar{y} = \sqrt{\frac{I_x}{M}} = \sqrt{\frac{4ka^5/15}{2ka^3/3}} = \sqrt{\frac{2a^2}{5}} = \frac{a\sqrt{10}}{5}. \quad \blacksquare$$

17.4 Exercises

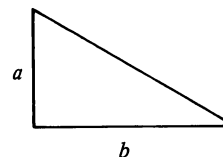
In Exercises 1–14 find the mass and the center of mass of the lamina that has the shape of the region bounded by the graphs of the given equations and having the indicated density.

- 1 $y = \sqrt{x}$, $x = 9$, $y = 0$; $\rho(x, y) = x + y$
- 2 $y = \sqrt[3]{x}$, $x = 8$, $y = 0$; $\rho(x, y) = y^2$
- 3 $y = x^2$, $y = 4$; density at the point $P(x, y)$ is directly proportional to the distance from the y -axis to P
- 4 $y = x^3$, $y = 2x$; density at the point $P(x, y)$ is directly proportional to the distance from the x -axis to P
- 5 $y = e^{-x^2}$, $y = 0$, $x = -1$, $x = 1$; $\rho(x, y) = |xy|$
- 6 $y = \sin x$, $y = 0$, $x = 0$, $x = \pi$; $\rho(x, y) = y$
- 7 $x = y^2$, $y - x = 2$, $y = -2$, $y = 3$; $\rho(x, y) = 1$
- 8 $y = x$, $y = 3x$, $x + y = 4$; $\rho(x, y) = 2$
- 9 $y = \sec x$, $x = -\pi/4$, $x = \pi/4$, $y = 1/2$; $\rho(x, y) = 4$
- 10 $xy^2 = 1$, $y = 1$, $y = 2$; $\rho(x, y) = x^2 + y^2$
- 11 $y = 1/x$, $y = x$, $x = 2$, $y = 0$; $\rho(x, y) = x$
- 12 $x = e^y$, $y = -1$, $x = 2$; $\rho(x, y) = 1$
- 13 $y = e^{-x}$, $y = 0$, $x = 0$, $x = 1$; $\rho(x, y) = y^2$
- 14 $y = \ln x$, $y = 0$, $x = 2$; $\rho(x, y) = 1/x$
- 15 Find I_x , I_y , and I_O for the lamina of Exercise 1.
- 16 Find I_x , I_y , and I_O for the lamina of Exercise 2.
- 17 Find I_x , I_y , and I_O for the lamina of Exercise 3.
- 18 Find I_x , I_y , and I_O for the lamina of Exercise 4.
- 19 A homogeneous lamina has the shape of a square of side a . Find the moment of inertia with respect to (a) a side; (b) a diagonal; and (c) the center of mass.
- 20 A homogeneous lamina has the shape of an equilateral triangle of side a . Find the moment of inertia with respect to (a) an altitude; (b) a side; and (c) a vertex.
- 21 Find the radius of gyration in (a) of Exercise 19.
- 22 Find the radius of gyration in (a) of Exercise 20.

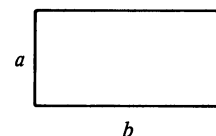
- 23 Find the moment of inertia and radius of gyration of a homogeneous circular disc of radius a with respect to a line along a diameter.
- 24 A lamina T has the shape of an isosceles right triangle with equal sides of length a . The density at a point P is directly proportional to the square of the distance from P to the vertex V that is opposite the hypotenuse. Find the moment of inertia and the radius of gyration with respect to a line along one of the equal sides (see Example 1).

Homogeneous laminas of density ρ are illustrated in Exercises 25–28, where $a < b$. Find the moment of inertia and radius of gyration with respect to the indicated side of length b .

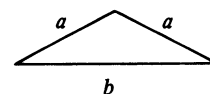
- 25 Right triangle



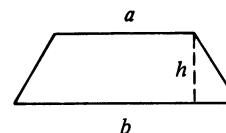
- 26 Rectangle



- 27 Isosceles triangle



- 28 Isosceles trapezoid



17.5 Double Integrals in Polar Coordinates

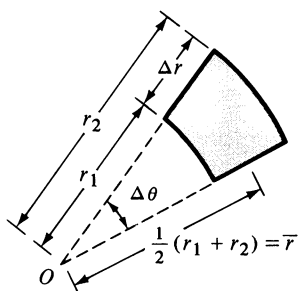


FIGURE 17.23 $\Delta A = \bar{r} \Delta r \Delta \theta$

A region in a polar coordinate plane of the type illustrated in Figure 17.23, which is bounded by arcs of circles of radii r_1 and r_2 with centers at the pole and by two rays from the origin, will be called an **elementary polar region**. If $\Delta\theta$ denotes the radian measure of the angle between the rays and $\Delta r = r_2 - r_1$, then the area ΔA of the region is given by

$$\Delta A = \frac{1}{2}r_2^2 \Delta\theta - \frac{1}{2}r_1^2 \Delta\theta.$$

This formula may also be written

$$\Delta A = \frac{1}{2}(r_2^2 - r_1^2) \Delta\theta = \frac{1}{2}(r_2 + r_1)(r_2 - r_1) \Delta\theta.$$

If we denote the **average radius** $\frac{1}{2}(r_2 + r_1)$ by $\bar{r}$, then

$$\Delta A = \bar{r} \Delta r \Delta\theta.$$

Next, consider a region R of the type illustrated in (i) of Figure 17.24, bounded by two rays that make positive angles α and β with the polar axis and by the graphs of two polar equations $r = g_1(\theta)$ and $r = g_2(\theta)$, where the functions g_1 and g_2 are continuous and $g_1(\theta) \leq g_2(\theta)$ for all θ in the interval $[\alpha, \beta]$. If R is subdivided by means of circular arcs and rays as shown in (ii) of Figure 17.24, then the collection of elementary polar regions $R_1, R_2, \dots, R_n$ that lie completely within R is called an **inner polar partition** P of R . The norm $\|P\|$ of P is the length of the longest diagonal of the R_i . If we choose a point (r_i, θ_i) in R_i such that r_i is the average radius, then as in Figure 17.23, the area ΔA_i of R_i is $r_i \Delta r_i \Delta\theta_i$.

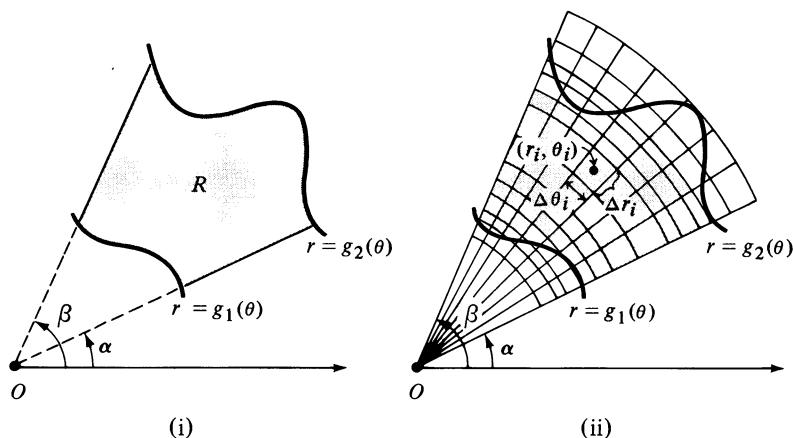


FIGURE 17.24

If f is a continuous function of the polar variables r and θ , then it can be proved that

$$(17.16) \quad \lim_{\|P\| \rightarrow 0} \sum_i f(r_i, \theta_i) r_i \Delta r_i \Delta\theta_i = \iint_R f(r, \theta) dA = \int_{\alpha}^{\beta} \int_{g_1(\theta)}^{g_2(\theta)} f(r, \theta) r dr d\theta.$$

It is important to note that the integrand on the right in (17.16) is the product of $f(r, \theta)$ and r . This is due to the fact that ΔA_i equals $r_i \Delta r_i \Delta \theta_i$.

It is sometimes convenient to regard the iterated integral in (17.16) in terms of limits of double sums in a manner similar to our discussion for rectangular coordinates at the end of Section 17.2. In the present situation, we first hold θ fixed and sum along one of the wedge-shaped regions shown in (ii) of Figure 17.24, from the graph of g_1 to the graph of g_2 . For the second summation we sweep out the region by letting θ vary from α to β . Intuitively, the iterated integral does this and simultaneously takes the limit as $\|P\|$ approaches zero.

If $f(r, \theta) = 1$ throughout R , then the integral in (17.16) equals the area of R . This also follows from our work in Chapter 13 since after the partial integration with respect to r , the formula given in Theorem (13.10) is obtained.

Example 1 A lamina has the shape of the region R that lies outside the graph of $r = a$ and inside the graph of $r = 2a \sin \theta$. Find the mass if the density at a point $P(r, \theta)$ is inversely proportional to the distance from the pole to P .

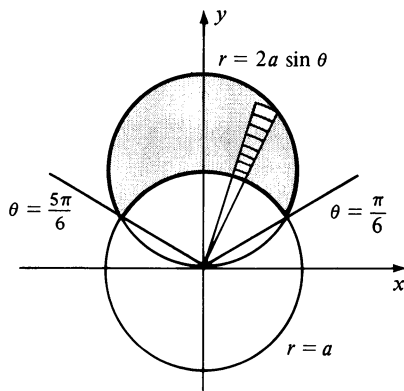


FIGURE 17.25

Solution The region is sketched in Figure 17.25, together with a typical strip of elementary polar regions involved in summing from one boundary of R to the other. By hypothesis, the density $f(r, \theta)$ at the point (r, θ) is k/r , where k is some real number. The mass M of the lamina is $\iint_R f(r, \theta) dA$. Thus, by (17.16),

$$\begin{aligned} M &= \iint_R f(r, \theta) dA = \int_{\pi/6}^{5\pi/6} \int_a^{2a \sin \theta} (k/r)r dr d\theta = k \int_{\pi/6}^{5\pi/6} r \Big|_a^{2a \sin \theta} d\theta \\ &= k \int_{\pi/6}^{5\pi/6} (2a \sin \theta - a) d\theta = ka \left[-2 \cos \theta - \theta \right]_{\pi/6}^{5\pi/6} \\ &= ka[(\sqrt{3} - 5\pi/6) - (-\sqrt{3} - \pi/6)] \\ &= ka(2\sqrt{3} - 2\pi/3). \end{aligned}$$

It can be shown that under suitable conditions an iterated double integral in rectangular coordinates can be transformed into a double integral in polar coordinates. First, the variables x and y in the integrand are replaced by $r \cos \theta$ and $r \sin \theta$. Next, $dy dx$ or $dx dy$ is replaced by $r dr d\theta$ or $r d\theta dr$ and the limits of integration are changed to polar coordinates. This gives us the following **change of variable formula**:

$$(17.17) \quad \iint_R f(x, y) dA = \iint_R f(r \cos \theta, r \sin \theta) r dr d\theta$$

This formula will be justified in Section 18.10. As an illustration, formula (17.13) for the moment of inertia with respect to the y -axis of a lamina having the shape of (i) in Figure 17.24 is

$$I_y = \iint_R x^2 \rho(x, y) dA = \int_{\alpha}^{\beta} \int_{g_1(\theta)}^{g_2(\theta)} (r \cos \theta)^2 \rho(r \cos \theta, r \sin \theta) r dr d\theta,$$

where $\rho(x, y)$ is the density at (x, y) . For the special case of the lamina in Example 1, the last equation takes on the form

$$I_y = \int_{\pi/6}^{5\pi/6} \int_a^{2a \sin \theta} (r \cos \theta)^2 (k/r) r \, dr \, d\theta = \int_{\pi/6}^{5\pi/6} \int_a^{2a \sin \theta} k r^2 \cos^2 \theta \, dr \, d\theta.$$

If the integrand $f(x, y)$ of a double integral $\iint_R f(x, y) \, dy \, dx$ contains the expression $x^2 + y^2$, or if the region R involves circular arcs, then the introduction of polar coordinates often leads to a simpler evaluation. This is illustrated in the following example.

Example 2 Use polar coordinates to evaluate

$$\int_{-a}^a \int_0^{\sqrt{a^2 - x^2}} (x^2 + y^2)^{3/2} \, dy \, dx.$$

Solution Evidently the region of integration is bounded by the graphs of $y = \sqrt{a^2 - x^2}$ and $y = 0$ illustrated in (ii) of Figure 17.22. Replacing $x^2 + y^2$ in the integrand by r^2 , $dy \, dx$ by $r \, dr \, d\theta$, and changing the limits, we have

$$\begin{aligned} \int_{-a}^a \int_0^{\sqrt{a^2 - x^2}} (x^2 + y^2)^{3/2} \, dy \, dx &= \int_0^\pi \int_0^a r^3 \, r \, dr \, d\theta \\ &= \int_0^\pi \left[\frac{r^5}{5} \right]_0^a d\theta = \frac{a^5}{5} \int_0^\pi d\theta \\ &= \frac{a^5}{5} \theta \Big|_0^\pi = \frac{\pi a^5}{5}. \quad \blacksquare \end{aligned}$$

Polar coordinates can also be used for double integrals over a region R of the type illustrated in (i) of Figure 17.26. In this case R is bounded by the arcs of two circles of radii a and b and by the graphs of polar equations $\theta = h_1(r)$ and $\theta = h_2(r)$, where the functions h_1 and h_2 are continuous and $h_1(r) \leq h_2(r)$ for all r in the interval $[a, b]$.

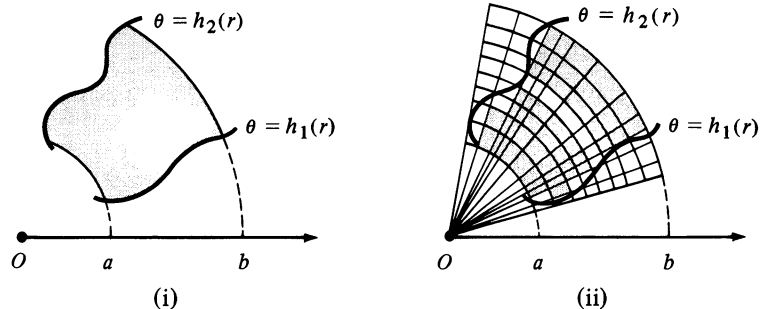


FIGURE 17.26

If f is a function of r and θ that is continuous on R , then the limit in (17.16) exists and the corresponding evaluation formula is

$$\iint_R f(r, \theta) dA = \int_a^b \int_{h_1(r)}^{h_2(r)} f(r, \theta) r d\theta dr.$$

The preceding iterated integral may be interpreted in terms of limits of double sums. In this case we first hold r fixed and sum along a circular arc, as illustrated by the colored region in (ii) of Figure 17.26. Secondly, we sweep out R by summing terms that correspond to such ring-shaped regions. This technique is illustrated in the next example.

Example 3 Find the polar moment of inertia of a homogeneous lamina that has the shape of the smaller of the regions bounded by the polar axis, the graphs of $r = 1$, $r = 2$, and the part of the spiral $r\theta = 1$ from $\theta = \frac{1}{2}$ to $\theta = 1$.

Solution The region, together with some elementary polar regions, is sketched in Figure 17.27.

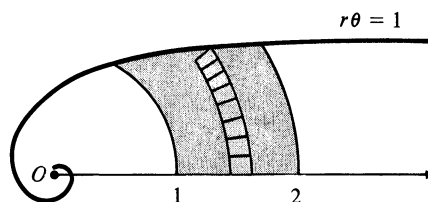


FIGURE 17.27

By hypothesis, the density at every point (r, θ) is a constant k . From the previous discussion,

$$\begin{aligned} I_0 &= \iint_R (x^2 + y^2) \rho(x, y) dA = \int_1^2 \int_0^{1/r} r^2 k r d\theta dr \\ &= k \int_1^2 r^3 \theta \Big|_0^{1/r} dr = k \int_1^2 r^2 dr = k \left[\frac{r^3}{3} \right]_1^2 = \frac{7}{3} k. \end{aligned}$$

■

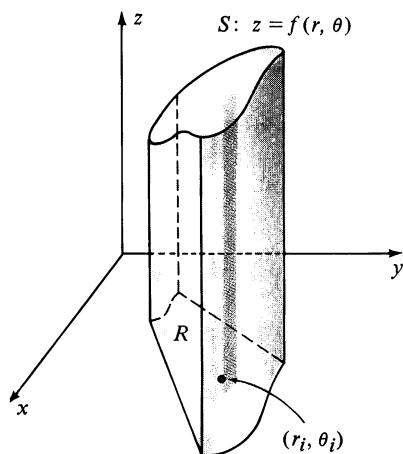


FIGURE 17.28

If $f(r, \theta) \geq 0$ throughout a polar region R , then the double integral $\iint_R f(r, \theta) dA$ in (17.16) may be regarded as the volume of a solid in a manner completely analogous to our formulation of Definition (17.4) or (17.10) in rectangular coordinates. The principal difference in the present situation is that we consider the graph S of $z = f(r, \theta)$ in cylindrical coordinates. The desired solid lies under the graph of S and over the region R . If we use a polar partition of R , then the expression $f(r_i, \theta_i) r_i \Delta r_i \Delta \theta_i$ in (17.16) may be interpreted as the volume of a prism of height $f(r_i, \theta_i)$ and base area $r_i \Delta r_i \Delta \theta_i$, as illustrated in Figure 17.28. The limit of a sum of these expressions equals the volume.

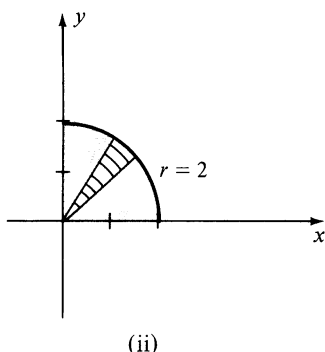
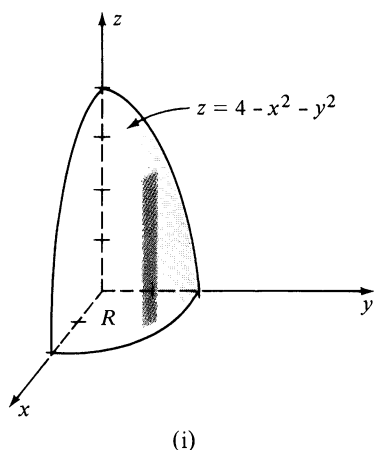


FIGURE 17.29

Example 4 Find the volume V of the solid bounded by the paraboloid $z = 4 - x^2 - y^2$ and the xy -plane.

Solution The first octant portion of the solid is sketched in (i) of Figure 17.29. By symmetry, it is sufficient to find the volume of this portion and multiply the result by 4. If we introduce cylindrical coordinates, then an equation for the paraboloid is $z = 4 - r^2$. The region R in the xy -plane is bounded by the coordinate axes and one-fourth of the circle $r = 2$. This is illustrated in (ii) of Figure 17.29 together with a wedge of elements of a polar partition.

By the discussion preceding this example, with $f(r, \theta) = 4 - r^2$, we have

$$\begin{aligned} V &= 4 \iint_R (4 - r^2) dA = 4 \int_0^{\pi/2} \int_0^2 (4 - r^2) r dr d\theta \\ &= 4 \int_0^{\pi/2} \left[2r^2 - \frac{r^4}{4} \right]_0^2 d\theta = 4 \int_0^{\pi/2} 4 d\theta = 16\theta \Big|_0^{\pi/2} = 8\pi. \end{aligned}$$

The same problem, using rectangular coordinates, leads to the following double integral:

$$V = 4 \iint_R (4 - x^2 - y^2) dA = 4 \int_0^2 \int_0^{\sqrt{4-x^2}} (4 - x^2 - y^2) dy dx.$$

After evaluating the last integral, the reader should be thoroughly convinced of the advantage of using cylindrical (or polar) coordinates in certain problems. ■

17.5 Exercises

In Exercises 1–4 use a double integral to find the area of the indicated region.

- 1 One loop of $r^2 = 9 \sin 2\theta$
- 2 One loop of $r = 2 \cos 4\theta$
- 3 Inside $r = 2 - 2 \cos \theta$ and outside $r = 3$
- 4 Bounded by $r = 3 + 2 \sin \theta$

In Exercises 5–8 find the mass and center of mass of the lamina that has the indicated shape and density.

- 5 Bounded by $r = 2 \cos \theta$; density at the point $P(r, \theta)$ is directly proportional to the distance from the pole to P .

- 6 Bounded by the loop of $r^2 = a \cos 2\theta$ from $\theta = -\pi/4$ to $\theta = \pi/4$; density at the point $P(r, \theta)$ is directly proportional to the square of the distance from the pole to P .
- 7 Inside the graph of $r = 3 \sin \theta$ and outside the graph of $r = 1 + \sin \theta$; density at the point $P(r, \theta)$ is inversely proportional to the distance from the pole to P .
- 8 Inside the graph of $r = 4 \cos \theta$ and outside the graph of $r = 2$; density at the point $P(r, \theta)$ is directly proportional to the distance from the polar axis to P .
- 9 Find the polar moment of inertia of a homogeneous lamina that has the shape of the region bounded by $r^2 = a \sin 2\theta$.

- 10 A homogeneous lamina has the shape of the region $R = \{(r, \theta): r^{-1} \leq \theta \leq 2r^{-1}, 1 \leq r \leq 2\}$. Find the polar moment of inertia.
- 11 Find the moment of inertia and radius of gyration of a homogeneous circular lamina of radius a with respect to a line through the center.
- 12 Find the moment of inertia and radius of gyration of a homogeneous circular lamina of radius a with respect to a tangent line.

Use polar coordinates to evaluate the integrals in Exercises 13–18.

- 13 $\iint_R (x^2 + y^2)^{3/2} dA$, where R is the region bounded by the graph of $x^2 + y^2 = 4$.
- 14 $\iint_R x^2(x^2 + y^2)^3 dA$, where R is the region bounded by the graphs of $y = \sqrt{1 - x^2}$ and $y = 0$.
- 15 $\iint_R x^2/(x^2 + y^2) dA$, where R is the region between the concentric circles $x^2 + y^2 = a^2$ and $x^2 + y^2 = b^2$, with $0 < a < b$.
- 16 $\iint_R (x^2 + y^2) dA$, where R is the region bounded by the graph of $(x^2 + y^2)^2 = x^2 - y^2$.
- 17 $\iint_R (x + y) dA$, where R is the region bounded by the graph of $x^2 + y^2 - 2y = 0$.
- 18 $\iint_R \sqrt{x^2 + y^2} dA$, where R is the triangular region with vertices $(0, 0)$, $(3, 0)$ and $(3, 3)$.

Evaluate the integrals in Exercises 19–24 by changing to polar coordinates.

- 19 $\int_{-a}^a \int_0^{\sqrt{a^2 - x^2}} e^{-(x^2 + y^2)} dy dx$
- 20 $\int_0^a \int_0^{\sqrt{a^2 - x^2}} (x^2 + y^2)^{3/2} dy dx$
- 21 $\int_1^2 \int_0^x \frac{1}{\sqrt{x^2 + y^2}} dy dx$

$$22 \int_0^1 \int_0^{\sqrt{1-x^2}} e^{\sqrt{x^2 + y^2}} dy dx$$

$$23 \int_0^2 \int_0^{\sqrt{4-y^2}} \cos(x^2 + y^2) dx dy$$

$$24 \int_0^2 \int_{-\sqrt{2y-y^2}}^{\sqrt{2y-y^2}} x dx dy$$

- 25 Find the volume of the solid that lies inside the sphere $x^2 + y^2 + z^2 = 25$ and outside the cylinder $x^2 + y^2 = 9$.
- 26 Find the volume of the solid that is cut out of the ellipsoid $4x^2 + 4y^2 + z^2 = 16$ by the cylinder $x^2 + y^2 = 1$.
- 27 Find the volume of the solid bounded by the cone $z = r$ and the cylinder $r = 2 \cos \theta$.
- 28 Find the volume of the solid bounded by the paraboloid $z = 4r^2$, the cylinder $r = 3 \sin \theta$, and the plane $z = 0$.
- 29 Find the volume of the solid that lies inside the graphs of both $x^2 + y^2 + z^2 = 16$ and $x^2 + y^2 - 4y = 0$.
- 30 Find the volume of the solid bounded by the graphs of $z = 9 - x^2 - y^2$ and $z = 5$.

- 31 (a) Let R_a be the region that is bounded by the circle $x^2 + y^2 = a^2$. If we define

$$\int_{-\infty}^{\infty} \int_{-\infty}^{\infty} e^{-(x^2 + y^2)} dx dy = \lim_{a \rightarrow \infty} \iint_{R_a} e^{-(x^2 + y^2)} dA,$$

evaluate this improper integral and interpret the result geometrically.

- (b) It can be proved, using advanced methods, that

$$\int_{-\infty}^{\infty} \int_{-\infty}^{\infty} e^{-(x^2 + y^2)} dx dy = \int_{-\infty}^{\infty} e^{-x^2} dx \int_{-\infty}^{\infty} e^{-y^2} dy.$$

Use this and part (a) to prove that $\int_{-\infty}^{\infty} e^{-x^2} dx = \sqrt{\pi}$. Interpret this result geometrically.

- (c) Use (b) to prove that

$$\frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} e^{-x^2/2} dx = 1.$$

(This result is important in the field of statistics.)

17.6 Triple Integrals

It is possible to define *triple integrals* for a function f of *three* variables x , y , and z in a manner similar to that used in Section 17.2 for functions of two variables. The simplest case occurs if f is continuous throughout a region of the form

$$Q = \{(x, y, z): a \leq x \leq b, c \leq y \leq d, k \leq z \leq l\},$$

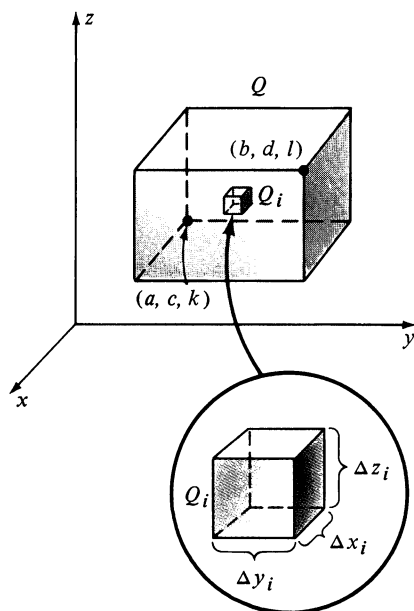


FIGURE 17.30

as illustrated in Figure 17.30. If Q is divided into subregions $Q_1, Q_2, \dots, Q_n$ by means of planes parallel to the three coordinate planes, then the collection $\{Q_i: i = 1, \dots, n\}$ is called a **partition** P of Q . The **norm** $\|P\|$ of the partition is the length of the longest diagonal of all the Q_i . As in Figure 17.30, if Δx_i , Δy_i , and Δz_i are the dimensions of Q_i , then the volume ΔV_i of Q_i is the product $\Delta x_i \Delta y_i \Delta z_i$.

If (u_i, v_i, w_i) is a point in Q_i , then the sum

$$\sum_{i=1}^n f(u_i, v_i, w_i) \Delta V_i$$

is called a **Riemann sum** of f for P .

The concept of a limit of Riemann sums for a function of three variables is defined in a manner similar to that given in Definition (17.2) for functions of two variables. If this limit exists it is called the **triple integral** of f over Q , denoted by

$$(17.18) \quad \iiint_Q f(x, y, z) dV = \lim_{\|P\| \rightarrow 0} \sum_i f(u_i, v_i, w_i) \Delta V_i.$$

Moreover, it can be shown that

$$\iiint_Q f(x, y, z) dV = \int_a^l \int_c^d \int_b^a f(x, y, z) dx dy dz,$$

where the **iterated integral** on the right means that the first integration is with respect to x (with y and z fixed), the second is with respect to y (with z fixed), and the third is with respect to z . There are five other iterated integrals that equal the triple integral of f over the parallelepiped Q . For example, if we integrate in the order y, z, x , then

$$\iiint_Q f(x, y, z) dV = \int_a^b \int_k^l \int_c^d f(x, y, z) dy dz dx.$$

Example 1 Evaluate $\iiint_Q 3xy^3z^2 dV$ if

$$Q = \{(x, y, z): -1 \leq x \leq 3, 1 \leq y \leq 4, 0 \leq z \leq 2\}.$$

Solution Of the six possible iterated integrals, we shall use the following:

$$\begin{aligned} \int_1^4 \int_{-1}^3 \int_0^2 3xy^3z^2 dz dx dy &= \int_1^4 \int_{-1}^3 xy^3 z^3 \Big|_0^2 dx dy \\ &= \int_1^4 \int_{-1}^3 8xy^3 dx dy = \int_1^4 4x^2 y^3 \Big|_{-1}^3 dy \\ &= \int_1^4 (36y^3 - 4y^3) dy = 8y^4 \Big|_1^4 = 2040. \end{aligned}$$

■

Triple integrals may be defined over regions other than those bounded by rectangular parallelepipeds. Suppose, for example, that R is a region in the xy -plane which can be divided into subregions of Types I and II, and that Q is the region in three dimensions defined by

$$Q = \{(x, y, z): (x, y) \text{ is in } R \text{ and } k_1(x, y) \leq z \leq k_2(x, y)\}$$

where the functions k_1 and k_2 have continuous first partial derivatives throughout R . Geometrically, Q lies between the graphs of $z = k_1(x, y)$ and $z = k_2(x, y)$ and over or under the region R . If Q is subdivided by means of planes parallel to the three coordinate planes, then the resulting parallelepipeds $Q_1, Q_2, \dots, Q_n$ that lie completely within Q form an **inner partition** P of Q . The sketch in Figure 17.31 shows a region Q of the type under consideration together with a typical element Q_i of an inner partition.

A **Riemann sum** of f for P is any sum of the form $\sum_i f(u_i, v_i, w_i) \Delta V_i$, where (u_i, v_i, w_i) is an arbitrary point in Q_i and ΔV_i is the volume of Q_i . The triple integral of f over Q is again defined as the limit (17.18). It can be shown that if f is continuous throughout Q , then

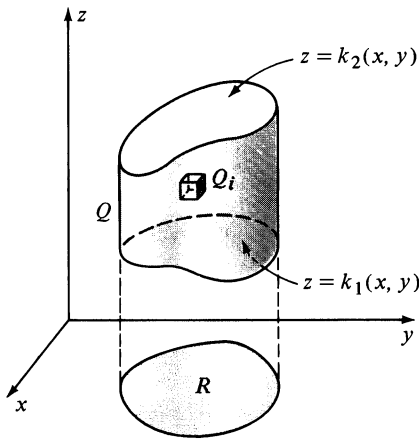


FIGURE 17.31

$$\iiint_Q f(x, y, z) dV = \iint_R \left[\int_{k_1(x, y)}^{k_2(x, y)} f(x, y, z) dz \right] dA.$$

In particular, if R is of Type I, as in (i) of Figure 17.1, then

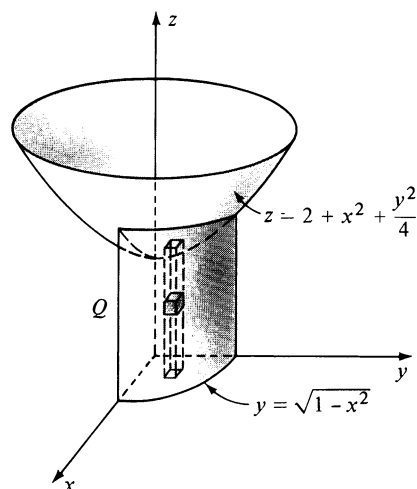
$$(17.19) \quad \iiint_Q f(x, y, z) dV = \int_a^b \int_{g_1(x)}^{g_2(x)} \int_{k_1(x, y)}^{k_2(x, y)} f(x, y, z) dz dy dx.$$

The symbol on the right-hand side of the last equation is called an **iterated triple integral**. It is evaluated by means of partial integrations of $f(x, y, z)$ in the order z , y , and x , substituting the indicated limits in the usual way. Similarly, if R is of Type II as in (ii) of Figure 17.1, then

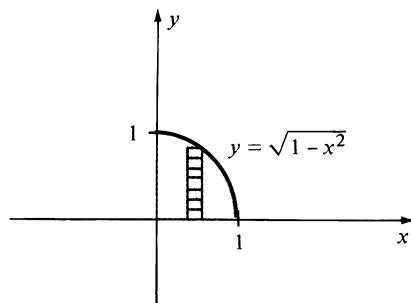
$$(17.20) \quad \iiint_Q f(x, y, z) dV = \int_c^d \int_{h_1(y)}^{h_2(y)} \int_{k_1(x, y)}^{k_2(x, y)} f(x, y, z) dz dx dy.$$

It is convenient to regard the iterated integral in either (17.19) or (17.20) as a limit of a *triple sum* of terms of the form $f(u_i, v_j, w_k) \Delta z_k \Delta y_j \Delta x_i$, where we first sum along a column of parallelepipeds in the direction of the z -axis from the lower surface (with equation $z = k_1(x, y)$) to the upper surface (with equation $z = k_2(x, y)$). The resulting double integral is then evaluated over R using the techniques discussed in Section 17.2.

Example 2 Express $\iiint_Q f(x, y, z) dV$ as an iterated integral, where Q is the region in the first octant bounded by the coordinate planes and the graphs of $z - 2 = x^2 + (y^2/4)$ and $x^2 + y^2 = 1$.



(i)



(ii)

FIGURE 17.32

Solution As illustrated in (i) of Figure 17.32, Q lies under the paraboloid $z = 2 + x^2 + (y^2/4)$ and over the graph of $z = 0$ (the xy -plane). The region R in the xy -plane is bounded by the coordinate axes and the graph of $y = \sqrt{1 - x^2}$. Also shown in (i) of the figure is a column that corresponds to a first summation over the parallelepipeds of volumes $\Delta z_k \Delta y_j \Delta x_i$ in the direction of the z -axis. Since the column extends from the xy -plane ($z = 0$) to the paraboloid, the lower limit on the innermost integral sign in (17.19) is 0 and the upper limit is $2 + x^2 + (y^2/4)$. The second and third integrations are taken over the region R (see (ii) of Figure 17.32). Consequently, the integral in (17.19) has the form

$$\int_0^1 \int_0^{\sqrt{1-x^2}} \int_0^{2+x^2+(y^2/4)} f(x, y, z) dz dy dx.$$

■

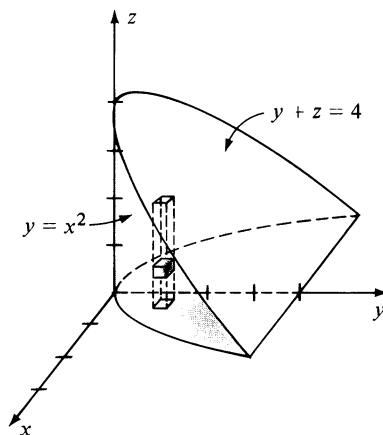
If $f(x, y, z) = 1$ throughout Q , then the triple integral of f over Q is written $\iiint_Q dV$ and its value is the volume of the region Q . It follows from Example 2 that the value of the integral

$$\int_0^1 \int_0^{\sqrt{1-x^2}} \int_0^{2+x^2+(y^2/4)} dz dy dx$$

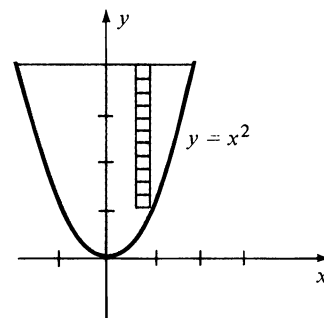
is the volume of the region Q shown in (i) of Figure 17.32.

Example 3 Find the volume of the solid that is bounded by the cylinder $y = x^2$ and by the planes $y + z = 4$ and $z = 0$.

Solution The solid is sketched in (i) of Figure 17.33 together with a column corresponding to a first summation in the z -direction. Note that the column extends from $z = 0$ to $z = 4 - y$. The region R in the xy -plane is shown in (ii)



(i)



(ii)

FIGURE 17.33

of the figure, together with a strip corresponding to the first (of a double) integration with respect to y . Applying (17.19) with $f(x, y, z) = 1$,

$$\begin{aligned} V &= \int_{-2}^2 \int_{x^2}^4 \int_0^{4-y} dz \, dy \, dx = \int_{-2}^2 \int_{x^2}^4 (4 - y) \, dy \, dx \\ &= \int_{-2}^2 \left[4y - \frac{1}{2} y^2 \right]_{x^2}^4 dx = \int_{-2}^2 \left(8 - 4x^2 + \frac{1}{2} x^4 \right) dx \\ &= \left[8x - \frac{4}{3} x^3 + \frac{1}{10} x^5 \right]_{-2}^2 = \frac{256}{15}. \end{aligned}$$

If we use (17.20), then the strip in (ii) of Figure 17.33 would be horizontal and

$$V = \int_0^4 \int_{-\sqrt{y}}^{\sqrt{y}} \int_0^{4-y} dz \, dx \, dy.$$

The reader may verify that the value of this integral is also $\frac{256}{15}$. ■

For certain regions, a triple integral may be evaluated by means of an iterated triple integral in which the first integration is with respect to y or x . Thus, consider

$$Q = \{(x, y, z): a \leq x \leq b, h_1(x) \leq z \leq h_2(x), k_1(x, z) \leq y \leq k_2(x, z)\}$$

where the given functions are sufficiently well behaved. A graph of this type of region is illustrated in Figure 17.34. It can be shown that

$$(17.21) \quad \iiint_Q f(x, y, z) \, dV = \int_a^b \int_{h_1(x)}^{h_2(x)} \int_{k_1(x, z)}^{k_2(x, z)} f(x, y, z) \, dy \, dz \, dx.$$

This iterated integral may be interpreted as a limit of a triple sum obtained by first summing over a row of parallelepipeds in the direction of the y -axis from the left surface (with equation $y = k_1(x, z)$) to the right surface (with equation $y = k_2(x, z)$), as indicated in Figure 17.34. The resulting double integral is then evaluated over the region R in the xz -plane.

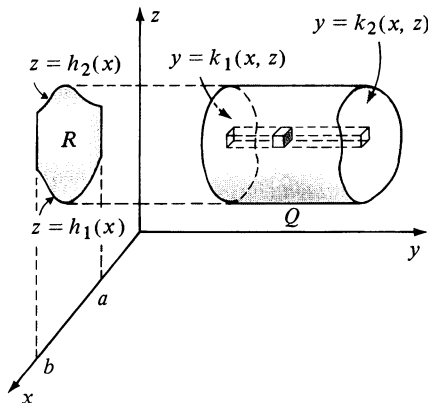


FIGURE 17.34

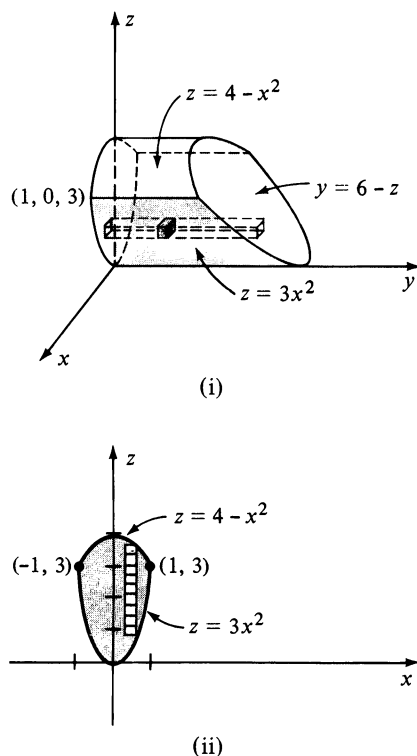


FIGURE 17.35

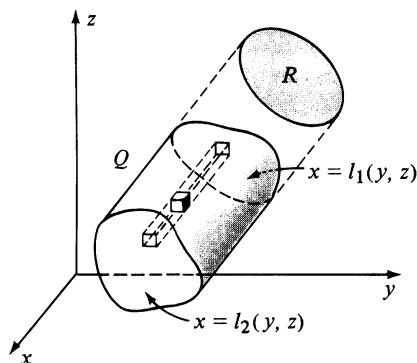


FIGURE 17.36

Example 4 Find the volume of the region Q bounded by the graphs of $z = 3x^2$, $z = 4 - x^2$, $y = 0$, and $z + y = 6$.

Solution As illustrated in (i) of Figure 17.35, Q lies under the cylinder $z = 4 - x^2$, over the cylinder $z = 3x^2$, to the right of the xz -plane, and to the left of the plane $z + y = 6$. Hence Q is a region of the type illustrated in Figure 17.34, where $k_1(x, z) = 0$ and $k_2(x, z) = 6 - z$. The region R in the xz -plane is sketched in (ii) of Figure 17.35. Applying (17.21),

$$\begin{aligned} V &= \iiint_Q dV = \int_{-1}^1 \int_{3x^2}^{4-x^2} \int_0^{6-z} dy \, dz \, dx = \int_{-1}^1 \int_{3x^2}^{4-x^2} (6 - z) \, dz \, dx \\ &= \int_{-1}^1 \left[6z - \frac{1}{2} z^2 \right]_{3x^2}^{4-x^2} dx = \int_{-1}^1 (16 - 20x^2 + 4x^4) \, dx = \frac{304}{15}. \end{aligned}$$

If a different order of integration is used it is necessary to use several triple integrals. (Why?) ■

Finally, if Q is a region of the type illustrated in Figure 17.36, where the functions l_1 and l_2 have continuous first partial derivatives on a suitable region R in the yz -plane, then

$$\iiint_Q f(x, y, z) \, dV = \iint_R \left[\int_{l_1(y, z)}^{l_2(y, z)} f(x, y, z) \, dx \right] dA.$$

In the final iterated double integral, dA will be replaced by $dz \, dy$ or $dy \, dz$.

Example 5 Rework Example 3 by integrating first with respect to x .

Solution The solid is bounded by the graphs of $y = x^2$, $y + z = 4$, and $z = 0$. Referring to (i) of Figure 17.37, a first integration with respect to x would correspond to a summation along a row of parallelepipeds extending from the graph of $x = -\sqrt{y}$ to the graph of $x = \sqrt{y}$. The region R in the yz -plane lies between these graphs and is bounded by the y - and z -axes and the line $y + z = 4$.

If the second integration is with respect to y , as indicated by the strip in (ii) of Figure 17.37, then

$$V = \int_0^4 \int_0^{4-z} \int_{-\sqrt{y}}^{\sqrt{y}} dx \, dy \, dz.$$

It may be verified that the value of this integral is $\frac{256}{15}$. If we take the second integration with respect to z instead of y , then the iterated integral is

$$V = \int_0^4 \int_0^{4-y} \int_{-\sqrt{y}}^{\sqrt{y}} dx \, dz \, dy.$$

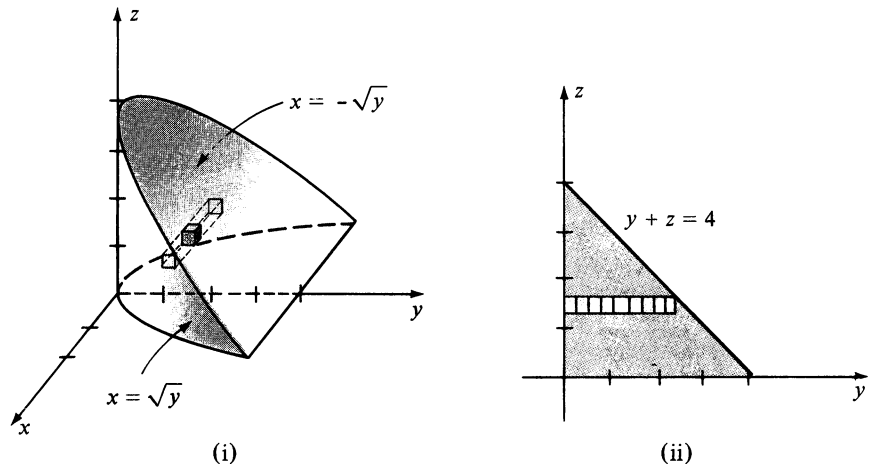


FIGURE 17.37

17.6 Exercises

- Evaluate the integral of Example 1 using five orders of integration different from that used in the solution.
- If $Q = \{(x, y, z) : 1 \leq x \leq 2, -1 \leq y \leq 0, 0 \leq z \leq 3\}$, evaluate

$$\iiint_Q (x + 2y + 4z) dV$$

in six different ways.

Evaluate the iterated integrals in Exercises 3–6.

- $\int_0^1 \int_{1+x}^{2x} \int_z^{x+z} x \, dy \, dz \, dx$
- $\int_1^2 \int_0^{z^2} \int_{x+z}^{x-z} z \, dy \, dx \, dz$
- $\int_{-1}^2 \int_1^{x^2} \int_0^{x+y} 2x^2y \, dz \, dy \, dx$
- $\int_2^3 \int_0^{3y} \int_1^{yz} (2x + y + z) \, dx \, dz \, dy$

In Exercises 7–10 sketch the region Q bounded by the graphs of the given equations and express $\iiint_Q f(x, y, z) dV$ as an iterated integral in six different ways.

- $x + 2y + 3z = 6; x = 0, y = 0, z = 0$
- $x^2 + y^2 = 9, z = 0, z = 2$
- $z = 9 - 4x^2 - y^2, z = 0$
- $36x^2 + 9y^2 + 4z^2 = 36$

In Exercises 11–20 sketch the region bounded by the graphs of the given equations and use a triple integral to find its volume.

- $z + x^2 = 4, y + z = 4, y = 0, z = 0$
- $x^2 + z^2 = 4, y^2 + z^2 = 4$
- $y = 2 - z^2, y = z^2, x + z = 4, x = 0$
- $z = 4y^2, z = 2, x = 2, x = 0$
- $y^2 + z^2 = 1, x + y + z = 2, x = 0$
- $z = x^2 + y^2, y + z = 2$
- $z = 9 - x^2, z = 0, y = -1, y = 2$
- $z = e^{x+y}, y = 3x, x = 2, y = 0, z = 0$
- $z = x^2, z = x^3, y = z^2, y = 0$
- $y = x^2 + z^2, z = x^2, z = 4, y = 0$

Each of the iterated integrals in Exercises 21–26 represents the volume of a region Q . Describe Q .

- $\int_0^1 \int_{\sqrt{1-z}}^{\sqrt{4-z}} \int_2^3 dx \, dy \, dz$
- $\int_0^1 \int_{z^3}^{\sqrt{z}} \int_0^{4-x} dy \, dx \, dz$
- $\int_0^2 \int_{x^2}^{2x} \int_0^{x+y} dz \, dy \, dx$
- $\int_0^1 \int_x^{3x} \int_0^{xy} dz \, dy \, dx$

$$25 \int_1^2 \int_{-\sqrt{z}}^{\sqrt{z}} \int_{-\sqrt{z-x^2}}^{\sqrt{z-x^2}} dy \, dx \, dz$$

$$26 \int_1^4 \int_{-z}^z \int_{-\sqrt{z^2-y^2}}^{\sqrt{z^2-y^2}} dx \, dy \, dz$$

27 Define the limit of Riemann sums for functions of three variables (see Definition (17.2)).

28 Define the integral $\iiint_U f(x, y, z, t) \, dW$, where

$$U = \{(x, y, z, t): a \leq x \leq b, c \leq y \leq d, m \leq z \leq n, p \leq t \leq q\}.$$

17.7 Applications of Triple Integrals

If a solid has the shape of a three-dimensional region Q , and the density (mass per unit volume) at (x, y, z) is $\rho(x, y, z)$, where ρ is continuous throughout Q , then as in Definition (17.11) the mass is defined by

$$(17.22) \quad M = \iiint_Q \rho(x, y, z) \, dV.$$

If a particle of mass m is at the point (x, y, z) , then its moments with respect to the xy -, xz -, and yz -planes are defined as zm , ym , and xm , respectively. By employing the usual techniques involving limits of sums, we are led to define the **moments** of the *solid* with respect to the coordinate planes as

$$(17.23) \quad \begin{aligned} M_{xy} &= \iiint_Q z\rho(x, y, z) \, dV \\ M_{xz} &= \iiint_Q y\rho(x, y, z) \, dV \\ M_{yz} &= \iiint_Q x\rho(x, y, z) \, dV. \end{aligned}$$

For example, if $\{Q_i\}$ is a partition of the region Q and if T_i is the part of the solid corresponding to Q_i , then the mass of T_i may be approximated by $\rho(x_i, y_i, z_i) \Delta V_i$ where (x_i, y_i, z_i) is a point in Q_i . The moment of T_i with respect to the xy -plane may be approximated by $z_i \rho(x_i, y_i, z_i) \Delta V_i$. Summing all such moments and taking the limit leads to the first integral in (17.23). The other formulas may be motivated in similar fashion. The **center of mass** is the point $(\bar{x}, \bar{y}, \bar{z})$, such that

$$(17.24) \quad \bar{x} = \frac{M_{yz}}{M}, \quad \bar{y} = \frac{M_{xz}}{M}, \quad \bar{z} = \frac{M_{xy}}{M}.$$

If Q is homogeneous, then the function ρ is constant and hence cancels in (17.24). Consequently, the center of mass of a homogeneous solid depends only on the shape of Q . As in two dimensions, the corresponding point for geometric solids is called the **centroid** of the solid.

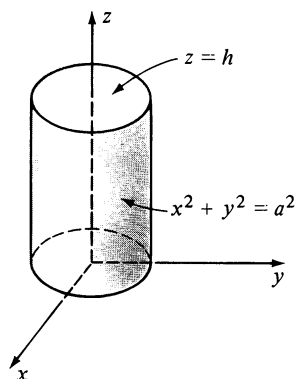


FIGURE 17.38

Example 1 A solid has the shape of a right circular cylinder of base radius a and height h . Find the center of mass if the density at a point P is directly proportional to the distance from one of the bases to P .

Solution If we introduce a coordinate system as in Figure 17.38, then the solid is bounded by the graphs of $x^2 + y^2 = a^2$, $z = 0$, and $z = h$. By hypothesis the density at (x, y, z) is $\rho(x, y, z) = kz$ for some constant k . Evidently, the center of mass is on the z -axis and, therefore, it is sufficient to find $\bar{z} = M_{xy}/M$. Moreover, by the form of ρ and the symmetry of the solid, we may calculate M and M_{xy} for the first octant portion and multiply by 4. Using (17.22),

$$\begin{aligned} M &= 4 \int_0^a \int_0^{\sqrt{a^2-x^2}} \int_0^h kz \, dz \, dy \, dx = 4k \int_0^a \int_0^{\sqrt{a^2-x^2}} \frac{h^2}{2} \, dy \, dx \\ &= 2kh^2 \int_0^a \sqrt{a^2-x^2} \, dx = 2kh^2 \left(\frac{\pi a^2}{4} \right) = \frac{k\pi h^2 a^2}{2}. \end{aligned}$$

Next using (17.23),

$$\begin{aligned} M_{xy} &= 4 \int_0^a \int_0^{\sqrt{a^2-x^2}} \int_0^h z(kz) \, dz \, dy \, dx = 4k \int_0^a \int_0^{\sqrt{a^2-x^2}} \frac{h^3}{3} \, dy \, dx \\ &= \frac{4kh^3}{3} \int_0^a \sqrt{a^2-x^2} \, dx = \frac{4kh^3}{3} \left(\frac{\pi a^2}{4} \right) = \frac{k\pi h^3 a^2}{3}. \end{aligned}$$

Finally, by (17.24),

$$\bar{z} = \frac{M_{xy}}{M} = \frac{k\pi h^3 a^2}{3} \cdot \frac{2}{k\pi h^2 a^2} = \frac{2h}{3}.$$

Hence the center of mass is on the axis of the cylinder, $\frac{2}{3}$ of the way from the lower base. ■

In the following example we shall merely *set up* the integrals necessary for the solution; that is, the integrals will be expressed in iterated form, but not evaluated.

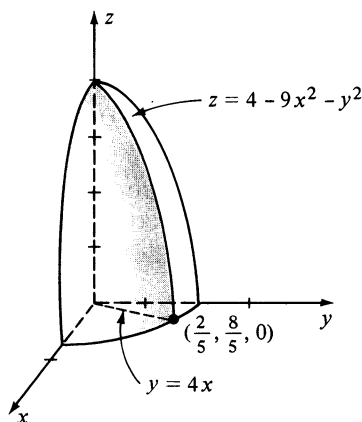


FIGURE 17.39

Example 2 A solid has the shape of the region in the first octant bounded by the graphs of $4 - z = 9x^2 + y^2$, $y = 4x$, $z = 0$, and $y = 0$. If the density at the point $P(x, y, z)$ is proportional to the distance from the origin to P , set up the integrals needed to find $\bar{x}$.

Solution The region is sketched in Figure 17.39. The density at (x, y, z) is $\rho(x, y, z) = k(x^2 + y^2 + z^2)^{1/2}$ for some k . Using (17.22) and (17.23), M and M_{yz} are given by

$$\begin{aligned} M &= \int_0^{8/5} \int_{y/4}^{\sqrt{4-y^2}/3} \int_0^{4-9x^2-y^2} k(x^2 + y^2 + z^2)^{1/2} \, dz \, dx \, dy \\ M_{yz} &= \int_0^{8/5} \int_{y/4}^{\sqrt{4-y^2}/3} \int_0^{4-9x^2-y^2} xk(x^2 + y^2 + z^2)^{1/2} \, dz \, dx \, dy. \end{aligned}$$

By (17.24), $\bar{x} = M_{yz}/M$. ■

If a particle of mass m is at the point (x, y, z) , then its distance to the z -axis is $(x^2 + y^2)^{1/2}$ and its **moment of inertia I_z with respect to the z -axis** is defined as $(x^2 + y^2)m$. Next, consider a solid T that has the shape of a region Q and density $\rho(x, y, z)$ at (x, y, z) . As before, let $\{Q_i\}$ be a partition of Q and choose any point (x_i, y_i, z_i) in Q_i . If T_i is the part of the solid corresponding to Q_i , then the moment of inertia of T_i with respect to the z -axis may be approximated by $(x_i^2 + y_i^2)\rho(x_i, y_i, z_i) \Delta V_i$. Taking the limit of a sum of such terms leads to the following definition of the moment of inertia of T with respect to the z -axis:

$$(17.25) \quad I_z = \iiint_Q (x^2 + y^2)\rho(x, y, z) dV.$$

Similarly, the moments of inertia with respect to the x - and y -axis, respectively, are

$$(17.26) \quad I_x = \iiint_Q (y^2 + z^2)\rho(x, y, z) dV, \quad I_y = \iiint_Q (x^2 + z^2)\rho(x, y, z) dV.$$

If a solid of mass M has moment of inertia I with respect to a line, then the **radius of gyration** is, by definition, the number d such that $I = Md^2$. This formula implies that the radius of gyration is the distance from the line at which all the mass could be concentrated without changing the moment of inertia of the solid.

Example 3 Find the moment of inertia and radius of gyration, with respect to the axis of symmetry, of the cylindrical solid described in Example 1.

Solution The solid is sketched in Figure 17.38, where $\rho(x, y, z) = kz$. Using (17.25) and symmetry,

$$\begin{aligned} I_z &= 4 \int_0^a \int_0^{\sqrt{a^2-x^2}} \int_0^h (x^2 + y^2)kz \, dz \, dy \, dx \\ &= 4k \int_0^a \int_0^{\sqrt{a^2-x^2}} (x^2 + y^2) \frac{h^2}{2} \, dy \, dx \\ &= 2kh^2 \int_0^a \left[x^2 y + \frac{y^3}{3} \right]_0^{\sqrt{a^2-x^2}} dx \\ &= 2kh^2 \int_0^a \left[x^2 \sqrt{a^2-x^2} + \frac{1}{3} \sqrt{(a^2-x^2)^3} \right] dx. \end{aligned}$$

The last integral may be evaluated by using a trigonometric substitution or a table of integrals. This results in $I_z = k\pi h^2 a^4/4$.

If d is the radius of gyration, then $d^2 = I_z/M$. Using the value for M found in Example 1, we obtain

$$d^2 = \frac{k\pi h^2 a^4}{4} \cdot \frac{2}{k\pi h^2 a^2} = \frac{a^2}{2}.$$

Hence $d = a/\sqrt{2} \approx 0.7a$; that is, the radius of gyration is a distance from the axis of the cylinder which is approximately $\frac{7}{10}$ of the radius of the cylinder. ■

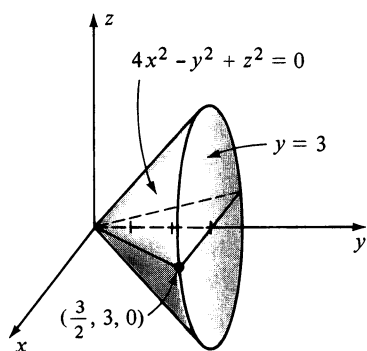


FIGURE 17.40

Example 4 A homogeneous solid has the shape of the region Q bounded by the graphs of $4x^2 - y^2 + z^2 = 0$ and $y = 3$. Set up a triple integral for finding the moment of inertia with respect to the y -axis.

Solution The region is part of an elliptic cone and is sketched in Figure 17.40. Note that the trace of the cone on the xy -plane is the pair of lines $x = \pm y/2$. If we denote the constant density by k , then applying (17.26),

$$I_y = \int_0^3 \int_{-y/2}^{y/2} \int_{-\sqrt{y^2 - 4x^2}}^{\sqrt{y^2 - 4x^2}} (x^2 + z^2)k \, dz \, dx \, dy.$$

We could also find I_y by multiplying by 4 the moment of inertia of that part of the solid that lies in the first octant. Thus

$$I_y = 4 \int_0^3 \int_0^{y/2} \int_0^{\sqrt{y^2 - 4x^2}} (x^2 + z^2)k \, dz \, dx \, dy. \quad \blacksquare$$

17.7 Exercises

In Exercises 1 and 2 set up the integrals needed to find the centroid of the given solid, and then find the centroid.

- The solid in the first octant bounded by the coordinate planes and the graphs of $z = 9 - x^2$ and $2x + y = 6$.
- The solid bounded by the graphs of $z = x^2$, $y = x^2$, $y = x^3$, and $z = 0$.

In Exercises 3 and 4 set up, but do not evaluate, the integrals needed to find the centroid of the given solid.

- The solid bounded by the graphs of $y = 4x^2 + 9z^2$, $z = x^2$, $z = 1$, and $y = 0$.
- The solid in the first octant bounded by the coordinate planes and the graphs of $x^2 + z^2 = 9$ and $y^2 + z^2 = 9$.
- The density at a point P in a cubical solid of edge a is directly proportional to the square of the distance from P to a fixed corner of the cube. Find the center of mass.
- Let Q be the tetrahedron bounded by the coordinate planes and the plane $2x + 5y + z = 10$. Find the center of mass if the density at the point $P(x, y, z)$ is directly proportional to the distance from the xz -plane to P .
- Let Q be bounded by the paraboloid $x = y^2 + 4z^2$ and the plane $x = 4$. If the density at (x, y, z) is $\rho(x, y, z) = x^2 + z^2$, set up, but do not evaluate, the integrals necessary for finding the center of mass of Q .

- Let Q be bounded by the hyperboloid $y^2 - x^2 - z^2 = 1$ and the plane $y = 2$. If the density at (x, y, z) is given

by $\rho(x, y, z) = x^2 y^2 z^2$, set up, but do not evaluate, the integrals necessary for finding the center of mass.

- Find the moment of inertia with respect to a line through one of its edges of a homogeneous cube of volume a^3 . What is the radius of gyration?
- Find the moment of inertia with respect to the z -axis of the tetrahedron described in Exercise 6. What is the radius of gyration?

In Exercises 11–16 set up the integral for the moment of inertia with respect to the z -axis of the indicated solid.

- The solid bounded by the graphs of $z = 36 - 4x^2 - 9y^2$ and $z = 0$; $\rho(x, y, z) = z$.
- The cylinder bounded by the graphs of $4x^2 + y^2 = a^2$, $z = 0$, and $z = h$; $\rho(x, y, z) = 1 + z$.
- A sphere of radius a with center at the origin; $\rho(x, y, z) = x^2 + y^2 + z^2$.
- The cone bounded by the graphs of $x^2 + 9y^2 - z^2 = 0$ and $z = 36$; $\rho(x, y, z) = x^2 + y^2$.
- The homogeneous tetrahedron bounded by the coordinate planes and the graph of $x/a + y/b + z/c = 1$, where a , b , and c are positive.
- The homogeneous solid bounded by the ellipsoid

$$\frac{x^2}{a^2} + \frac{y^2}{b^2} + \frac{z^2}{c^2} = 1.$$

17.8 Triple Integrals in Cylindrical and Spherical Coordinates

Triple integrals may sometimes be expressed in terms of cylindrical coordinates. The simplest case occurs if a function of r , θ , and z is continuous throughout a region of the form

$$Q = \{(r, \theta, z) : a \leq r \leq b, c \leq \theta \leq d, k \leq z \leq l\}.$$

We first divide Q into subregions $Q_1, Q_2, \dots, Q_n$ that have the same shape as Q by using graphs of equations $r = a_i$, $\theta = c_i$, and $z = k_i$, where a_i , c_i , and k_i are numbers in the intervals $[a, b]$, $[c, d]$, and $[k, l]$, respectively. A typical subregion Q_i , appropriately labeled, is sketched in Figure 17.41, where $\bar{r}_i$ is the average radius of the base of Q_i . The volume ΔV_i of Q_i is the product of the area of the base $\bar{r}_i \Delta r_i \Delta \theta_i$ and the altitude Δz_i . Thus

$$\Delta V_i = \bar{r}_i \Delta r_i \Delta \theta_i \Delta z_i.$$

If (r_i, θ_i, z_i) is any point in Q_i , then the triple integral of f over Q is given by

$$\iiint_Q f(r, \theta, z) dV = \lim_{\|P\| \rightarrow 0} \sum_i f(r_i, \theta_i, z_i) \Delta V_i$$

where the norm $\|P\|$ is the length of the longest diagonal of the Q_i . Furthermore, it can be proved that

$$\iiint_Q f(r, \theta, z) dV = \int_k^l \int_c^d \int_a^b f(r, \theta, z) r dr d\theta dz.$$

There are five other possible orders of integration for this iterated integral.

Triple integrals in cylindrical coordinates may be defined over more complicated regions by employing inner partitions. For example, it can be shown that if R is a polar region of the type discussed in Section 17.5 and if

$$Q = \{(r, \theta, z) : (r, \theta) \text{ is in } R \text{ and } k_1(r, \theta) \leq z \leq k_2(r, \theta)\}$$

where the functions k_1 and k_2 have continuous first partial derivatives throughout R , then

$$\iiint_Q f(r, \theta, z) dV = \iint_R \left[\int_{k_1(r, \theta)}^{k_2(r, \theta)} f(r, \theta, z) dz \right] dA.$$

In particular, suppose R is a region of the type illustrated in Figure 17.42. We may then evaluate the triple integral of f over Q as follows.

$$(17.27) \quad \iiint_Q f(r, \theta, z) dV = \int_\alpha^\beta \int_{g_1(\theta)}^{g_2(\theta)} \int_{k_1(r, \theta)}^{k_2(r, \theta)} f(r, \theta, z) r dz dr d\theta.$$

The iterated integral in (17.27) may be interpreted as a limit of a triple sum, in which the first summation takes place along a column of the subregions

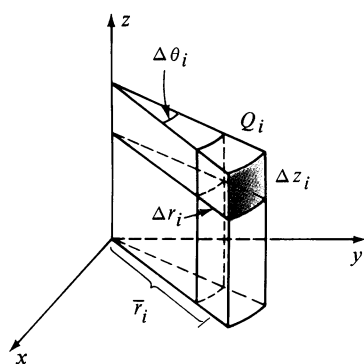


FIGURE 17.41 $\Delta V_i = \bar{r}_i \Delta r_i \Delta \theta_i \Delta z_i$

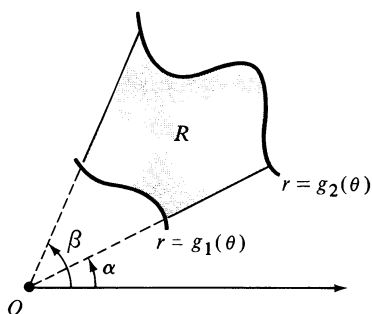


FIGURE 17.42

Q_i from the lower surface (with equation $z = k_1(r, \theta)$) to the upper surface (with equation $z = k_2(r, \theta)$) (see Figure 17.43). The second summation extends over a row of these columns, where θ is held fixed and r is allowed to vary. At this stage we have summed over a slice of Q as illustrated in Figure 17.43. Finally, we sweep out Q by letting θ vary from α to β .

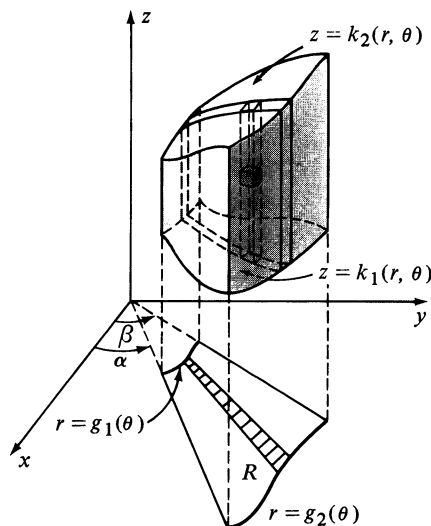


FIGURE 17.43

Example 1 A solid has the shape of the region Q that lies inside the cylinder $r = a$, within the sphere $r^2 + z^2 = 4a^2$ and above the xy -plane. Find the center of mass and the moment of inertia I_z if the density at a point P is directly proportional to the distance from the xy -plane to P .

Solution The region Q is sketched in Figure 17.44 together with a column that corresponds to a first summation in the direction of the z -axis. Since the density at the point $P(r, \theta, z)$ is given by kz for some constant k , the mass M may be found by applying (17.27) with $f(r, \theta, z) = kz$. Thus

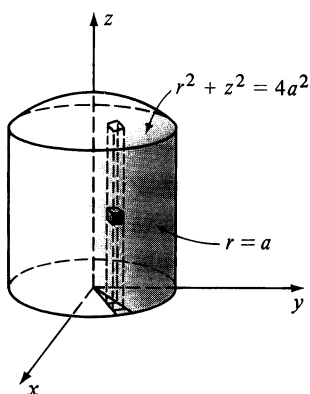


FIGURE 17.44

$$\begin{aligned} M &= \int_0^{2\pi} \int_0^a \int_0^{\sqrt{4a^2-r^2}} kzr \, dz \, dr \, d\theta = \frac{k}{2} \int_0^{2\pi} \int_0^a z^2 r \, dr \, d\theta \\ &= \frac{k}{2} \int_0^{2\pi} \int_0^a (4a^2r - r^3) \, dr \, d\theta = \frac{k}{2} \int_0^{2\pi} \left[2a^2r^2 - \frac{1}{4}r^4 \right]_0^a d\theta \\ &= \frac{7a^4k}{8} \int_0^{2\pi} d\theta = \frac{7a^4\pi k}{4}. \end{aligned}$$

To find I_z we employ (17.25) and use cylindrical coordinates, obtaining

$$\begin{aligned} I_z &= \int_0^{2\pi} \int_0^a \int_0^{\sqrt{4a^2-r^2}} r^2 kzr \, dz \, dr \, d\theta = \frac{k}{2} \int_0^{2\pi} \int_0^a r^3 z^2 \, dr \, d\theta \\ &= \frac{k}{2} \int_0^{2\pi} \int_0^a (4a^2r^3 - r^5) \, dr \, d\theta = \frac{5a^6\pi k}{6}. \end{aligned}$$

Triple integrals in spherical coordinates may also be considered. Suppose a function f of ρ , ϕ , and θ is continuous throughout a region of the form

$$Q = \{(\rho, \phi, \theta): a \leq \rho \leq b, c \leq \phi \leq d, k \leq \theta \leq l\}.$$

We may divide Q into subregions $Q_1, Q_2, \dots, Q_n$ having the same shape as Q by using graphs of equations of the form $\rho = a_i$, $\phi = c_i$, and $\theta = k_i$. A typical subregion Q_i is sketched in (i) of Figure 17.45, where $\Delta\rho_i$, $\Delta\phi_i$, and $\Delta\theta_i$ are the indicated increments.

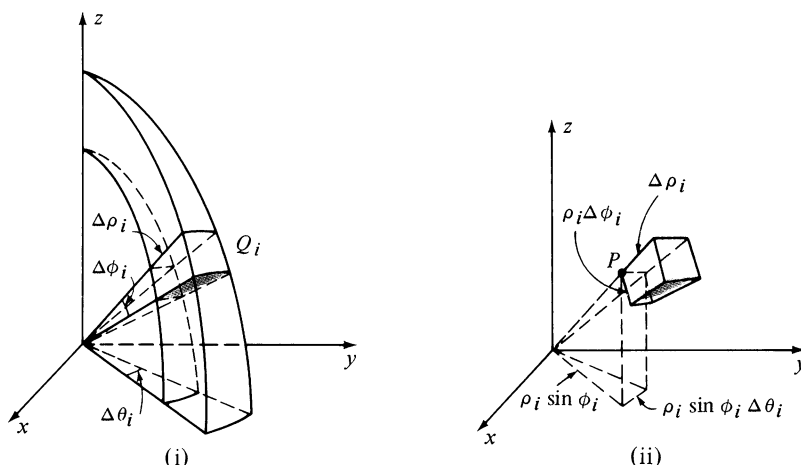


FIGURE 17.45

The volume of the subregion Q_i may be approximated by regarding it as a rectangular box. If $P(\rho_i, \phi_i, \theta_i)$ is a corner of the box, then as illustrated in (ii) of Figure 17.45, the dimensions of the box are approximately $\Delta\rho_i$, $\rho_i \Delta\phi_i$, and $\rho_i \sin \phi_i \Delta\theta_i$. Hence, if ΔV_i is the volume of Q_i , then

$$(17.28) \quad \Delta V_i \approx \rho_i^2 \sin \phi_i \Delta\rho_i \Delta\phi_i \Delta\theta_i.$$

If the triple integral of f over Q is defined as a limit of a sum in the usual way, then it can be proved that

$$(17.29) \quad \iiint_Q f(\rho, \phi, \theta) dV = \int_k^l \int_c^d \int_a^b f(\rho, \phi, \theta) \rho^2 \sin \phi d\rho d\phi d\theta.$$

There are five other possible orders of integration. Spherical coordinates may also be employed over more complicated regions by using inner partitions. In this case the limits of integration in (17.29) must be suitably adjusted.

Example 2 Find the volume and the centroid of the region Q that is bounded above by the sphere $\rho = a$ and below by the cone $\phi = m$ where $0 < m < \pi/2$.

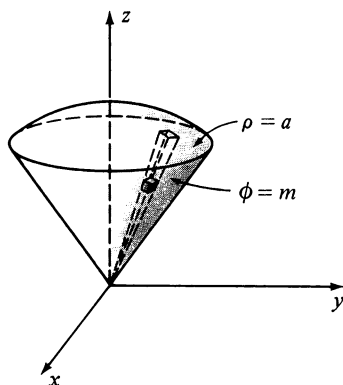


FIGURE 17.46

Solution The region Q is sketched in Figure 17.46. The volume V is

$$\begin{aligned} V &= \int_0^{2\pi} \int_0^m \int_0^a \rho^2 \sin \phi \, d\rho \, d\phi \, d\theta \\ &= \int_0^{2\pi} \int_0^m \frac{a^3}{3} \sin \phi \, d\phi \, d\theta = \frac{a^3}{3} \int_0^{2\pi} \left[-\cos \phi \right]_0^m d\theta \\ &= \frac{a^3}{3} \int_0^{2\pi} (1 - \cos m) \, d\theta = \frac{2\pi a^3}{3} (1 - \cos m). \end{aligned}$$

By symmetry, the centroid is on the z -axis. If (x, y, z) are rectangular coordinates of a point, then by Theorem (14.50) $z = \rho \cos \phi$. Consequently,

$$\begin{aligned} M_{xy} &= \iiint_Q z \, dV = \int_0^{2\pi} \int_0^m \int_0^a \rho \cos \phi \, \rho^2 \sin \phi \, d\rho \, d\phi \, d\theta \\ &= \int_0^{2\pi} \int_0^m \frac{a^4}{4} \sin \phi \cos \phi \, d\phi \, d\theta = \frac{a^4}{4} \int_0^{2\pi} \left[\frac{1}{2} \sin^2 \phi \right]_0^m d\theta \\ &= \frac{a^4}{8} \sin^2 m \int_0^{2\pi} d\theta = \frac{\pi a^4}{4} \sin^2 m. \end{aligned}$$

The centroid, in rectangular coordinates, is $(0, 0, \bar{z})$, where

$$\bar{z} = \frac{M_{xy}}{V} = \frac{3}{8}a(1 + \cos m).$$

The letter ρ used in spherical coordinates should not be confused with the density function. For problems that involve spherical coordinates *and* a density function, a different symbol should be used to denote density.

17.8 Exercises

Use cylindrical coordinates in Exercises 1–10.

- Find the volume and the centroid of the solid bounded by the graphs of $z = x^2 + y^2$, $x^2 + y^2 = 4$, and $z = 0$.
- Find the volume and the centroid of the solid bounded by the graphs of $x^2 + y^2 - z^2 = 0$ and $x^2 + y^2 = 4$.
- Find the moment of inertia of a homogeneous right circular cylinder of altitude h and radius of base a with respect to (a) the axis of the cylinder; (b) a diameter of the base.
- A homogeneous solid is bounded by the graphs of $z = \sqrt{x^2 + y^2}$ and $z = x^2 + y^2$. Find (a) the center of mass; (b) the moment of inertia with respect to the z -axis.

- The density at a point P of a spherical solid of radius a is directly proportional to the distance from P to a fixed line l through the center of the solid. Find the mass of the solid.
- Find the mass of the conical solid bounded by the graphs of $z = \sqrt{x^2 + y^2}$ and $z = 4$ if the density at a point P is directly proportional to the distance from the z -axis to P .
- Find the moment of inertia with respect to l for the solid described in Exercise 5. What is the radius of gyration?
- Find the moment of inertia with respect to the z -axis for the solid described in Exercise 6. What is the radius of gyration?

- 9 Let $Q = \{(x, y, z): 1 \leq z \leq 5 - x^2 - y^2, x^2 + y^2 \geq 1\}$. If the density at the point $P(x, y, z)$ is directly proportional to the distance from the xy -plane to P , find the mass and the center of mass of Q .
- 10 Find the mass and center of mass of the solid that lies inside both the cylinder $x^2 + y^2 - 2y = 0$ and the sphere $x^2 + y^2 + z^2 = 4$ if the density at the point $P(x, y, z)$ is directly proportional to the distance from the xy -plane to P .

Use spherical coordinates in Exercises 11–18.

- 11 Find the mass and center of mass of a solid hemisphere of radius a if the density at a point P is directly proportional to the distance from the center of the base to P .
- 12 Rework Exercise 2.
- 13 Find the moment of inertia with respect to the axis for the hemisphere in Exercise 11.
- 14 Find the moment of inertia with respect to a diameter of the base of a homogeneous solid hemisphere of radius a .
- 15 Find the volume of the solid that lies above the cone $z^2 = x^2 + y^2$ and inside the sphere $x^2 + y^2 + z^2 = 4z$.
- 16 Find the volume of the solid that lies outside the cone $z^2 = x^2 + y^2$ and inside the sphere $x^2 + y^2 + z^2 = 1$.

- 17 Find the mass of the solid that lies outside the sphere $x^2 + y^2 + z^2 = 1$ and inside the sphere $z^2 + y^2 + z^2 = 2$ if the density at a point P is directly proportional to the square of the distance from the center of the spheres to P .
- 18 A homogeneous spherical shell has inner radius a and outer radius b . Find its moment of inertia with respect to a line through the center.

Evaluate the integrals in Exercises 19 and 20 by changing to cylindrical coordinates.

- 19
$$\int_0^1 \int_0^{\sqrt{1-y^2}} \int_0^{\sqrt{4-x^2-y^2}} z \, dz \, dx \, dy$$
- 20
$$\int_0^2 \int_{-\sqrt{2x-x^2}}^{\sqrt{2x-x^2}} \int_0^{\sqrt{x^2+y^2}} \sqrt{x^2+y^2} \, dz \, dy \, dx$$

Evaluate the integrals in Exercises 21 and 22 by changing to spherical coordinates.

- 21
$$\int_{-2}^2 \int_{-\sqrt{4-x^2}}^{\sqrt{4-x^2}} \int_{\sqrt{4-x^2-y^2}}^{\sqrt{8-x^2-y^2}} (x^2 + y^2 + z^2) \, dz \, dy \, dx$$
- 22
$$\int_0^{\sqrt{2}} \int_1^{\sqrt{4-y^2}} \int_0^{\sqrt{4-x^2-y^2}} \sqrt{x^2 + y^2 + z^2} \, dz \, dx \, dy$$
- 23 Interpret (17.29) as a limit of a triple sum.

17.9 Surface Area

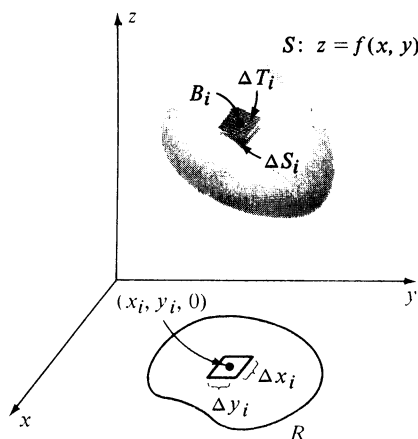


FIGURE 17.47

In Chapter 13 we obtained formulas for the area of a surface of revolution (see (13.13)–(13.15)). We shall now discuss a method for finding areas of surfaces of the type illustrated in Figure 17.47. Suppose that $f(x, y) \geq 0$ throughout a region R in the xy -plane and that f has continuous first partial derivatives on R . Let S denote the part of the graph of f whose projection on the xy -plane is R . To simplify the discussion we shall assume that no normal vector to S is parallel to the xy -plane. Our objective is to define the area A of S and to find a formula that can be used to calculate A .

Let $P = \{R_i\}$ be an inner partition of R , where the dimensions of the rectangle R_i are Δx_i and Δy_i . We choose a point $(x_i, y_i, 0)$ in each R_i and let $B_i(x_i, y_i, f(x_i, y_i))$ be the corresponding point on S . Next consider the tangent plane to S at B_i . Let ΔT_i and ΔS_i be the areas of the regions on the tangent plane and S , respectively, obtained by projecting R_i vertically upward (see Figure 17.47). If the norm $\|P\|$ of the partition is small, then ΔT_i is an approximation to ΔS_i , and $\sum_i \Delta T_i$ is an approximation to the area A of S . Since this approximation should improve as $\|P\|$ approaches 0, we define the **surface area** by

$$(17.30) \quad A = \lim_{\|P\| \rightarrow 0} \sum_i \Delta T_i.$$

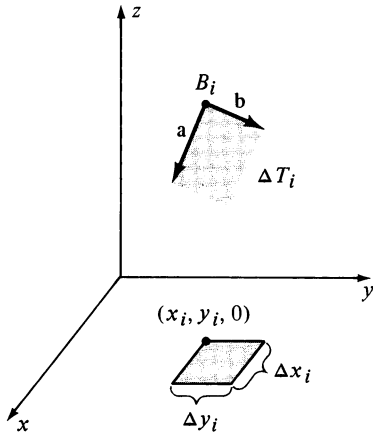


FIGURE 17.48

To find an integral formula for A , let us choose $(x_i, y_i, 0)$ as the corner of R_i closest to the origin. Let $\mathbf{a}$ and $\mathbf{b}$ be the vectors with initial point $B_i(x_i, y_i, f(x_i, y_i))$ that are tangent to the traces of S on the planes $y = y_i$ and $x = x_i$, respectively, as illustrated in the isolated view shown in Figure 17.48.

Using the geometric interpretation for partial derivatives given in Section 16.3, the slopes of the lines determined by $\mathbf{a}$ and $\mathbf{b}$ in these planes are $f_x(x_i, y_i)$ and $f_y(x_i, y_i)$, respectively. It follows that

$$\mathbf{a} = \Delta x_i \mathbf{i} + f_x(x_i, y_i) \Delta x_i \mathbf{k}$$

$$\mathbf{b} = \Delta y_i \mathbf{j} + f_y(x_i, y_i) \Delta y_i \mathbf{k}.$$

The area ΔT_i of the parallelogram determined by $\mathbf{a}$ and $\mathbf{b}$ is $|\mathbf{a} \times \mathbf{b}|$. Since

$$\mathbf{a} \times \mathbf{b} = \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ \Delta x_i & 0 & f_x(x_i, y_i) \Delta x_i \\ 0 & \Delta y_i & f_y(x_i, y_i) \Delta y_i \end{vmatrix}$$

we have

$$\mathbf{a} \times \mathbf{b} = -f_x(x_i, y_i) \Delta x_i \Delta y_i \mathbf{i} - f_y(x_i, y_i) \Delta x_i \Delta y_i \mathbf{j} + \Delta x_i \Delta y_i \mathbf{k}.$$

Consequently,

$$\Delta T_i = |\mathbf{a} \times \mathbf{b}| = \sqrt{(f_x(x_i, y_i))^2 + (f_y(x_i, y_i))^2 + 1} \Delta x_i \Delta y_i$$

where $\Delta x_i \Delta y_i = \Delta A_i$. If, as in (17.30), we take the limit of a sum of the ΔT_i and apply the definition of the double integral, we obtain

$$(17.31) \quad A = \iint_R \sqrt{(f_x(x, y))^2 + (f_y(x, y))^2 + 1} \, dA.$$

This formula may also be used if $f(x, y) \leq 0$ on R .

Example 1 Let R be the triangular region in the xy -plane with vertices $(0, 0, 0)$, $(0, 1, 0)$, and $(1, 1, 0)$. Find the surface area of that part of the graph of $z = 3x + y^2$ that lies over R .

Solution The region R is bounded by the graphs of $y = x$, $x = 0$, and $y = 1$ as shown in Figure 17.49. Letting $f(x, y) = 3x + y^2$ and applying (17.31) gives us

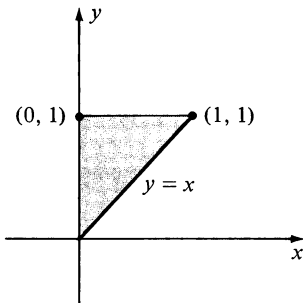


FIGURE 17.49

$$\begin{aligned} A &= \iint_R \sqrt{3^2 + (2y)^2 + 1} \, dA = \int_0^1 \int_0^y (10 + 4y^2)^{1/2} \, dx \, dy \\ &= \int_0^1 (10 + 4y^2)^{1/2} x \Big|_0^y \, dy = \int_0^1 (10 + 4y^2)^{1/2} y \, dy \\ &= \frac{1}{12} (10 + 4y^2)^{3/2} \Big|_0^1 = \frac{14^{3/2} - 10^{3/2}}{12} \approx 1.7. \end{aligned}$$

■

Example 2 Find the surface area of the part of the graph of $z = 9 - x^2 - y^2$ that lies above or on the xy -plane.

Solution The graph is sketched in Figure 16.5. By (17.31),

$$A = \iint_R \sqrt{4x^2 + 4y^2 + 1} \, dx \, dy$$

where R is the region in the xy -plane bounded by the circle $x^2 + y^2 = 9$. If we use polar coordinates to evaluate the double integral, then

$$A = \int_0^{2\pi} \int_0^3 (4r^2 + 1)^{1/2} r \, dr \, d\theta.$$

It is left to the reader to show that $A = \pi(37^{3/2} - 1)/6 \approx 117.3$. ■

Formulas similar to (17.31) may be stated if the surface S has suitable projections on the yz - or xz -planes. Thus, if S is the graph of an equation $y = g(x, z)$ and if the projection on the xz -plane is R_1 , then

$$A = \iint_{R_1} \sqrt{(g_x(x, z))^2 + (g_z(x, z))^2 + 1} \, dx \, dz.$$

Another formula of this type may be stated if S is given by $x = h(y, z)$.

17.9 Exercises

- Find the surface area of that part of the graph of $z = y + (x^2/2)$ that lies over the square region in the xy -plane having vertices $(0, 0, 0)$, $(1, 0, 0)$, $(1, 1, 0)$, and $(0, 1, 0)$.
- Find the area of the region on the plane $z = y + 1$ that lies inside the cylinder $x^2 + y^2 = 1$.
- Find the area of the part of the plane $x/a + y/b + z/c = 1$ cut out by the cylinder $x^2 + y^2 = d^2$, where a, b, c , and d are positive.
- Set up an integral for finding the surface area of that part of the sphere $x^2 + y^2 + z^2 = 4$ that lies over the square region in the xy -plane having vertices $(1, 1, 0)$, $(1, -1, 0)$, $(-1, 1, 0)$, and $(-1, -1, 0)$.
- Find the surface area of that part of the sphere with equation $x^2 + y^2 + z^2 = a^2$ that lies inside the cylinder $x^2 + y^2 - ay = 0$.
- Find the surface area of that part of the cylinder $y^2 + z^2 = a^2$ that lies inside the cylinder $x^2 + y^2 = a^2$.
- Find the surface area of the part of the paraboloid $z = x^2 + y^2$ that is cut off by the plane $z = 1$.
- Let R be the triangular region in the xy -plane with vertices $(0, 0, 0)$, $(0, 2, 0)$, and $(2, 2, 0)$. Find the surface area of that part of the graph of $z = y^2$ that lies over R .
- Find the surface area of the part of the graph of $z = xy$ that is inside the cylinder $x^2 + y^2 = 4$.
- Find the surface area of that part of the hyperbolic paraboloid $z = x^2 - y^2$ that lies in the first octant and is inside the cylinder $x^2 + y^2 = 1$.

17.10 Review: Concepts

Define or discuss each of the following.

- 1 Inner partitions
- 2 Riemann sums
- 3 Double integral
- 4 Iterated double integral
- 5 Evaluation of double integrals
- 6 Areas and volumes using double integrals
- 7 Center of mass of a lamina
- 8 Moments of inertia of a lamina
- 9 Radius of gyration
- 10 Double integrals in polar coordinates
- 11 Triple integral
- 12 Iterated triple integral
- 13 Evaluation of triple integrals
- 14 Applications of triple integrals
- 15 Triple integrals in cylindrical and spherical coordinates
- 16 Surface area

Review: Exercises

Evaluate the integrals in Exercises 1–6.

- 1 $\int_{-1}^0 \int_{x+1}^{x^3} (x^2 - 2y) dy dx$
- 2 $\int_1^2 \int_1^{\ln y} \frac{1}{y} dx dy$
- 3 $\int_0^3 \int_r^{r^2+1} r d\theta dr$
- 4 $\int_2^0 \int_0^{z^2} \int_x^z (x + z) dy dx dz$
- 5 $\int_0^2 \int_{\sqrt{y}}^1 \int_{z^2}^y xy^2 z^3 dx dz dy$
- 6 $\int_0^{\pi/2} \int_0^{\pi/4} \int_0^{a \cos \phi} \rho^2 \sin \phi d\rho d\phi d\theta$

In Exercises 7–10 express $\iint_R f(x, y) dA$ as an iterated integral if R is the region bounded by the graphs of the given equations.

- 7 $x^2 - y^2 = 4, x = 4$
- 8 $x^2 - y^2 = 4, y = 4, y = 0$
- 9 $y^2 = 4 + x, y^2 = 4 - x$
- 10 $y = -x^2 + 4, y = 3x^2$

Each of the integrals in Exercises 11 and 12 represents the area of a region R in the xy -plane. Describe R .

- 11 $\int_{-1}^1 \int_{e^y}^{y^3} dx dy$
- 12 $\int_{-1}^0 \int_x^{-x^2} dy dx$

In Exercises 13 and 14 reverse the order of integration and evaluate the resulting integral.

- 13 $\int_0^3 \int_{y^2}^9 ye^{-x^2} dx dy$
- 14 $\int_0^1 \int_x^{\sqrt{x}} e^{x/y} dy dx$

In Exercises 15 and 16 find the mass and the center of mass of the lamina that has the shape of the region bounded by the graphs of the given equations and having the indicated density.

- 15 $y = x, y = 2x, x = 3$; density at the point $P(x, y)$ is directly proportional to the distance from the y -axis to P .
- 16 $y^2 = x, x = 4$; density at the point $P(x, y)$ is directly proportional to the distance from the line with equation $x = -1$ to P .
- 17 A lamina has the shape of the region that lies inside the graph of $r = 2 + \sin \theta$ and outside the graph of $r = 1$. Find the mass if the density at the point $P(r, \theta)$ is inversely proportional to the distance from the pole to P .
- 18 Find the area of the region bounded by the polar axis and the graphs of $r = e^\theta$ and $r = 2$ from $\theta = 0$ to $\theta = \ln 2$.
- 19 Use polar coordinates to evaluate

$$\int_{-a}^0 \int_{-\sqrt{a^2-x^2}}^0 \sqrt{x^2+y^2} dy dx$$

- 20 Find I_x, I_y , and I_O for the lamina bounded by the graphs of $y = x^2$ and $y = x^3$ if the density at the point $P(x, y)$ is directly proportional to the distance from the y -axis to P .

- 21 A lamina has the shape of a right triangle with sides of lengths a , b , and $\sqrt{a^2 + b^2}$. The density is directly proportional to the distance from the side of length a . Find the moment of inertia and the radius of gyration with respect to a line along the side of length a .
- 22 A lamina has the shape of the region between concentric circles of radii a and b , where $a < b$. If the density at a point P is directly proportional to the distance from the center to P , use polar coordinates to find the moment of inertia with respect to a line through the center.
- 23 Find the volume of the solid that lies under the graph of $z = xy^2$ and over the rectangle in the xy -plane with vertices $(1, 1, 0)$, $(2, 1, 0)$, $(1, 3, 0)$, and $(2, 3, 0)$.
- 24 Express $\iiint_Q f(x, y, z) dV$ as an iterated integral in six different ways, where Q is bounded by the graphs of $y = x^2 + 4z^2$ and $y = 4$.
- 25 Use triple integrals to find the volume and centroid of the solid bounded by the graphs of $z = x^2$, $z = 4$, $y = 0$, and $y + z = 4$.
- 26 Set up a triple integral for the moment of inertia with respect to the z -axis of the solid bounded by the graphs of $z = 9x^2 + y^2$ and $z = 9$ if the density at the point $P(x, y, z)$ is inversely proportional to the square of the distance from the point $(0, 0, -1)$ to P .
- 27 Set up a triple integral for the moment of inertia with respect to the y -axis of the solid bounded by the graphs of $x^2 - y^2 + z^2 = 1$, $y = 0$, and $y = 4$ if the density at the point $P(x, y, z)$ is directly proportional to the distance from the y -axis to P .
- 28 A homogeneous solid is bounded by the graph of $z = 9 - x^2 - y^2$, the interior of the cylinder $x^2 + y^2 = 4$, and the xy -plane. Use cylindrical coordinates to find (a) the mass; (b) the center of mass; (c) the moment of inertia with respect to the z -axis.
- 29 A solid has the shape of a sphere of radius a . Use spherical coordinates to find the mass if the density at a point P is directly proportional to the distance from the center to P .
- 30 Find the surface area of that part of the cone $z = (x^2 + y^2)^{1/2}$ that is inside the cylinder $x^2 + y^2 = 4x$.

18 Topics in Vector Calculus

The concepts introduced in this chapter have many applications in the physical sciences. After discussing *vector fields* and defining *line integrals*, we consider evaluation theorems and determine conditions for independence of path. *Green's Theorem* is then proved for elementary plane regions. This important result provides a connection between line integrals and double

integrals. We also investigate the *divergence* and *curl* of a vector field and discuss two of the major theorems in vector calculus: the *Divergence Theorem* and *Stokes' Theorem*. The chapter ends with a study of *transformations of coordinates*, *Jacobians*, and *change of variables in multiple integrals*.

18.1 Vector Fields

If, to each point K in a region, there is associated a unique vector having initial point K , then the totality of such vectors is called a **vector field**. Vector fields are common in everyday life. The diagram in (i) of Figure 18.1 illustrates a vector field determined by a wheel rotating about an axle. To each point on the wheel there corresponds a velocity vector. A vector field of this type is called a **velocity field**.

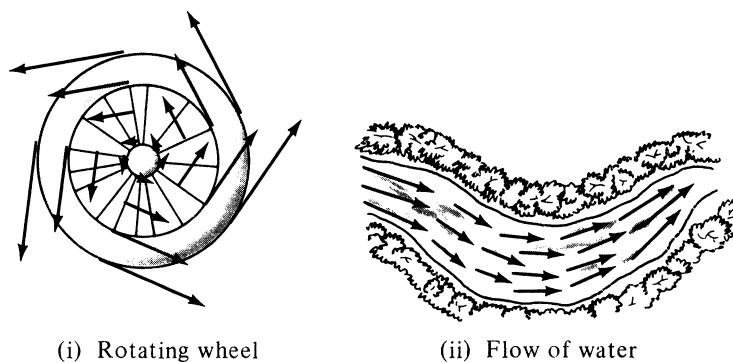


FIGURE 18.1 Velocity fields

Other examples of velocity fields are those determined by the flow of water or of wind. Thus, (ii) of Figure 18.1 could indicate the velocity vectors associated with fluid particles moving in a stream. Sketches of this type show only a few vectors of the vector field. It is important to remember that a vector is associated with *every* point in the region under consideration. Other common vector fields are **force fields**, which arise in the study of mechanics or electricity. In each of these illustrations it was assumed that the vectors were independent of time. These are called **steady vector fields** and are the only types we shall consider in this chapter.

If a rectangular coordinate system is introduced, then the vector associated with the point $K(x, y, z)$ may be denoted by $\mathbf{F}(x, y, z)$. Since the components of $\mathbf{F}(x, y, z)$ depend on the coordinates of K , we may write

$$(18.1) \quad \mathbf{F}(x, y, z) = M(x, y, z)\mathbf{i} + N(x, y, z)\mathbf{j} + P(x, y, z)\mathbf{k}$$

where M , N , and P are functions of x , y , and z . Conversely, every equation of the form (18.1) determines a vector field. It follows that a vector field may be defined as a **vector function** $\mathbf{F}$ whose domain D is a subset of $\mathbb{R} \times \mathbb{R} \times \mathbb{R}$ and whose range is a subset of V_3 . As a special case, if the domain corresponds to a region in a plane and the range is a subset of V_2 , then a vector field (in two dimensions) is given by

$$(18.2) \quad \mathbf{F}(x, y) = M(x, y)\mathbf{i} + N(x, y)\mathbf{j}$$

where M and N are functions of x and y . For convenience we shall use the terms “vector field” and “vector function” interchangeably. The functions discussed in Chapter 16, which associate a *number* with each point K , will be called **scalar functions** (or **scalar fields**).

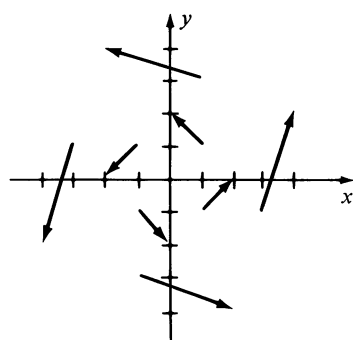
Example 1 Describe the vector field $\mathbf{F}$ if $\mathbf{F}(x, y) = -y\mathbf{i} + x\mathbf{j}$.

Solution The vectors $\mathbf{F}(x, y)$ associated with several points (x, y) are listed in the following table and are sketched in (i) of Figure 18.2.

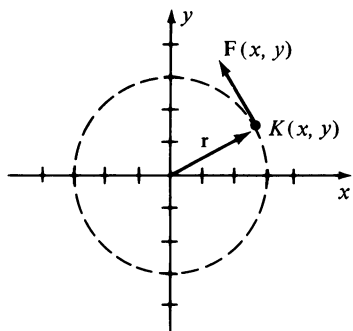
(x, y)	$\mathbf{F}(x, y)$	(x, y)	$\mathbf{F}(x, y)$
$(1, 1)$	$-\mathbf{i} + \mathbf{j}$	$(1, 3)$	$-3\mathbf{i} + \mathbf{j}$
$(-1, 1)$	$-\mathbf{i} - \mathbf{j}$	$(-3, 1)$	$-\mathbf{i} - 3\mathbf{j}$
$(-1, -1)$	$\mathbf{i} - \mathbf{j}$	$(-1, -3)$	$3\mathbf{i} - \mathbf{j}$
$(1, -1)$	$\mathbf{i} + \mathbf{j}$	$(3, -1)$	$\mathbf{i} + 3\mathbf{j}$

The vectors are much like those associated with the rotating wheel in (i) of Figure 18.1. To arrive at a more general description of the vector field $\mathbf{F}$, consider an arbitrary point $K(x, y)$ and let $\mathbf{r} = x\mathbf{i} + y\mathbf{j}$ be the position vector of $K(x, y)$ (see (ii) of Figure 18.2). Since

$$\begin{aligned} \mathbf{r} \cdot \mathbf{F}(x, y) &= (x\mathbf{i} + y\mathbf{j}) \cdot (-y\mathbf{i} + x\mathbf{j}) \\ &= -xy + yx = 0, \end{aligned}$$



(i)



(ii)

FIGURE 18.2

it follows that $\mathbf{F}(x, y)$ is orthogonal to the position vector $\mathbf{r}$ and, therefore, is tangent to the circle of radius $|\mathbf{r}|$ with center at the origin, as shown in (ii) of the figure. Note that

$$|\mathbf{F}(x, y)| = \sqrt{y^2 + x^2} = |\mathbf{r}|$$

and hence the magnitude of $\mathbf{F}(x, y)$ equals the radius of this circle. Thus the vector field is, indeed, similar to the velocity field of the rotating wheel illustrated in Figure 18.1, where the origin is at the point of rotation. ■

The next definition introduces one of the most important vector fields that occurs in the physical sciences.

Definition (18.3)

Let $\mathbf{r} = x\mathbf{i} + y\mathbf{j} + z\mathbf{k}$ be the position vector of a point $K(x, y, z)$. A vector field $\mathbf{F}$ is called an **inverse square field** if

$$\mathbf{F}(x, y, z) = \frac{c}{|\mathbf{r}|^2} \mathbf{u}$$

where c is a scalar and $\mathbf{u} = (1/|\mathbf{r}|)\mathbf{r}$ is a unit vector having the same direction as $\mathbf{r}$.

Example 2 Describe the inverse square field $\mathbf{F}(x, y, z)$ of (18.3) if $c < 0$.

Solution Since $\mathbf{u} = (1/|\mathbf{r}|)\mathbf{r}$, where $\mathbf{r} = x\mathbf{i} + y\mathbf{j} + z\mathbf{k}$, we have

$$\mathbf{F}(x, y, z) = \frac{c}{|\mathbf{r}|^3} \mathbf{r} = \frac{c}{(x^2 + y^2 + z^2)^{3/2}} (x\mathbf{i} + y\mathbf{j} + z\mathbf{k}).$$

The last expression may be written in form (18.1). It is simpler, however, to analyze the vectors in the field by using the expression containing $\mathbf{r}$. Since $\mathbf{F}(x, y, z)$ is a negative scalar multiple of $\mathbf{r}$, the direction of $\mathbf{F}(x, y, z)$ is toward the origin O . Also,

$$|\mathbf{F}(x, y, z)| = \frac{|c|}{|\mathbf{r}|^2} |\mathbf{u}| = \frac{|c|}{|\mathbf{r}|^2}$$

and hence the magnitude of $\mathbf{F}(x, y, z)$ is inversely proportional to the square of the distance from O to the point (x, y, z) . This means that as the point $K(x, y, z)$ moves away from the origin, the length of the associated vector $\mathbf{F}(x, y, z)$ decreases. Some typical vectors in the field are sketched in Figure 18.3. ■

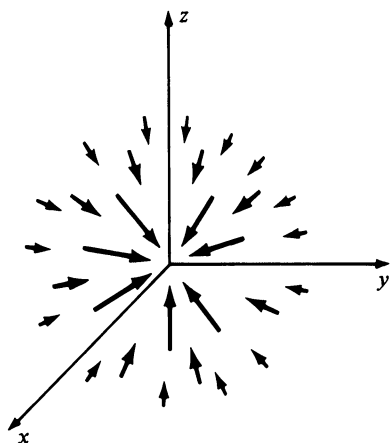


FIGURE 18.3 Inverse square vector field

The force of gravity determines an inverse square vector field. According to Newton's Law of Universal Gravitation, if a particle of mass M is located

at the origin of a rectangular coordinate system, then the force exerted on a particle of mass m located at $K(x, y, z)$ is given by

$$\mathbf{F}(x, y, z) = -G \frac{Mm}{|\mathbf{r}|^2} \mathbf{u}$$

where G is a gravitational constant, $\mathbf{r}$ is the position vector of the point K , and $\mathbf{u} = (1/|\mathbf{r}|)\mathbf{r}$.

Inverse square fields also occur in electrical theory. In particular, Coulomb's Law states that if a point charge of Q coulombs is at the origin, then the force $\mathbf{F}(x, y, z)$ it exerts on a point charge of q coulombs located at $K(x, y, z)$ is given by

$$\mathbf{F}(x, y, z) = c \frac{Qq}{|\mathbf{r}|^2} \mathbf{u}$$

for some constant c , where $\mathbf{r}$ and $\mathbf{u}$ are as in Definition (18.3). Note that Coulomb's Law has the same form as Newton's Law of Universal Gravitation.

Important types of vector fields may be obtained by using the gradient of a scalar function f of two or three variables. For example, if $w = f(x, y, z)$, then as in (16.27), the *gradient* ∇w of w is given by

$$\nabla w = f_x(x, y, z)\mathbf{i} + f_y(x, y, z)\mathbf{j} + f_z(x, y, z)\mathbf{k}.$$

We know from our work in Chapter 16 that the direction of the vector ∇w at any point K is orthogonal to the level surface associated with f that contains K . Moreover, the magnitude of ∇w equals the maximum rate of change of f at the point K .

Definition (18.4)

A vector field $\mathbf{F}$ is called a **conservative vector field** if it is the gradient of a scalar function, that is, if

$$\mathbf{F}(x, y, z) = \nabla f(x, y, z)$$

for some function f .

If $\mathbf{F}$ is conservative, then the function f in Definition (18.4) is called a **potential function** for $\mathbf{F}$ and $f(x, y, z)$ is called the **potential** at the point $K(x, y, z)$.

Inverse square fields are conservative. In particular, if $\mathbf{F}$ is the field given in Definition (18.3), then

$$\mathbf{F}(x, y, z) = \nabla \left(\frac{-c}{r} \right)$$

where $r = |\mathbf{r}| = (x^2 + y^2 + z^2)^{1/2}$. The verification of this fact is left to the reader (see Exercise 11). In physics, the potential function of a conservative vector field $\mathbf{F}$ is defined as the function p such that $\mathbf{F}(x, y, z) = -\nabla p(x, y, z)$. In this event, the field in Definition (18.3) is $\mathbf{F}(x, y, z) = \nabla(c/r)$. We shall have more to say about conservative fields later in this chapter.

The vector differential operator ∇ in three dimensions is

$$\nabla = \mathbf{i} \frac{\partial}{\partial x} + \mathbf{j} \frac{\partial}{\partial y} + \mathbf{k} \frac{\partial}{\partial z}.$$

It has no practical use standing alone; however, if it operates on a scalar function f it produces the gradient of f , as given by

$$\text{grad } f = \nabla f = \frac{\partial f}{\partial x} \mathbf{i} + \frac{\partial f}{\partial y} \mathbf{j} + \frac{\partial f}{\partial z} \mathbf{k}.$$

We shall next use ∇ as an operator on *vector* functions.

In certain applications a vector function $\mathbf{F}$ is used to obtain another vector function called the **curl** of $\mathbf{F}$, denoted by $\text{curl } \mathbf{F}$ or $\nabla \times \mathbf{F}$, and defined as follows.

Definition (18.5)

Let a vector function $\mathbf{F}$ in three dimensions be given by

$$\mathbf{F}(x, y, z) = M(x, y, z)\mathbf{i} + N(x, y, z)\mathbf{j} + P(x, y, z)\mathbf{k}$$

where M , N , and P have partial derivatives in some region. The **curl** of $\mathbf{F}$ is

$$\text{curl } \mathbf{F} = \nabla \times \mathbf{F} = \left(\frac{\partial P}{\partial y} - \frac{\partial N}{\partial z} \right) \mathbf{i} + \left(\frac{\partial M}{\partial z} - \frac{\partial P}{\partial x} \right) \mathbf{j} + \left(\frac{\partial N}{\partial x} - \frac{\partial M}{\partial y} \right) \mathbf{k}.$$

As indicated in Definition (18.5), it is customary to use either the symbol $\text{curl } \mathbf{F}$ or $\nabla \times \mathbf{F}$ to denote the *value* of this function at (x, y, z) . The formula for $\text{curl } \mathbf{F}$ may be regarded as the determinant “expansion” (by the first row) of the following expression:

$$(18.6) \quad \text{curl } \mathbf{F} = \nabla \times \mathbf{F} = \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ M & N & P \end{vmatrix}.$$

This determinant notation is ambiguous, since the first row is made up of vectors, the second row of partial derivative operators, and the third row of scalar functions; however, it is an extremely useful device for remembering the cumbersome formula given in Definition (18.5).

Example 3 Find $\nabla \times \mathbf{F}$ if $\mathbf{F}(x, y, z) = xy^2z^4\mathbf{i} + (2x^2y + z)\mathbf{j} + y^3z^2\mathbf{k}$.

Solution Applying (18.6),

$$\begin{aligned}\nabla \times \mathbf{F} &= \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ xy^2z^4 & (2x^2y + z) & y^3z^2 \end{vmatrix} \\ &= (3y^2z^2 - 1)\mathbf{i} + 4xy^2z^3\mathbf{j} + (4xy - 2xyz^4)\mathbf{k}. \quad \blacksquare\end{aligned}$$

The operator ∇ is also used to obtain a certain *scalar* field from a vector field F as follows.

Definition (18.7)

Let $\mathbf{F}(x, y, z) = M(x, y, z)\mathbf{i} + N(x, y, z)\mathbf{j} + P(x, y, z)\mathbf{k}$. The **divergence** of $\mathbf{F}$, denoted by $\text{div } \mathbf{F}$ or $\nabla \cdot \mathbf{F}$, is defined by

$$\text{div } \mathbf{F} = \nabla \cdot \mathbf{F} = \frac{\partial M}{\partial x} + \frac{\partial N}{\partial y} + \frac{\partial P}{\partial z}.$$

The reason for using the symbol $\nabla \cdot \mathbf{F}$ is evident from the fact that the formula for $\text{div } \mathbf{F}$ may be obtained by taking the formal dot product of ∇ and $\mathbf{F}$.

Example 4 Find $\nabla \cdot \mathbf{F}$ if $\mathbf{F}(x, y, z) = xy^2z^4\mathbf{i} + (2x^2y + z)\mathbf{j} + y^3z^2\mathbf{k}$.

Solution By Definition (18.7),

$$\begin{aligned}\nabla \cdot \mathbf{F} &= \frac{\partial}{\partial x}(xy^2z^4) + \frac{\partial}{\partial y}(2x^2y + z) + \frac{\partial}{\partial z}(y^3z^2) \\ &= y^2z^4 + 2x^2 + 2y^3z. \quad \blacksquare\end{aligned}$$

We shall discuss the physical significance of $\text{curl } \mathbf{F}$ and $\text{div } \mathbf{F}$ later in this chapter. In this section let us concentrate on algebraic properties of these functions. A property that is analogous to the product rule for derivatives is given in the next example. Several other properties are stated in exercises at the end of this section.

Example 5 If f and $\mathbf{F}$ are scalar and vector functions, respectively, that possess partial derivatives, show that

$$\nabla \cdot [f\mathbf{F}] = f[\nabla \cdot \mathbf{F}] + [\nabla f] \cdot \mathbf{F}.$$

Solution If we write $\mathbf{F} = M\mathbf{i} + N\mathbf{j} + P\mathbf{k}$, where M, N , and P are functions of x, y , and z , then

$$f\mathbf{F} = fM\mathbf{i} + fN\mathbf{j} + fP\mathbf{k}.$$

Applying Definition (18.7),

$$\begin{aligned}\nabla \cdot [f\mathbf{F}] &= \frac{\partial}{\partial x}(fM) + \frac{\partial}{\partial y}(fN) + \frac{\partial}{\partial z}(fP) \\ &= f \frac{\partial M}{\partial x} + \frac{\partial f}{\partial x} M + f \frac{\partial N}{\partial y} + \frac{\partial f}{\partial y} N + f \frac{\partial P}{\partial z} + \frac{\partial f}{\partial z} P.\end{aligned}$$

Rearranging terms gives us

$$\begin{aligned}\nabla \cdot [f\mathbf{F}] &= f \left[\frac{\partial M}{\partial x} + \frac{\partial N}{\partial y} + \frac{\partial P}{\partial z} \right] + \left[\frac{\partial f}{\partial x} M + \frac{\partial f}{\partial y} N + \frac{\partial f}{\partial z} P \right] \\ &= f[\nabla \cdot \mathbf{F}] + [\nabla f] \cdot \mathbf{F}.\end{aligned}$$

■

Finally, if a vector field $\mathbf{F}$ is described as in (18.1), then we may define limits, continuity, partial derivatives, and multiple integrals by using the components of $\mathbf{F}(x, y, z)$ in a fashion similar to that used for vector-valued functions in Chapter 15. For example, if we wish to differentiate or integrate $\mathbf{F}(x, y, z)$, we differentiate or integrate each component. The usual theorems may be established. Thus, $\mathbf{F}$ is continuous if and only if the component functions M, N , and P are continuous, $\mathbf{F}$ has partial derivatives if and only if the same is true for M, N , and P ; etc.

18.1 Exercises

In Exercises 1–10 sketch a sufficient number of vectors $\mathbf{F}(x, y, z)$ to illustrate the pattern of the vectors in the field $\mathbf{F}$.

- 1 $\mathbf{F}(x, y) = x\mathbf{i} - y\mathbf{j}$
- 2 $\mathbf{F}(x, y) = -x\mathbf{i} + y\mathbf{j}$
- 3 $\mathbf{F}(x, y) = 2x\mathbf{i} + 3y\mathbf{j}$
- 4 $\mathbf{F}(x, y) = 3\mathbf{i} + x\mathbf{j}$
- 5 $\mathbf{F}(x, y) = (x^2 + y^2)^{-1/2}(x\mathbf{i} + y\mathbf{j})$
- 6 $\mathbf{F}(x, y, z) = x\mathbf{i} + z\mathbf{k}$
- 7 $\mathbf{F}(x, y, z) = -x\mathbf{i} - y\mathbf{j} - z\mathbf{k}$
- 8 $\mathbf{F}(x, y, z) = x\mathbf{i} + y\mathbf{j} + z\mathbf{k}$
- 9 $\mathbf{F}(x, y, z) = \mathbf{i} + \mathbf{j} + \mathbf{k}$
- 10 $\mathbf{F}(x, y, z) = 2\mathbf{k}$
- 11 If $\mathbf{F}$ is the inverse square vector field of Definition (18.3), prove that $\mathbf{F}(x, y, z) = \nabla(-c/r)$, where $r = |\mathbf{r}|$.

- 12 Find a potential function for $\mathbf{F}(x, y, z) = -(GMm/|\mathbf{r}|^2)\mathbf{u}$.

In Exercises 13–16 find a conservative vector field that has the given potential.

- 13 $f(x, y, z) = x^2 - 3y^2 + 4z^2$
- 14 $f(x, y, z) = \sin(x^2 + y^2 + z^2)$
- 15 $f(x, y) = \arctan(xy)$
- 16 $f(x, y) = y^2 e^{-3x}$
- 17 If $\mathbf{F}$ is given by (18.1), define

$$\lim_{(x, y, z) \rightarrow (x_0, y_0, z_0)} \mathbf{F}(x, y, z) = \mathbf{a}$$

in a manner analogous to Definition (15.3). Give an ε - δ definition for this limit. What is the geometric significance of the limit?

- 18 If $\mathbf{F}$ is given by (18.1), define the notion of *continuity* at (x_0, y_0, z_0) in a manner similar to Definition (15.4). What is the geometric significance of a continuous vector function?

In Exercises 19–22 find $\nabla \times \mathbf{F}$ and $\nabla \cdot \mathbf{F}$.

- 19 $\mathbf{F}(x, y, z) = x^2z\mathbf{i} + y^2x\mathbf{j} + (y + 2z)\mathbf{k}$
 20 $\mathbf{F}(x, y, z) = (3x + y)\mathbf{i} + xy^2z\mathbf{j} + xz^2\mathbf{k}$
 21 $\mathbf{F}(x, y, z) = 3xyz^2\mathbf{i} + y^2 \sin z\mathbf{j} + xe^{2x}\mathbf{k}$
 22 $\mathbf{F}(x, y, z) = x^3 \ln z\mathbf{i} + xe^{-y}\mathbf{j} - (y^2 + 2z)\mathbf{k}$

Verify the identities in Exercises 23–26.

- 23 $\nabla \times (\mathbf{F} + \mathbf{G}) = \nabla \times \mathbf{F} + \nabla \times \mathbf{G}$
 24 $\nabla \cdot (\mathbf{F} + \mathbf{G}) = \nabla \cdot \mathbf{F} + \nabla \cdot \mathbf{G}$
 25 $\nabla \times (f\mathbf{F}) = f(\nabla \times \mathbf{F}) + (\nabla f) \times \mathbf{F}$
 26 $\nabla \cdot (\mathbf{F} \times \mathbf{G}) = (\nabla \times \mathbf{F}) \cdot \mathbf{G} - (\nabla \times \mathbf{G}) \cdot \mathbf{F}$

If f and $\mathbf{F}$ have continuous second partial derivatives, and $\mathbf{a}$ is a constant vector, verify the identities in Exercises 27–30.

- 27 $\text{curl grad } f = \mathbf{0}$
 28 $\text{div curl } \mathbf{F} = 0$
 29 $\text{curl (grad } f + \text{curl } \mathbf{F}) = \text{curl curl } \mathbf{F}$
 30 $\text{curl } \mathbf{a} = \mathbf{0}$
 31 If $\mathbf{r} = x\mathbf{i} + y\mathbf{j} + z\mathbf{k}$ prove that $\nabla \cdot \mathbf{r} = 3$, $\nabla \times \mathbf{r} = \mathbf{0}$, and $\nabla|\mathbf{r}| = \mathbf{r}/|\mathbf{r}|$.

- 32 If $\mathbf{r} = x\mathbf{i} + y\mathbf{j} + z\mathbf{k}$ and $\mathbf{a}$ is a constant vector, prove that $\text{curl } (\mathbf{a} \times \mathbf{r}) = 2\mathbf{a}$ and $\text{div } (\mathbf{a} \times \mathbf{r}) = 0$.

- 33 Prove that both the curl and the divergence of an inverse square vector field are zero.

- 34 Let $\mathbf{r} = x\mathbf{i} + y\mathbf{j} + z\mathbf{k}$ and $r = |\mathbf{r}|$. If $\mathbf{F}(x, y, z) = (c/r^k)\mathbf{r}$, where c is a constant and k is any positive real number, prove that the curl of $\mathbf{F}$ is $\mathbf{0}$. (Hint: Use Exercise 25.)

- 35 If a vector field $\mathbf{F}$ is conservative and has continuous partial derivatives, prove that $\text{curl } \mathbf{F} = \mathbf{0}$.

- 36 The differential operator Lap is denoted by $\nabla^2 = \nabla \cdot \nabla = \partial^2/\partial x^2 + \partial^2/\partial y^2 + \partial^2/\partial z^2$. If it operates on $f(x, y, z)$, it produces a scalar function called the **Laplacian** of f , that is,

$$\text{Lap } f = \nabla^2 f = \frac{\partial^2 f}{\partial x^2} + \frac{\partial^2 f}{\partial y^2} + \frac{\partial^2 f}{\partial z^2}.$$

If f and g are scalar functions that have second partial derivatives, prove that

$$(a) \text{ div grad } f = \text{Lap } f.$$

$$(b) \text{ Lap } (fg) =$$

$$f \text{ Lap } g + g \text{ Lap } f + 2(\text{grad } f) \cdot (\text{grad } g).$$

Using the notation of Exercise 36, prove that the functions defined in Exercises 37 and 38 satisfy **Laplace's equation** $\nabla^2 f = 0$. (Functions of this type are called **harmonic** and are very important in physical applications.)

$$37 \quad f(x, y, z) = (x^2 + y^2 + z^2)^{-1/2}$$

$$38 \quad f(x, y, z) = ax^2 + by^2 + cz^2, \text{ where } a + b + c = 0.$$

18.2 Line Integrals

To define $\int_a^b f(x) dx$ we began by dividing the interval $[a, b]$ into n subintervals of lengths $\Delta x_1, \Delta x_2, \dots, \Delta x_n$. We then chose a number w_i in each subinterval and took the limit of the Riemann sum $\sum_i f(w_i) \Delta x_i$ as all the Δx_i approached 0. A similar process can be carried out for functions of several variables. For example, if $f(x, y)$ is defined on a finite plane curve C , we could divide C into n subarcs of lengths $\Delta s_1, \Delta s_2, \dots, \Delta s_n$. After choosing a point (u_i, v_i) in each subarc we could then consider the limit of the sum $\sum_i f(u_i, v_i) \Delta s_i$ as all the Δs_i approach 0. This technique produces what is known as a *line integral*. Let us now make these remarks more precise. We shall begin by recalling the definition of *smooth curve* that was introduced in Section 13.1.

Definition (18.8)

A plane curve C is **smooth** if it has a parametric representation

$$x = g(t), \quad y = h(t), \quad \text{where} \quad a \leq t \leq b,$$

such that g' and h' are continuous and not simultaneously 0 on $[a, b]$.

The preceding definition can be extended to space curves in the obvious way, by considering a third function k such that $z = k(t)$.

The parametric representation of C in Definition (18.8) is sometimes called a **smooth parameterization**. For example, if C is the parabola $y = x^2$, then $x = t, y = t^2$ is a smooth parameterization; however, $x = t^3, y = t^6$ is not, since dx/dt and dy/dt are simultaneously 0 at $t = 0$. Hereafter, whenever we refer to a smooth curve C , it will be assumed that a smooth parameterization $x = f(t), y = g(t)$ has been chosen.

The cycloid $x = a(t - \sin t), y = a(1 - \cos t)$ sketched in Figure 13.4 is not smooth. Note that both $dx/dt = 0$ and $dy/dt = 0$ if $t = 2\pi n$, where n is any integer. Moreover, there is no smooth parameterization, since the cycloid has a corner at every point $(2\pi n, 0)$ on the x -axis.

Consider $f(x, y)$, where f is continuous on a region D containing a smooth curve C . Let A and B be the points on C determined by the parameter values a and b , respectively. We shall assign a positive direction, or **orientation**, along C as that determined by increasing values of t . Let us partition the parameter interval $[a, b]$ by choosing

$$a = t_0 < t_1 < t_2 < \cdots < t_n = b.$$

This leads to a subdivision of C into subarcs $\widehat{P_{i-1}P_i}$, where $P_i(x_i, y_i)$ is the point on C corresponding to t_i . As in Figure 18.4, let $\Delta x_i = x_i - x_{i-1}$, $\Delta y_i = y_i - y_{i-1}$, and let Δs_i denote the length of the subarc $\widehat{P_{i-1}P_i}$. The **norm** $\|\Delta\|$ of the subdivision of C is, by definition, the largest of the Δs_i .

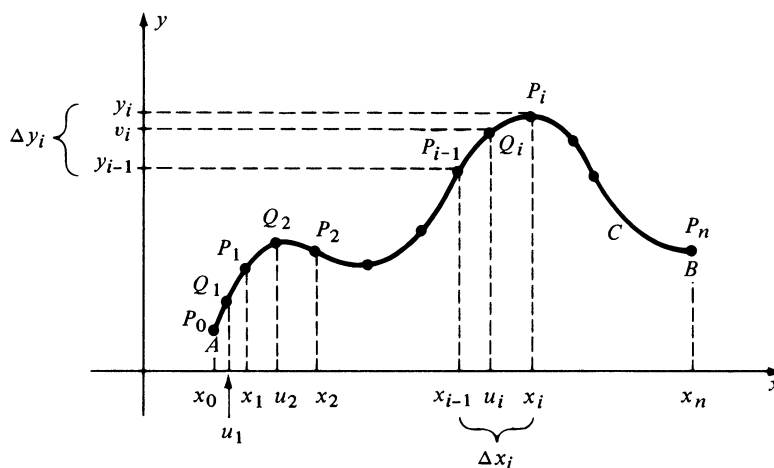


FIGURE 18.4

We next choose a point $Q_i(u_i, v_i)$ in each subarc $\widehat{P_{i-1}P_i}$, as illustrated in Figure 18.4. For each i , we evaluate the function f at (u_i, v_i) , multiply this number by Δs_i , and form the sum

$$\sum_{i=1}^n f(u_i, v_i) \Delta s_i.$$

If, in a manner similar to Definition (5.7), this sum has a limit L as $n \rightarrow \infty$ and $\|\Delta\| \rightarrow 0$ that is independent of the partition of $[a, b]$ and the points Q_i , then L is called the **line integral of f along C from A to B** and is denoted by

$$(18.9) \quad \int_C f(x, y) ds = \lim_{\|\Delta\| \rightarrow 0} \sum_i f(u_i, v_i) \Delta s_i.$$

The terminology *line integral* is a misnomer. It would be more descriptive to use *curve integral* for (18.9).

It can be shown that if f is continuous on D , then the limit in (18.9) exists and is the same for all parametric representations of C (provided the same orientation is used). Moreover, the integral may be evaluated as follows.

Theorem (18.10)

If a smooth curve C is given by $x = g(t)$, $y = h(t)$, where $a \leq t \leq b$, and if $f(x, y)$ is continuous on a region D containing C , then

$$\int_C f(x, y) ds = \int_a^b f(g(t), h(t)) \sqrt{[g'(t)]^2 + [h'(t)]^2} dt.$$

The form of the radical in Theorem (18.10) is a consequence of the discussion of arc length in Chapter 13 (see Theorem (13.11)). It is worth noting that if C is an interval on the x -axis, then $\Delta s_i = \Delta x_i$, all $v_i = 0$, and (18.10) reduces to a definite integral of the type considered in Chapter 5.

The preceding discussion can be extended to more complicated curves. In particular, suppose C is a **piecewise smooth curve**, in the sense that it can be expressed as the union of a finite number of smooth curves $C_1, C_2, \dots, C_n$, where the terminal point B_i of C_i is the initial point A_{i+1} of C_{i+1} for $i = 1, 2, \dots, n-1$. In this case the line integral of f along C is defined as the sum of the line integrals along the individual curves.

Example 1 Evaluate $\int_C xy^2 ds$ if C has parametric equations $x = \cos t$, $y = \sin t$ where $0 \leq t \leq \pi/2$.

Solution The curve C is that part of the unit circle with center O that lies in the first quadrant. Applying Theorem (18.10),

$$\begin{aligned} \int_C xy^2 ds &= \int_0^{\pi/2} \cos t \sin^2 t \sqrt{\sin^2 t + \cos^2 t} dt \\ &= \int_0^{\pi/2} \sin^2 t \cos t dt = \left[\frac{1}{3} \sin^3 t \right]_0^{\pi/2} = \frac{1}{3}. \end{aligned}$$

■

It is possible to establish properties of line integrals that are similar to those obtained for the definite integrals of Chapter 5. For example, it can be shown that reversing the direction of integration changes the sign of the integral, that the integral of a sum of two functions is the sum of the integrals of the individual functions, etc.

An elementary physical application of the line integral (18.9) may be obtained by regarding the curve as a thin wire of variable density. If the wire is represented by the curve C in Figure 18.4, and if the linear density (the mass per unit length) at the point (x, y) is given by $f(x, y)$, then $f(u_i, v_i) \Delta s_i$ is an approximation to the mass Δm_i of the part of the wire between P_{i-1} and P_i . The sum

$$\sum_i \Delta m_i = \sum_i f(u_i, v_i) \Delta s_i$$

is an approximation to the total mass M of the wire. To define M we take the limit of this sum, obtaining

$$M = \int_C f(x, y) ds.$$

Example 2 A thin wire is bent into the shape of a semicircle of radius a . If the density at a point P is directly proportional to its distance from the line through the endpoints, find the mass of the wire.

Solution If we introduce a coordinate system such that the shape of the wire coincides with the upper half of a circle of radius a and center O , then parametric equations for C are

$$x = a \cos t, \quad y = a \sin t; \quad 0 \leq t \leq \pi.$$

By hypothesis, the density function f is given by $f(x, y) = ky$ for some constant k . Hence, by the previous discussion, the mass of the wire is

$$\begin{aligned} M &= \int_C ky \, ds = \int_0^\pi ka \sin t \sqrt{a^2 \sin^2 t + a^2 \cos^2 t} \, dt \\ &= \int_0^\pi ka(\sin t)a \, dt = -ka^2 \cos t \Big|_0^\pi = 2ka^2. \quad \blacksquare \end{aligned}$$

Two other types of line integrals may be obtained by using Δx_i and Δy_i in place of Δs_i in (18.9). They are called the **line integrals of f along C with respect to x and y** , respectively. Thus, by definition,

$$\begin{aligned} \int_C f(x, y) dx &= \lim_{\|\Delta\| \rightarrow 0} \sum_i f(u_i, v_i) \Delta x_i \\ (18.11) \quad \int_C f(x, y) dy &= \lim_{\|\Delta\| \rightarrow 0} \sum_i f(u_i, v_i) \Delta y_i. \end{aligned}$$

If C is given parametrically by $x = g(t)$, $y = h(t)$, where $a \leq t \leq b$, then the integrals in (18.11) may be evaluated by substituting $g(t)$ and $h(t)$ for x and y , respectively, letting $dx = g'(t) dt$, $dy = h'(t) dt$, and using a and b for the limits of integration.

Example 3 Evaluate $\int_C xy^2 dx$ and $\int_C xy^2 dy$ if C is the part of the parabola $y = x^2$ from $A(0, 0)$ to $B(2, 4)$.

Solution Parametric equations for C are $x = t$, $y = t^2$, where $0 \leq t \leq 2$. Hence $dx = dt$, $dy = 2t dt$, and

$$\begin{aligned}\int_C xy^2 dx &= \int_0^2 t^5 dt = \left[\frac{1}{6} t^6 \right]_0^2 = \frac{32}{3} \\ \int_C xy^2 dy &= \int_0^2 t^5 (2t) dt = \left[\frac{2}{7} t^7 \right]_0^2 = \frac{256}{7}. \quad \blacksquare\end{aligned}$$

If C is the graph of a rectangular equation $y = g(x)$ where $a \leq x \leq b$, then parametric equations for C can be found by letting

$$x = t, \quad y = g(t) \quad \text{where} \quad a \leq t \leq b.$$

The line integrals in (18.11) may then be evaluated as follows:

$$\begin{aligned}\int_C f(x, y) dx &= \int_a^b f(t, g(t)) dt = \int_a^b f(x, g(x)) dx \\ \int_C f(x, y) dy &= \int_a^b f(t, g(t)) g'(t) dt = \int_a^b f(x, g(x)) g'(x) dx.\end{aligned}$$

This shows that for curves given in the rectangular form $y = g(x)$ where $a \leq x \leq b$, we may bypass the parametric equations by substituting $y = g(x)$, $dy = g'(x) dx$, and then using a and b for the limits of integration. To illustrate, in Example 3, where $y = x^2$, we could have written

$$\begin{aligned}\int_C xy^2 dx &= \int_0^2 x(x^4) dx = \int_0^2 x^5 dx = \frac{32}{3} \\ \int_C xy^2 dy &= \int_0^2 x(x^4) 2x dx = \int_0^2 2x^6 dx = \frac{256}{7}.\end{aligned}$$

In applications, line integrals often occur in the combination

$$\int_C M(x, y) dx + \int_C N(x, y) dy$$

where M and N are continuous functions on a domain D containing C . This sum is usually abbreviated by writing it in the form

$$\int_C M(x, y) dx + N(x, y) dy.$$

If C is given parametrically by $x = g(t)$, $y = h(t)$, where $a \leq t \leq b$, then the integral may be evaluated by substituting for x , y , dx , and dy as was done previously.

Example 4 Evaluate $\int_C xy \, dx + x^2 \, dy$ if:

- (a) C consists of line segments from $(2, 1)$ to $(4, 1)$ and from $(4, 1)$ to $(4, 5)$.
- (b) C is the line segment from $(2, 1)$ to $(4, 5)$.
- (c) parametric equations for C are $x = 3t - 1$, $y = 3t^2 - 2t$; $1 \leq t \leq \frac{5}{3}$.

Solution The curves for parts (a)–(c) are sketched in Figure 18.5.

- (a) If C is subdivided into two parts C_1 and C_2 as shown in (i) of Figure 18.5, then parametric equations for these curves are

$$C_1: x = t, y = 1; \quad 2 \leq t \leq 4$$

$$C_2: x = 4, y = t; \quad 1 \leq t \leq 5.$$

The line integral along C may be expressed as a sum of two line integrals, the first along C_1 and the second along C_2 . On C_1 we have $dy = 0$, $dx = dt$, and hence

$$\int_{C_1} xy \, dx + x^2 \, dy = \int_2^4 t(1) \, dt + 0 = \frac{1}{2} t^2 \Big|_2^4 = 6.$$

On C_2 we have $dx = 0$, $dy = dt$, and therefore

$$\int_{C_2} xy \, dx + x^2 \, dy = \int_1^5 0 + 16 \, dt = 16t \Big|_1^5 = 64.$$

Consequently, the line integral along C equals $6 + 64$ or 70 .

- (b) The graph of C is sketched in (ii) of Figure 18.5. A rectangular equation for C is $y = 2x - 3$, where $2 \leq x \leq 4$. In this case $dy = 2 \, dx$ and we may write

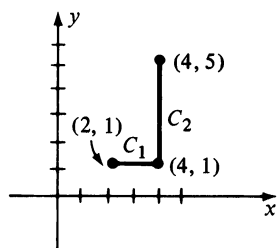
$$\begin{aligned} \int_C xy \, dx + x^2 \, dy &= \int_2^4 x(2x - 3) \, dx + x^2 2 \, dx \\ &= \int_2^4 (4x^2 - 3x) \, dx = \frac{170}{3} = 56\frac{2}{3}. \end{aligned}$$

- (c) The graph of C is part of a parabola (see (iii) of Figure 18.5). In this case $dx = 3 \, dt$, $dy = (6t - 2) \, dt$, and the given integral equals

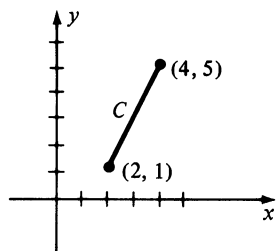
$$\int_1^{5/3} (3t - 1)(3t^2 - 2t)3 \, dt + (3t - 1)^2(6t - 2) \, dt.$$

It is left to the reader to show that the value is 58. Another method of solution is to use the rectangular equation $y = \frac{1}{3}(x^2 - 1)$ for the parabola, where $2 \leq x \leq 4$. In this event we obtain the following integral:

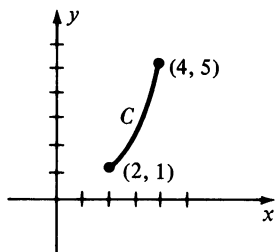
$$\int_C xy \, dx + x^2 \, dy = \int_2^4 x \frac{1}{3} (x^2 - 1) \, dx + x^2 \left(\frac{2}{3} x \right) \, dx = 58. \quad \blacksquare$$



(i)



(ii)



(iii)

FIGURE 18.5

In the preceding example we obtained three different values for the line integral along three different paths from $(2, 1)$ to $(4, 5)$. In Section 18.3 we shall consider line integrals that have the same value along every curve joining two points A and B . For such integrals we use the phrase *independent of path*.

If a smooth curve C in three dimensions is given parametrically by

$$x = g(t), \quad y = h(t), \quad z = k(t)$$

where $a \leq t \leq b$, then line integrals of a function f of *three* variables are defined in a manner similar to those for two variables. In this case, instead of using (x_i, y_i) and (u_i, v_i) as the coordinates of P_i and Q_i on C (see Figure 18.4), we use (x_i, y_i, z_i) and (u_i, v_i, w_i) , respectively. Instead of (18.9) we now have

$$\int_C f(x, y, z) \, ds = \lim_{\|\Delta\| \rightarrow 0} \sum_i f(u_i, v_i, w_i) \Delta s_i.$$

This integral may be evaluated by using the formula

$$\int_a^b f(g(t), h(t), k(t)) \sqrt{(g'(t))^2 + (h'(t))^2 + (k'(t))^2} \, dt.$$

In addition to line integrals $\int_C f(x, y, z) \, dx$ and $\int_C f(x, y, z) \, dy$, there is a line integral with respect to z in three dimensions, given by

$$\int_C f(x, y, z) \, dz = \lim_{\|\Delta\| \rightarrow 0} \sum_i f(u_i, v_i, w_i) \Delta z_i$$

where $\Delta z_i = z_i - z_{i-1}$. As in two dimensions, these line integrals often occur in the form

$$\int_C M(x, y, z) \, dx + N(x, y, z) \, dy + P(x, y, z) \, dz$$

where M , N , and P are functions of x , y , and z that are continuous throughout a region containing C . If C is given parametrically, then this line integral may be evaluated by substituting for x , y , and z in the same manner as in the two-variable case.

Example 5 Evaluate $\int_C yz \, dx + xz \, dy + xy \, dz$ where C is the twisted cubic

$$x = t, \quad y = t^2, \quad z = t^3; \quad 0 \leq t \leq 2.$$

Solution Substituting for x , y , and z and using $dx = dt$, $dy = 2t \, dt$, $dz = 3t^2 \, dt$, we obtain

$$\int_0^2 t^5 \, dt + 2t^5 \, dt + 3t^5 \, dt = \int_0^2 6t^5 \, dt = t^6 \Big|_0^2 = 64. \quad \blacksquare$$

One of the most important physical applications of line integrals has to do with force fields. Let us suppose that the force acting at the point (x, y, z) is

$$\mathbf{F}(x, y, z) = M(x, y, z)\mathbf{i} + N(x, y, z)\mathbf{j} + P(x, y, z)\mathbf{k}$$

where the functions M , N , and P are continuous. Our objective is to define the work done as the point of application of $\mathbf{F}(x, y, z)$ moves along a smooth curve C , where C is given parametrically by $x = g(t)$, $y = h(t)$, $z = k(t)$; $a \leq t \leq b$. We shall assume that the motion is in the direction determined by increasing values of t .

Let us subdivide C by choosing points $P_0, P_1, P_2, \dots, P_n$ and then, for each i , choose a point Q_i in the subarc $\widehat{P_{i-1}P_i}$, as illustrated in Figure 18.6. Let the coordinates of P_i and Q_i be (x_i, y_i, z_i) and (u_i, v_i, w_i) , respectively. If the norm $\|\Delta\|$ is small, then the work done by $\mathbf{F}(x, y, z)$ along the arc $\widehat{P_{i-1}P_i}$ can be approximated by the work Δw_i done by the constant force $\mathbf{F}(u_i, v_i, w_i)$ as its point of application moves along $\widehat{P_{i-1}P_i}$.

If $\Delta x_i = x_i - x_{i-1}$, $\Delta y_i = y_i - y_{i-1}$, and $\Delta z_i = z_i - z_{i-1}$, then $\widehat{P_{i-1}P_i}$ corresponds to the vector $\Delta x_i \mathbf{i} + \Delta y_i \mathbf{j} + \Delta z_i \mathbf{k}$ of V_3 . By (14.29) the work Δw_i done by $\mathbf{F}(u_i, v_i, w_i)$ along $\widehat{P_{i-1}P_i}$ is

$$\begin{aligned}\Delta w_i &= \mathbf{F}(u_i, v_i, w_i) \cdot (\Delta x_i \mathbf{i} + \Delta y_i \mathbf{j} + \Delta z_i \mathbf{k}) \\ &= M(u_i, v_i, w_i) \Delta x_i + N(u_i, v_i, w_i) \Delta y_i + P(u_i, v_i, w_i) \Delta z_i.\end{aligned}$$

The work W done by $\mathbf{F}$ along C is, by definition,

$$W = \lim_{\|\Delta\| \rightarrow 0} \sum_i \Delta w_i$$

that is,

$$(18.12) \quad W = \int_C M(x, y, z) dx + N(x, y, z) dy + P(x, y, z) dz.$$

In applications, the preceding integral is usually expressed in vector form. If we let

$$\mathbf{r}(t) = x\mathbf{i} + y\mathbf{j} + z\mathbf{k}$$

where $x = g(t)$, $y = h(t)$, and $z = k(t)$, then $\mathbf{r}(t)$ is the position vector of the point $Q(x, y, z)$ on C . If s denotes arc length measured along C , then as in Section 15.4, a unit tangent vector $\mathbf{T}(s)$ to C at Q is

$$\mathbf{T}(s) = \frac{d}{ds} \mathbf{r}(t) = \frac{dx}{ds} \mathbf{i} + \frac{dy}{ds} \mathbf{j} + \frac{dz}{ds} \mathbf{k}.$$

The vectors $\mathbf{r}(t)$, $\mathbf{F}(x, y, z)$, and $\mathbf{T}(s)$ are illustrated in Figure 18.7.

The tangential component of $\mathbf{F}$ at Q (see Figure 18.7), is

$$\mathbf{F}(x, y, z) \cdot \mathbf{T}(s) = M(x, y, z) \frac{dx}{ds} + N(x, y, z) \frac{dy}{ds} + P(x, y, z) \frac{dz}{ds}.$$

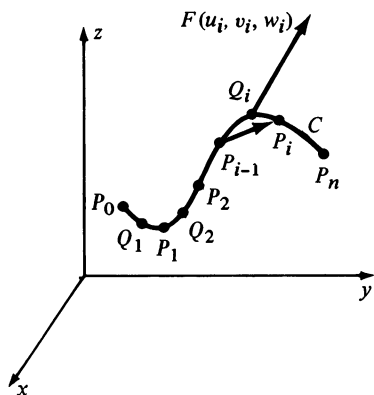


FIGURE 18.6

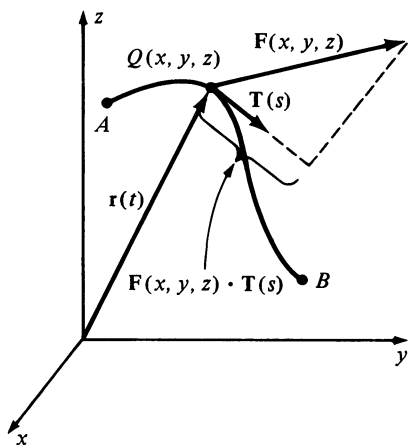


FIGURE 18.7

Note that this component is positive if the angle between $\mathbf{F}(x, y, z)$ and $\mathbf{T}(s)$ is acute, whereas it is negative if this angle is obtuse. Formula (18.12) may now be rewritten

$$W = \int_C \mathbf{F}(x, y, z) \cdot \mathbf{T}(s) \, ds$$

that is, *the work done as the point of application of $\mathbf{F}(x, y, z)$ moves along C equals the line integral of the tangential component of $\mathbf{F}$ along C* . To simplify the notation we shall denote $\mathbf{F}(x, y, z)$ by $\mathbf{F}$, $\mathbf{T}(s)$ by $\mathbf{T}$, and formally let

$$d\mathbf{r} = dx\mathbf{i} + dy\mathbf{j} + dz\mathbf{k} = \mathbf{T} \, ds.$$

With this understanding, our discussion may be summarized as follows.

Definition (18.13)

Let C be a smooth space curve, and let $\mathbf{T}$ be a unit tangent vector to C at (x, y, z) . If $\mathbf{F}$ is the force acting at (x, y, z) , then the **work W done by $\mathbf{F}$ along C** is

$$W = \int_C \mathbf{F} \cdot \mathbf{T} \, ds = \int_C \mathbf{F} \cdot d\mathbf{r}$$

where $\mathbf{r} = x\mathbf{i} + y\mathbf{j} + z\mathbf{k}$.

Intuitively, we may regard $\mathbf{F} \cdot d\mathbf{r}$ in Definition (18.13) as representing the work done as the point of application of $\mathbf{F}$ moves along the tangent vector $d\mathbf{r}$ to C . The integral sign represents the limit of a sum of such elements of work.

Example 6 If an inverse force field $\mathbf{F}$ is given by

$$\mathbf{F}(x, y, z) = \frac{k}{|\mathbf{r}|^3} \mathbf{r}$$

find the work done by $\mathbf{F}$ as its point of application moves along the x -axis from $A(1, 0, 0)$ to $B(2, 0, 0)$.

Solution Let C denote the line segment from A to B . Parametric equations for C are $x = t$, $y = 0$, $z = 0$, where $1 \leq t \leq 2$. Since

$$\mathbf{F}(x, y, z) = \frac{k}{(x^2 + y^2 + z^2)^{3/2}} (x\mathbf{i} + y\mathbf{j} + z\mathbf{k}),$$

it follows from Definition (18.13) that

$$W = \int_C \mathbf{F} \cdot d\mathbf{r} = \int_C \frac{k}{(x^2 + y^2 + z^2)^{3/2}} (x \, dx + y \, dy + z \, dz).$$

Substituting for x , y , and z from the parametric equations for C gives us

$$W = \int_1^2 \frac{k}{(t^2)^{3/2}} t \, dt = \int_1^2 \frac{k}{t^2} \, dt = -\frac{k}{t} \Big|_1^2 = \frac{k}{2}.$$

The units for W depend on those for distance and $|\mathbf{F}(x, y, z)|$. ■

If we restrict our discussion to two dimensions, then a force field may be expressed in the form

$$\mathbf{F}(x, y) = M(x, y)\mathbf{i} + N(x, y)\mathbf{j}.$$

If C is a finite piecewise smooth plane curve, then the work done as the point of application of $\mathbf{F}(x, y)$ moves along C is

$$W = \int_C M(x, y) \, dx + N(x, y) \, dy = \int_C \mathbf{F} \cdot d\mathbf{r}.$$

Example 7 Let C be the part of the parabola $y = x^2$ between $A(0, 0)$ and $B(3, 9)$. If $\mathbf{F}(x, y) = -y\mathbf{i} + x\mathbf{j}$, find the work done by $\mathbf{F}$ along C from A to B .

Solution The vector field $\mathbf{F}$ was discussed in Example 1 of Section 18.1. Some typical vectors are sketched in (i) of Figure 18.2. Parametric equations for C (from A to B) are $x = t$, $y = t^2$, where $0 \leq t \leq 3$. Hence

$$\begin{aligned} W &= \int_C \mathbf{F} \cdot d\mathbf{r} = \int_C -y \, dx + x \, dy \\ &= \int_0^3 -t^2 \, dt + t(2t) \, dt = \int_0^3 t^2 \, dt = \frac{t^3}{3} \Big|_0^3 = 9. \end{aligned}$$

If, for example, $|\mathbf{F}|$ is measured in pounds and s is in feet, then $W = 9$ ft-lb. ■

18.2 Exercises

In Exercises 1 and 2 evaluate the line integrals $\int_C f(x, y) \, ds$, $\int_C f(x, y) \, dx$, and $\int_C f(x, y) \, dy$ if C is defined parametrically as indicated.

- 1 $f(x, y) = x^3 + y$; $x = 3t$, $y = t^3$; $0 \leq t \leq 1$
- 2 $f(x, y) = xy^{2/5}$; $x = \frac{1}{2}t$, $y = t^{5/2}$; $0 \leq t \leq 1$

In Exercises 3 and 4 evaluate the line integral along the given curve C and then verify directly that reversing the direction of integration changes the sign of the integral.

- 3 $\int_C 6x^2y \, dx + xy \, dy$ where C is the graph of $y = x^3 + 1$ from $(-1, 0)$ to $(1, 2)$.
- 4 $\int_C y \, dx + (x + y) \, dy$ where C is the graph of $y = x^2 + 2x$ from $(0, 0)$ to $(2, 8)$.

- 5 Evaluate $\int_C (x - y) \, dx + x \, dy$ where C is the graph of $y^2 = x$ from $(4, -2)$ to $(4, 2)$.

- 6 Evaluate $\int_C xy \, dx + x^2y^3 \, dy$ where C is the graph of $x = y^3$ from $(0, 0)$ to $(1, 1)$.

- 7 Evaluate $\int_C xy \, dx + (x + y) \, dy$ for each of the following curves C from $(0, 0)$ to $(1, 3)$.

- (a) C consists of line segments from $(0, 0)$ to $(1, 0)$ and from $(1, 0)$ to $(1, 3)$.
- (b) C consists of line segments from $(0, 0)$ to $(0, 3)$ and from $(0, 3)$ to $(1, 3)$.
- (c) C is the line segment from $(0, 0)$ to $(1, 3)$.
- (d) C is the part of the parabola $y = 3x^2$ from $(0, 0)$ to $(1, 3)$.

- 8 Evaluate $\int_C (x^2 + y^2) dx + 2x dy$ for each of the following curves C from $(1, 2)$ to $(-2, 8)$.
- C consists of line segments from $(1, 2)$ to $(1, 8)$ and from $(1, 8)$ to $(-2, 8)$.
 - C consists of line segments from $(1, 2)$ to $(-2, 2)$ and from $(-2, 2)$ to $(-2, 8)$.
 - C is the line segment from $(1, 2)$ to $(-2, 8)$.
 - C is the graph of $y = 2x^2$ from $(1, 2)$ to $(-2, 8)$.
- 9 Evaluate $\int_C xz dx + (y + z) dy + x dz$ if C is given by $x = e^t$, $y = e^{-t}$, $z = e^{2t}$; $0 \leq t \leq 1$.
- 10 Evaluate $\int_C y dx + z dy + x dz$ if C is given by $x = \sin t$, $y = 2 \sin t$, $z = \sin^2 t$; $0 \leq t \leq \pi/2$.
- 11 Evaluate

$$\int_C (x + y + z) dx + (x - 2y + 3z) dy + (2x + y - z) dz$$

for each of the following curves C from $(0, 0, 0)$ to $(2, 3, 4)$.

- C consists of three line segments, the first parallel to the x -axis, the second parallel to the y -axis, and the third parallel to the z -axis.
 - C consists of three line segments, the first parallel to the z -axis, the second parallel to the x -axis, and the third parallel to the y -axis.
 - C is a line segment.
- 12 Evaluate $\int_C (x - y) dx + (y - z) dy + x dz$ where the curve C from $(1, -2, 3)$ to $(-4, 5, 2)$ is of the type described in parts (a)–(c) of Exercise 11.
- 13 Evaluate $\int_C xyz ds$, if C is the line segment from $(0, 0, 0)$ to $(1, 2, 3)$.
- 14 Evaluate $\int_C (xy + z) ds$ if C is the helix $x = a \cos t$, $y = a \sin t$, $z = bt$, $0 \leq t \leq 2\pi$.
- 15 If a thin wire has the shape of a plane curve C , and if the (linear) density at (x, y) is $f(x, y)$, then the **moments with respect to the x - and y -axes** are defined by

$$M_x = \int_C yf(x, y) ds \quad \text{and} \quad M_y = \int_C xf(x, y) ds.$$

Use limits of sums to show that these are “natural” definitions for M_x and M_y . The center of mass $(\bar{x}, \bar{y})$ of the wire is defined by $\bar{x} = M_y/M$ and $\bar{y} = M_x/M$, where M is the mass of the wire. Find the center of mass of the wire in Example 2 of this section.

- 16 Refer to Exercise 15. A thin wire is situated in a coordinate plane such that its shape coincides with the part of the parabola $y = 4 - x^2$ between $(-2, 0)$ and $(2, 0)$. Find the

mass and center of mass if the density at the point (x, y) is directly proportional to its distance from the y -axis.

- 17 Extend the definitions of mass and center of mass of a wire to three dimensions (see Exercise 15).
- 18 A wire of constant density is bent into the shape of the helix $x = a \cos t$, $y = a \sin t$, $z = bt$, where $0 \leq t \leq 3\pi$. Find the mass and center of mass of the wire (see Exercise 17).
- 19 If the force at (x, y) is $\mathbf{F}(x, y) = xy^2\mathbf{i} + x^2y\mathbf{j}$, find the work done by $\mathbf{F}$ along the curves described in parts (a)–(d) of Exercise 7.
- 20 If the force at (x, y) is $\mathbf{F}(x, y) = (2x + y)\mathbf{i} + (x + 2y)\mathbf{j}$, find the work done by $\mathbf{F}$ along the curves described in parts (a)–(d) of Exercise 8.
- 21 The force acting at a point (x, y) in a coordinate plane is $\mathbf{F}(x, y) = (4/|\mathbf{r}|^3)\mathbf{r}$, where $\mathbf{r} = x\mathbf{i} + y\mathbf{j}$. Find the work done by $\mathbf{F}$ along the upper half of the circle $x^2 + y^2 = a^2$ from $(-a, 0)$ to $(a, 0)$.
- 22 The force at a point (x, y) in a coordinate plane is given by $\mathbf{F}(x, y) = (x^2 + y^2)\mathbf{i} + xy\mathbf{j}$. Find the work done by $\mathbf{F}(x, y)$ along the graph of $y = x^3$ from $(0, 0)$ to $(2, 8)$.
- 23 The force at a point (x, y, z) in three dimensions is given by $\mathbf{F}(x, y, z) = y\mathbf{i} + z\mathbf{j} + x\mathbf{k}$. Find the work done by $\mathbf{F}(x, y, z)$ along the curve $x = t$, $y = t^2$, $z = t^3$ from $(0, 0, 0)$ to $(2, 4, 8)$.
- 24 Solve Exercise 23 if $\mathbf{F}(x, y, z) = e^x\mathbf{i} + e^y\mathbf{j} + e^z\mathbf{k}$.
- 25 If a thin wire of variable density is represented by a curve C in the xy -plane, define the moments of inertia I_x and I_y with respect to the x - and y -axes, respectively. Find I_x and I_y for the wire in Example 2.
- 26 Find the moments of inertia I_x and I_y for the wire in Exercise 16.
- 27 If a thin wire of variable density is represented by a curve in three dimensions, define the moments of inertia with respect to the x -, y -, and z -axes.
- 28 Given the wire in Exercise 18, find the moment of inertia with respect to the z -axis (see also Exercise 27).
- 29 If an object moves through a force field $\mathbf{F}$ such that at each point (x, y, z) its velocity vector is orthogonal to $\mathbf{F}(x, y, z)$, show that the work done by $\mathbf{F}$ on the object is 0.
- 30 If a constant force $\mathbf{c}$ acts on a moving object as it travels once around a circle, show that the work done by $\mathbf{c}$ on the object is 0.
- 31 Discuss the significance of the case where the work integral $\int_C \mathbf{F} \cdot d\mathbf{r}$ in Definition (18.13) is negative.
- 32 Show that (14.29) is a special case of (18.13).

18.3 Independence of Path

A piecewise smooth curve that connects two points A and B is sometimes called a **path** from A to B . We shall now obtain conditions under which a line integral is **independent of path** in a region, in the sense that if A and B are arbitrary points, then the same value is obtained for *every* path from A to B . The results will be established for line integrals in two dimensions. Proofs for the three-dimensional case are similar and will be omitted.

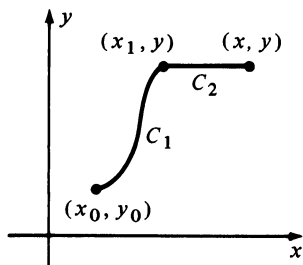
If the line integral $\int_C f(x, y) ds$ is independent of path we sometimes denote it by $\int_A^B f(x, y) ds$, since the value of the integral depends only on the end-points A and B of the curve C . A similar notation is used for $\int_C f(x, y) dx$, $\int_C f(x, y) dy$, and for line integrals in three dimensions.

It will be assumed throughout this section that all regions in the plane are **connected**. This means that any two points in a region can be joined by a piecewise smooth curve that lies in the region. We shall also assume that regions are *open*, that is, for every point A in a region D there exists a circle with center A that lies completely in D .

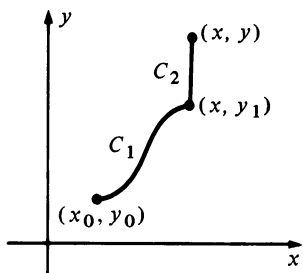
The next theorem gives us the fundamental result that if a vector function $\mathbf{F}$ is continuous on D , then the line integral $\int_C \mathbf{F} \cdot d\mathbf{r}$ is *independent of path* if and only if $\mathbf{F}$ is *conservative*.

Theorem (18.14)

If $\mathbf{F}(x, y) = M(x, y)\mathbf{i} + N(x, y)\mathbf{j}$ is continuous on an open connected region, then the line integral $\int_C \mathbf{F} \cdot d\mathbf{r}$ is independent of path if and only if $\mathbf{F}(x, y) = \nabla f(x, y)$ for some function f .



(i)



(ii)

FIGURE 18.8

Proof Suppose the integral is independent of path. If (x_0, y_0) is a fixed point in D , let f be defined by

$$f(x, y) = \int_{(x_0, y_0)}^{(x, y)} \mathbf{F} \cdot d\mathbf{r}$$

where (x, y) is any point in D . Since the integral is independent of path, f depends only on x and y , and not on the path C from (x_0, y_0) to (x, y) . Choose a circle in D with center (x, y) and let (x_1, y) be a point within the circle such that $x_1 \neq x$. Let C_1 be any path from (x_0, y_0) to (x_1, y) and let C_2 be the horizontal segment from (x_1, y) to (x, y) , as illustrated in (i) of Figure 18.8. We may, therefore, write

$$f(x, y) = \int_{C_1} \mathbf{F} \cdot d\mathbf{r} + \int_{C_2} \mathbf{F} \cdot d\mathbf{r} = \int_{(x_0, y_0)}^{(x_1, y)} \mathbf{F} \cdot d\mathbf{r} + \int_{(x_1, y)}^{(x, y)} \mathbf{F} \cdot d\mathbf{r}.$$

Since the first integral does not depend on x ,

$$\frac{\partial}{\partial x} f(x, y) = 0 + \frac{\partial}{\partial x} \int_{(x_1, y)}^{(x, y)} \mathbf{F} \cdot d\mathbf{r}.$$

Writing $\mathbf{F} \cdot d\mathbf{r} = M(x, y) dx + N(x, y) dy$ and using the fact that $dy = 0$ on C_2 gives us

$$\frac{\partial}{\partial x} f(x, y) = \frac{\partial}{\partial x} \int_{(x_1, y)}^{(x, y)} M(x, y) dx.$$

Since y is fixed in this partial differentiation we may regard the integrand as a function of one variable x . Applying, (5.25),

$$\frac{\partial}{\partial x} f(x, y) = M(x, y).$$

Similarly, if we choose the path shown in (ii) of Figure 18.8 and differentiate with respect to y , we obtain

$$\frac{\partial}{\partial y} f(x, y) = N(x, y).$$

This proves that $\nabla f(x, y) = M(x, y)\mathbf{i} + N(x, y)\mathbf{j} = \mathbf{F}(x, y)$.

Conversely, if there exists a function f such that $\mathbf{F}(x, y) = \nabla f(x, y)$, then

$$M(x, y)\mathbf{i} + N(x, y)\mathbf{j} = f_x(x, y)\mathbf{i} + f_y(x, y)\mathbf{j}$$

and hence

$$M(x, y) = f_x(x, y) \quad \text{and} \quad N(x, y) = f_y(x, y).$$

Consequently, if $A(x_1, y_1)$ and $B(x_2, y_2)$ are points in D , then

$$\int_C \mathbf{F} \cdot d\mathbf{r} = \int_C f_x(x, y) dx + f_y(x, y) dy$$

where C is any path from A to B . If C is smooth and is given parametrically by $x = g(t)$, $y = h(t)$ where $t_1 \leq t \leq t_2$, then substituting in the integrand,

$$\int_C \mathbf{F} \cdot d\mathbf{r} = \int_{t_1}^{t_2} [f_x(g(t), h(t))g'(t) + f_y(g(t), h(t))h'(t)] dt.$$

Applying the Chain Rule (16.19) and the Fundamental Theorem of Calculus,

$$\begin{aligned} \int_C \mathbf{F} \cdot d\mathbf{r} &= \int_{t_1}^{t_2} \frac{d}{dt} [f(g(t), h(t))] dt \\ &= f(g(t_2), h(t_2)) - f(g(t_1), h(t_1)) \\ &= f(x_2, y_2) - f(x_1, y_1). \end{aligned}$$

Thus the line integral depends only on the coordinates of A and B , not on the path C , which is what we wished to prove. The proof can be extended to piecewise smooth curves by subdividing C into a finite number of smooth curves. $\square$

The proof of the converse of Theorem (18.14) exhibits a technique for evaluating line integrals that are independent of path. We may state this result as follows.

Theorem (18.15)

Let $\mathbf{F}(x, y) = M(x, y)\mathbf{i} + N(x, y)\mathbf{j}$ be continuous on an open connected region D , and let C be a piecewise smooth curve in D with endpoints $A(x_1, y_1)$ and $B(x_2, y_2)$. If $\mathbf{F}(x, y) = \nabla f(x, y)$, then

$$\begin{aligned} \int_C M(x, y) dx + N(x, y) dy &= \int_{(x_1, y_1)}^{(x_2, y_2)} \mathbf{F} \cdot d\mathbf{r} \\ &= f(x_2, y_2) - f(x_1, y_1) = f(x, y) \Big|_{(x_1, y_1)}^{(x_2, y_2)} \end{aligned}$$

This evaluation theorem for line integrals is analogous to the Fundamental Theorem of Calculus.

In terms of conservative force fields, Theorem (18.15) implies that the work done along any path C from A to B equals the difference in the potentials between A and B . It follows that if C is a closed curve, that is, if $A = B$, then the work done in traversing C is 0. It can be shown that a converse of this result is also true, namely, if $\int_C \mathbf{F} \cdot d\mathbf{r}$ is 0 for every simple closed curve C , then the line integral is independent of path and hence the field is conservative. These facts are important in applications, since many of the vector fields that occur in nature (such as inverse square fields) are conservative. In physical terms, if a unit particle moves completely around a closed curve in a conservative force field, and if there is no loss of energy due to friction, then the work done is 0. More will be said about conservative fields at the end of this section.

Before considering an example, let us note another important byproduct of Theorem (18.14). If the integral $\int_C M(x, y) dx + N(x, y) dy$ is independent of path, then by Theorem (18.14), there is a function f such that

$$M = \frac{\partial f}{\partial x} \quad \text{and} \quad N = \frac{\partial f}{\partial y}.$$

Consequently,

$$\frac{\partial M}{\partial y} = \frac{\partial^2 f}{\partial y \partial x} \quad \text{and} \quad \frac{\partial N}{\partial x} = \frac{\partial^2 f}{\partial x \partial y}.$$

If M and N have continuous first partial derivatives, then f has continuous *second* partial derivatives, and hence the order of differentiation is immaterial, that is

$$\frac{\partial M}{\partial y} = \frac{\partial N}{\partial x}.$$

The converse of this result is false, unless we place additional restrictions on the domain D of $\mathbf{F}(x, y)$. In particular, if D is a **simply connected region**, in

the sense that every simple closed curve C in D encloses only points in D (that is, there are no “holes” in the region), then the condition $\partial M/\partial y = \partial N/\partial x$ implies that the line integral $\int_C M(x, y) dx + N(x, y) dy$ is independent of path (a proof may be found in books on advanced calculus). Our discussion can be summarized as follows.

Theorem (18.16)

If $M(x, y)$ and $N(x, y)$ have continuous first partial derivatives on a simply connected region D , then

$$\int_C M(x, y) dx + N(x, y) dy$$

is independent of path in D if and only if

$$\frac{\partial M}{\partial y} = \frac{\partial N}{\partial x}.$$

Example 1 If $\mathbf{F}(x, y) = (2x + y^3)\mathbf{i} + (3xy^2 + 4)\mathbf{j}$, show that $\int_C \mathbf{F} \cdot d\mathbf{r}$ is independent of path and evaluate $\int_{(0,1)}^{(2,3)} \mathbf{F} \cdot d\mathbf{r}$.

Solution The vector function $\mathbf{F}$ has continuous first partial derivatives for all (x, y) , and hence Theorem (18.16) is applicable. Letting $M = 2x + y^3$ and $N = 3xy^2 + 4$, we see that

$$\frac{\partial M}{\partial y} = 3y^2 = \frac{\partial N}{\partial x}.$$

Thus the line integral is independent of path. It follows from Theorem (18.14) that there exists a (potential) function f such that

$$f_x(x, y) = 2x + y^3 \quad \text{and} \quad f_y(x, y) = 3xy^2 + 4.$$

If we (partially) integrate $f_x(x, y)$ with respect to x we obtain

$$f(x, y) = x^2 + xy^3 + k(y)$$

where k is a function of y alone. Differentiating with respect to y and comparing with $f_y(x, y) = 3xy^2 + 4$ gives us

$$f_y(x, y) = 3xy^2 + k'(y) = 3xy^2 + 4.$$

Consequently, $k'(y) = 4$, or $k(y) = 4y + c$ for some constant c . Thus

$$f(x, y) = x^2 + xy^3 + 4y + c$$

defines a function of the desired type. Applying Theorem (18.15),

$$\begin{aligned} \int_{(0,1)}^{(2,3)} (2x + y^3) dx + (3xy^2 + 4) dy &= x^2 + xy^3 + 4y \Big|_{(0,1)}^{(2,3)} \\ &= (4 + 54 + 12) - 4 = 66. \end{aligned}$$

where we have dropped the constant c , since *any* potential function f may be used to evaluate the integral. ■

Example 2 Determine whether $\int_C x^2y dx + 3xy^2 dy$ is independent of path.

Solution If we let $M = x^2y$ and $N = 3xy^2$, then

$$\frac{\partial M}{\partial y} = x^2 \quad \text{and} \quad \frac{\partial N}{\partial x} = 3y^2.$$

Since $\partial M/\partial y \neq \partial N/\partial x$, the integral is not independent of path by Theorem (18.16). ■

In Example 1 we determined a potential function f for a two-dimensional conservative vector field. The solution of the next example illustrates a technique that can be used in three dimensions.

Example 3 If $\mathbf{F}(x, y, z) = y^2 \cos x \mathbf{i} + (2y \sin x + e^{2z}) \mathbf{j} + 2ye^{2z} \mathbf{k}$, prove that $\int_C \mathbf{F} \cdot d\mathbf{r}$ is independent of path and find a potential function f for $\mathbf{F}$. Find the work done by $\mathbf{F}$ along any curve C from $(0, 1, \frac{1}{2})$ to $(\pi/2, 3, 2)$.

Solution The integral is independent of path if there exists a differentiable function f of x , y , and z such that $\nabla f(x, y, z) = \mathbf{F}(x, y, z)$; that is, if

$$\begin{aligned} (a) \quad f_x(x, y, z) &= y^2 \cos x \\ f_y(x, y, z) &= 2y \sin x + e^{2z} \\ f_z(x, y, z) &= 2ye^{2z}. \end{aligned}$$

If we (partially) integrate $f_x(x, y, z)$ with respect to x we see that

$$(b) \quad f(x, y, z) = y^2 \sin x + g(y, z)$$

for some function g of y and z . Differentiating with respect to y and comparing with the equation for f_y in (a) gives us

$$f_y(x, y, z) = 2y \sin x + g_y(y, z) = 2y \sin x + e^{2z}$$

and hence

$$g_y(y, z) = e^{2z}.$$

Integrating with respect to y , we obtain

$$g(y, z) = ye^{2z} + k(z)$$

where k is a function of z alone. Consequently, from equation (b),

$$f(x, y, z) = y^2 \sin x + ye^{2z} + k(z).$$

Differentiating with respect to z , and using the equation for f_z in (a),

$$f_z(x, y, z) = 2ye^{2z} + k'(z) = 2ye^{2z}.$$

It follows that $k'(z) = 0$ or $k(z) = c$ for some constant c . Thus

$$f(x, y, z) = y^2 \sin x + ye^{2z} + c$$

defines a potential function for $\mathbf{F}$.

Using the extension of Theorem (18.15) to three dimensions, the work done by $\mathbf{F}$ along any curve C from $(0, 1, \frac{1}{2})$ to $(\pi/2, 3, 2)$ is

$$\begin{aligned} \int_C \mathbf{F} \cdot d\mathbf{r} &= y^2 \sin x + ye^{2z} \Big|_{(0, 1, 1/2)}^{(\pi/2, 3, 2)} \\ &= (9 + 3e^4) - (0 + e) = 9 + 3e^4 - e. \end{aligned} \quad \blacksquare$$

Let $\mathbf{F}(x, y, z)$ be a conservative vector field with potential function f . In physics, the potential energy $p(x, y, z)$ of a particle at the point (x, y, z) is defined by $p(x, y, z) = -f(x, y, z)$ and, therefore,

$$\mathbf{F}(x, y, z) = -\nabla p(x, y, z).$$

If A and B are two points, then as in Theorem (18.15), the work done by $\mathbf{F}$ along a smooth curve C with endpoints A and B is

$$W = \int_A^B \mathbf{F} \cdot d\mathbf{r} = -p(x, y, z) \Big|_A^B = p(A) - p(B)$$

where $p(A)$ and $p(B)$ denote the potential energy at A and B , respectively. Thus, the work W equals the difference in the potential energies at A and B . In particular, if B is a point at which the potential is 0, then $W = p(A)$. This is in agreement with the classical physical description of potential energy as the type of energy that a body has by virtue of its position.

Now suppose that the particle moves from A to B along the curve C such that its position at time t is given by $x = g(t)$, $y = h(t)$, $z = k(t)$, where $a \leq t \leq b$. If $\mathbf{r} = x\mathbf{i} + y\mathbf{j} + z\mathbf{k}$ is the position vector of the particle, then the velocity and acceleration at time t are

$$\mathbf{v} = \frac{d\mathbf{r}}{dt} \quad \text{and} \quad \mathbf{a} = \frac{d\mathbf{v}}{dt}$$

where we have used the abbreviations $\mathbf{v}$ and $\mathbf{a}$ for $\mathbf{v}(t)$ and $\mathbf{a}(t)$, respectively. The speed of the particle is $v(t) = v = |\mathbf{v}|$. In this event, the work done by $\mathbf{F}(x, y, z)$ along C is

$$W = \int_A^B \mathbf{F} \cdot d\mathbf{r} = \int_a^b \left(\mathbf{F} \cdot \frac{d\mathbf{r}}{dt} \right) dt = \int_a^b (\mathbf{F} \cdot \mathbf{v}) dt.$$

By Newton's Second Law of Motion, $\mathbf{F} = m\mathbf{a} = m(d\mathbf{v}/dt)$ and hence

$$\begin{aligned} W &= \int_a^b m \left(\frac{d\mathbf{v}}{dt} \cdot \mathbf{v} \right) dt = \int_a^b m \frac{1}{2} \frac{d}{dt} (\mathbf{v} \cdot \mathbf{v}) dt \\ &= \frac{1}{2} m \int_a^b \frac{d}{dt} (v^2) dt = \frac{1}{2} m v^2 \Big|_a^b \\ &= \frac{1}{2} m [v(b)]^2 - \frac{1}{2} m [v(a)]^2. \end{aligned}$$

Since the kinetic energy of a particle of mass m and velocity v is $\frac{1}{2}mv^2$, the last equality may be written

$$W = K(B) - K(A)$$

where $K(A)$ and $K(B)$ denote the kinetic energy of the particle at A and B , respectively. We have shown that

$$W = p(A) - p(B) = K(B) - K(A)$$

or

$$p(A) + K(A) = p(B) + K(B).$$

The last formula states that if a particle moves from one point to another in a conservative vector field, then the sum of the potential and kinetic energies remains constant; that is, the *total* energy does not change. This is known as the *Law of Conservation of Energy*, and is the reason for calling the vector field *conservative*.

18.3 Exercises

In Exercises 1–10 determine whether or not $\int_C \mathbf{F} \cdot d\mathbf{r}$ is independent of path. If it is, find a potential function f for $\mathbf{F}$.

1 $\mathbf{F}(x, y) = (3x^2y + 2)\mathbf{i} + (x^3 + 4y^3)\mathbf{j}$

2 $\mathbf{F}(x, y) = (6x^2 - 2xy^2)\mathbf{i} + (2x^2y + 5)\mathbf{j}$

3 $\mathbf{F}(x, y) = e^x\mathbf{i} + (3 - e^x \sin y)\mathbf{j}$

4 $\mathbf{F}(x, y) = (x^2 + y^2)^{-1/2}(-y\mathbf{i} + x\mathbf{j})$

5 $\mathbf{F}(x, y) = 4xy^3\mathbf{i} + 2xy^3\mathbf{j}$

6 $\mathbf{F}(x, y) = y^3 \cos x\mathbf{i} - 3y^2 \sin x\mathbf{j}$

7 $\mathbf{F}(x, y, z) = (y \sec^2 x - ze^x)\mathbf{i} + \tan x\mathbf{j} - e^x\mathbf{k}$

8 $\mathbf{F}(x, y, z) = (y + z)\mathbf{i} + (x + z)\mathbf{j} + (x + y)\mathbf{k}$

9 $\mathbf{F}(x, y, z) = 8xz\mathbf{i} + (1 - 6yz^3)\mathbf{j} + (4x^2 - 9y^2z^2)\mathbf{k}$

10 $\mathbf{F}(x, y, z) = e^{-z}\mathbf{i} + 2y\mathbf{j} + xe^{-z}\mathbf{k}$

In Exercises 11–14 prove that the line integral is independent of path and find its value.

11 $\int_{(-1, 2)}^{(3, 1)} (y^2 + 2xy) dx + (x^2 + 2xy) dy$

12 $\int_{(0, 0)}^{(1, \pi/2)} e^x \sin y dx + e^x \cos y dy$

13 $\int_{(1, 0, 2)}^{(-2, 1, 3)} (6xy^3 + 2z^2) dx + 9x^2y^2 dy + (4xz + 1) dz$

14 $\int_{(4, 0, 3)}^{(-1, 1, 2)} (yz + 1) dx + (xz + 1) dy + (xy + 1) dz$

15 Suppose a force $\mathbf{F}(x, y, z)$ is directed toward the origin with a magnitude that is inversely proportional to the distance from the origin. Prove that $\mathbf{F}$ is conservative and find a potential function for $\mathbf{F}$.

16 Suppose a force $\mathbf{F}(x, y, z)$ is directed away from the origin with a magnitude that is directly proportional to the distance from the origin. Prove that $\mathbf{F}$ is conservative and find a potential function for $\mathbf{F}$.

- 17 If $\int_C M(x, y, z) dx + N(x, y, z) dy + P(x, y, z) dz$ is independent of path and M , N , and P have continuous first partial derivatives, prove that

$$\frac{\partial M}{\partial y} = \frac{\partial N}{\partial x}, \quad \frac{\partial M}{\partial z} = \frac{\partial P}{\partial x}, \quad \frac{\partial N}{\partial z} = \frac{\partial P}{\partial y}.$$

- 18 Prove the analogue of Theorem (18.14) for line integrals in three dimensions.

Use Exercise 17 to prove that the line integrals in Exercises 19 and 20 are not independent of path.

19 $\int_C 2xy dx + (x^2 + z^2) dy + yz dz$

20 $\int_C e^y \cos z dx + xe^y \cos z dy + xe^y \sin z dz$

- 21 If $\mathbf{F}$ is a constant force, prove that the work done along any curve with endpoints P and Q is $\mathbf{F} \cdot \overline{PQ}$.
- 22 If $\mathbf{F}(x, y, z) = g(x)\mathbf{i} + h(y)\mathbf{j} + k(z)\mathbf{k}$, where g , h , and k are continuous functions of one variable, show that $\mathbf{F}$ is conservative.

- 23 Let $\mathbf{F}(x, y) = M(x, y)\mathbf{i} + N(x, y)\mathbf{j}$, where

$$M(x, y) = -y/(x^2 + y^2) \quad \text{and} \quad N(x, y) = x/(x^2 + y^2).$$

Show that $\partial M/\partial y = \partial N/\partial x$ for all (x, y) in the domain D of $\mathbf{F}$, but that $\int_C \mathbf{F} \cdot d\mathbf{r}$ is not independent of path in D . (Hint: Consider two different semi-circles with endpoints $(-1, 0)$ and $(1, 0)$.) Why doesn't this contradict Theorem (18.16)?

- 24 Suppose a satellite of mass m is orbiting the earth at a constant altitude of h miles, and that it completes one revolution every k minutes. If the radius of the earth is taken as 4,000 miles, find the work done by the gravitational field of the earth during any time interval.

- 25 Let $\mathbf{F}$ be an inverse square field such that $\mathbf{F}(x, y, z) = (c/|\mathbf{r}|^3)\mathbf{r}$, where $\mathbf{r} = x\mathbf{i} + y\mathbf{j} + z\mathbf{k}$ and c is a constant. Let P_1 and P_2 be any points whose distances from the origin are d_1 and d_2 , respectively. Express the work done by $\mathbf{F}$ along any piecewise smooth curve joining P_1 to P_2 in terms of d_1 and d_2 .

- 26 Suppose that as a particle moves through a conservative vector field, its potential energy is decreasing at a rate of k units per second. At what rate is its kinetic energy changing?

18.4 Green's Theorem

Let a smooth plane curve C have parametric representation $x = g(t)$, $y = h(t)$, where $a \leq t \leq b$. Recall that if $A = (g(a), h(a)) = (g(b), h(b)) = B$, then C is a smooth *closed* curve. If, in addition, $(g(t_1), h(t_1)) \neq (g(t_2), h(t_2))$ for all other numbers t_1, t_2 in $[a, b]$, then C does not cross itself between A and B and it is referred to as *simple*. (See Section 13.1.) Some common examples of smooth simple closed curves are circles and ellipses. A *piecewise smooth simple closed curve* consists of a finite union of smooth curves C_i such that as t varies from a to b , the point $P(t)$ obtained from parametric representations of the C_i traces C exactly one time, with the exception that $P(a) = P(b)$. A curve of this type forms the boundary of a region R in the plane and, by definition, the positive direction along C is such that R is on the left as $P(t)$ traces C . This is illustrated in Figure 18.9, where the arrows indicate the positive direction along C . The symbol

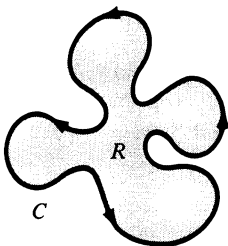


FIGURE 18.9

$$\oint_C M(x, y) dx + N(x, y) dy$$

will denote a line integral along C once in the positive direction.

The next theorem, named after the English mathematical physicist George Green (1793–1841), indicates the relationship between the line integral around C and a double integral over R . To simplify the statement, the symbols M , N , $\partial M/\partial x$, and $\partial N/\partial y$ are used in the integrands to denote the values of these functions at (x, y) .

Green's Theorem (18.17)

Let C be a piecewise smooth simple closed curve and let R be the region consisting of C and its interior. If M and N are functions that are continuous and have continuous first partial derivatives throughout an open region D containing R , then

$$\oint_C M dx + N dy = \iint_R \left(\frac{\partial N}{\partial x} - \frac{\partial M}{\partial y} \right) dA.$$

Partial proof We shall prove the theorem for a region R that is both of Type I and of Type II. Thus we may write

$$R = \{(x, y): a \leq x \leq b, g_1(x) \leq y \leq g_2(x)\}$$

and

$$R = \{(x, y): c \leq y \leq d, h_1(y) \leq x \leq h_2(y)\}$$

where the functions g_1, g_2, h_1 , and h_2 are smooth. An illustration of a region of this type appears in Figure 18.10. It is sufficient to show that each of the following is true:

$$\begin{aligned} \text{(a)} \quad \oint_C M dx &= - \iint_R \frac{\partial M}{\partial y} dA \\ \text{(b)} \quad \oint_C N dy &= \iint_R \frac{\partial N}{\partial x} dA. \end{aligned}$$

Note that addition of these integrals produces the desired conclusion.

To prove (a) we refer to (i) of Figure 18.10 and note that C consists of two curves C_1 and C_2 that have equations $y = g_1(x)$ and $y = g_2(x)$, respectively. The line integral $\oint_C M dx$ may be written

$$\begin{aligned} \oint_C M dx &= \int_{C_1} M(x, y) dx + \int_{C_2} M(x, y) dx \\ &= \int_a^b M(x, g_1(x)) dx + \int_b^a M(x, g_2(x)) dx \\ &= \int_a^b M(x, g_1(x)) dx - \int_a^b M(x, g_2(x)) dx. \end{aligned}$$

Applying (i) of Theorem (17.9) to the double integral $\iint_R (\partial M / \partial y) dA$ gives us

$$\begin{aligned} \iint_R \frac{\partial M}{\partial y} dA &= \int_a^b \int_{g_1(x)}^{g_2(x)} \frac{\partial M}{\partial y} dy dx \\ &= \int_a^b M(x, y) \Big|_{g_1(x)}^{g_2(x)} dx \\ &= \int_a^b [M(x, g_2(x)) - M(x, g_1(x))] dx. \end{aligned}$$

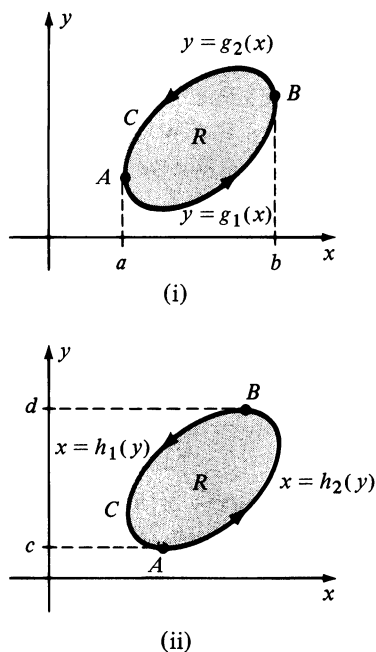


FIGURE 18.10

Comparing the last expression with that obtained for $\oint_C M dx$ gives us (a). The formula in (b) may be established in similar fashion by referring to (ii) of Figure 18.10. This is left as an exercise. $\square$

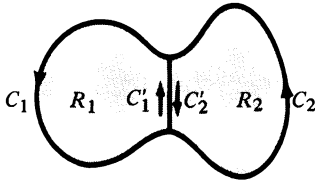


FIGURE 18.11

Although we shall not prove it, Green's Theorem is true for regions where part of the boundary consists of horizontal or vertical line segments. The theorem may then be extended to the case where R is a finite union of such regions. For example if, as illustrated in Figure 18.11, $R = R_1 \cup R_2$, where the boundary of R_1 is $C_1 \cup C_1'$ and the boundary of R_2 is $C_2 \cup C_2'$, then

$$\iint_{R_1} \left(\frac{\partial N}{\partial x} - \frac{\partial M}{\partial y} \right) dy dx = \oint_{C_1 \cup C_1'} M dx + N dy$$

$$\iint_{R_2} \left(\frac{\partial N}{\partial x} - \frac{\partial M}{\partial y} \right) dy dx = \oint_{C_2 \cup C_2'} M dx + N dy.$$

The sum of the two double integrals equals a double integral over R . We next observe that a line integral along C_1' in the direction indicated in Figure 18.11 is the negative of that along C_2' since the curve is the same but the directions are opposite one another. It follows that the sum of the two line integrals reduces to a line integral along $C_1 \cup C_2$, which is the boundary C of R . Thus

$$\iint_R \left(\frac{\partial N}{\partial x} - \frac{\partial M}{\partial y} \right) dy dx = \oint_C M dx + N dy.$$

We may now proceed, by mathematical induction, to any finite union. The proof of Green's Theorem for the most general case is beyond the scope of this book.

Example 1 Use Green's Theorem to evaluate $\oint_C 5xy dx + x^3 dy$, where C is the closed curve consisting of the graphs of $y = x^2$ and $y = 2x$ between the points $(0, 0)$ and $(2, 4)$.

Solution The region R bounded by C is illustrated in Figure 18.12. Applying Green's Theorem, with $M(x, y) = 5xy$ and $N(x, y) = x^3$,

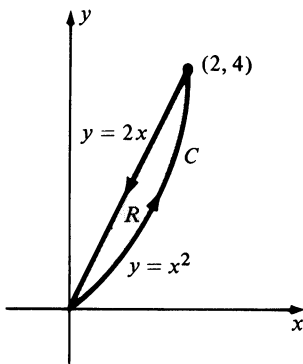


FIGURE 18.12

$$\begin{aligned} \oint_C 5xy dx + x^3 dy &= \iint_R \left[\frac{\partial}{\partial x} (x^3) - \frac{\partial}{\partial y} (5xy) \right] dA \\ &= \int_0^2 \int_{x^2}^{2x} (3x^2 - 5x) dy dx \\ &= \int_0^2 \left[3x^2 y - 5xy \right]_{x^2}^{2x} dx \\ &= \int_0^2 (11x^3 - 10x^2 - 3x^4) dx = -\frac{28}{15}. \end{aligned}$$

Of course, the line integral can also be evaluated directly. $\blacksquare$

Example 2 Use Green's Theorem to evaluate the line integral

$$\oint_C 2xy \, dx + (x^2 + y^2) \, dy$$

where C is the ellipse $4x^2 + 9y^2 = 36$.

Solution If R is the region bounded by C , then applying Green's Theorem with $M(x, y) = 2xy$ and $N(x, y) = x^2 + y^2$,

$$\begin{aligned} \oint_C 2xy \, dx + (x^2 + y^2) \, dy &= \iint_R (2x - 2x) \, dA \\ &= \iint_R 0 \, dA = 0. \end{aligned}$$

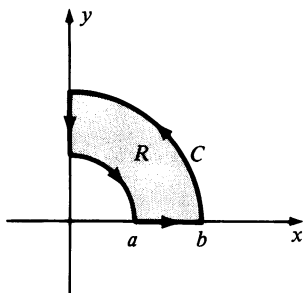


FIGURE 18.13

Example 3 Evaluate $\oint_C (4 + e^{\sqrt{x}}) \, dx + (\sin y + 3x^2) \, dy$, where C is the boundary of the region R in Fig. 18.13, between quarter circles of radii a and b , and the indicated segments on the x - and y -axes.

Solution Applying Green's Theorem, and then changing the variables in the double integral to polar coordinates, the given line integral equals

$$\begin{aligned} \iint_R (6x - 0) \, dA &= \int_0^{\pi/2} \int_a^b (6r \cos \theta) r \, dr \, d\theta \\ &= \int_0^{\pi/2} 2r^3 \cos \theta \Big|_a^b \, d\theta = \int_0^{\pi/2} 2(b^3 - a^3) \cos \theta \, d\theta \\ &= 2(b^3 - a^3) \sin \theta \Big|_0^{\pi/2} = 2(b^3 - a^3). \end{aligned}$$

The reader will gain a deeper appreciation of Green's Theorem by attempting to evaluate the line integral in this example directly. ■

Green's Theorem may be used to derive an interesting and useful formula for finding the area A of a region R that is bounded by a piecewise smooth simple closed curve C . On the one hand, if we let $M = 0$ and $N = x$ in (18.17) we obtain

$$A = \iint_R dA = \oint_C x \, dy.$$

On the other hand, if we let $M = -y$ and $N = 0$ the result is

$$A = \iint_R dA = -\oint_C y \, dx.$$

We may combine these two formulas for A by adding both sides of the equations and dividing by 2. This gives us the following result.

Theorem (18.18)

If a region R in the xy -plane is bounded by a piecewise smooth simple closed curve C , then the area A of R is

$$A = \frac{1}{2} \oint_C x \, dy - y \, dx.$$

Example 4 Use Theorem (18.18) to find the area of the ellipse $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$.

Solution Parametric equations for the ellipse C are $x = a \cos t$, $y = b \sin t$, where $0 \leq t \leq 2\pi$. Hence

$$\begin{aligned} A &= \frac{1}{2} \oint_C x \, dy - y \, dx \\ &= \frac{1}{2} \int_0^{2\pi} (a \cos t)(b \cos t) \, dt - (b \sin t)(-a \sin t) \, dt \\ &= \frac{1}{2} \int_0^{2\pi} ab(\cos^2 t + \sin^2 t) \, dt \\ &= \frac{ab}{2} \int_0^{2\pi} dt = \frac{ab}{2} (2\pi) = \pi ab. \end{aligned}$$

Green's Theorem can be extended to a region R that contains holes, provided we integrate over the entire boundary and always keep the region R to the left. This is illustrated in Figure 18.14, where it can be shown that the double integral over R equals the sum of the line integrals along C_1 and C_2 in the indicated directions. The proof consists of making a slit in R as illustrated in the figure and then noting that the sum of two line integrals in opposite directions along the same curve is zero. A similar argument can be used if the region has several holes. We shall use this observation in the solution of the next example.

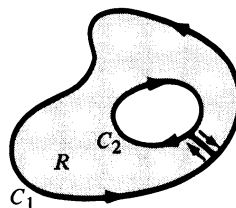


FIGURE 18.14

Example 5 Let C_1 and C_2 be two nonintersecting piecewise smooth simple closed curves having the origin O as an interior point. If $M = -y/(x^2 + y^2)$ and $N = x/(x^2 + y^2)$, prove that

$$\oint_{C_1} M dx + N dy = \oint_{C_2} M dx + N dy.$$

Solution If R denotes the region between C_1 and C_2 , then we have a situation similar to that illustrated in Figure 18.14, where O is inside of C_2 . By the remarks preceding this example,

$$\oint_{C_1} M dx + N dy + \oint_{C_2} M dx + N dy = \iint_R \left(\frac{\partial N}{\partial x} - \frac{\partial M}{\partial y} \right) dA,$$

where $\bigcirc$ indicates the positive direction along C_2 with respect to R . Since

$$\frac{\partial N}{\partial x} = \frac{(x^2 + y^2)(1) - x(2x)}{(x^2 + y^2)^2} = \frac{y^2 - x^2}{(x^2 + y^2)^2} = \frac{\partial M}{\partial y}$$

the double integral over R is zero. Consequently,

$$\oint_{C_1} M dx + N dy = - \oint_{C_2} M dx + N dy.$$

Since the positive direction along C_2 with respect to the region in the interior of C_2 is opposite to that indicated in the last integral we have

$$\oint_{C_1} M dx + N dy = \oint_{C_2} M dx + N dy$$

which is what we wished to prove. ■

Example 6 If $\mathbf{F}(x, y) = \frac{1}{x^2 + y^2} (-y\mathbf{i} + x\mathbf{j})$, prove that $\oint_C \mathbf{F} \cdot d\mathbf{r} = 2\pi$ for every piecewise smooth simple closed curve C having the origin in its interior.

Solution The line integral has the same form as those considered in Example 5. Hence, if we choose a circle C_1 of radius a with center at the origin that lies entirely within C , then by Example 5,

$$\oint_C \mathbf{F} \cdot d\mathbf{r} = \oint_{C_1} \mathbf{F} \cdot d\mathbf{r}.$$

Since parametric equations for C_1 are

$$x = a \cos t, \quad y = a \sin t; \quad 0 \leq t \leq 2\pi,$$